



PayPal Integration Guide v25.3.0

FOR SALESFORCE COMMERCE CLOUD



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1 Summary

This Implementation Guide describes how to integrate `int_paypal`, `bm_paypal` cartridges version 25.3.0 into Commerce Cloud reference application Store Front Reference Architecture (7.0.0 SFRA).

Cartridges include:

- PayPal Checkout from cart and payment pages
- Apple Pay Checkout from product, cart and payment pages
- Google Pay Checkout from cart and payment pages
- Vault flow for Credit Card and PayPal
- Review and post-process PayPal transactions in the Business Manager
- Dynamic Smart Button styling management from the Business Manager
- Configure custom site preferences in the Business Manager

PayPal integration uses PayPal REST Order API for checkout and Payment API for transaction post processing in Business Manager.

For more information, contact your PayPal support manager.

2 Component Overview

2.1 Supported Features

2.1.1 Smart Payment Button

PayPal Checkout with Smart Payment Buttons gives your buyers a simplified and secure checkout experience. PayPal intelligently presents the most relevant payment types to your shoppers, automatically, making it easier for them to complete their purchase using methods like Pay with Venmo, PayPal Credit, credit card payments.

[Button Demo](#)

NOTE: An authorized PayPal user has their own separate session and saves data separately for the domain from which the SDK is loaded (cookies, localStorage). The session is not cleared automatically when the user performs login/logout from their account.

2.1.2 Billing Agreement and Reference Transactions (Deprecated)

From version 24.2.0 Billing Agreement is deprecated and new Vault flow is introduced.

2.1.3 Alternative Payment Methods

Alternative payment methods allow customers across the globe to pay with their bank accounts, wallets, and local payment methods. Relevant alternative payment methods are automatically presented with Smart Payment Buttons.

To learn more, see [Alternative Payment Methods overview](#).

Current solution uses unbranded experience. See an example here:

[Unbranded APM](#)

UPDATE: Now the integration supports alternative payment methods:

- Apple Pay
- Google Pay
- Blik - for Polish locale.
- Bancontact - for Belgian locale.
- EPS - for Austrian locale.
- iDEAL - for Dutch locale.
- MyBank - for Italian locale.
- P24 - for Polish locale.

2.1.4 PayPal Credit Card Payment Processing

Card Fields is a way to accept credit card payments securely using PayPal JavaScript SDK. PayPal's payment gateway connects your storefront to any processor or bank that it supports.

Important: PayPal Card Fields are applicable for the merchants who enabled advanced credit and debit card payments.

Check: To confirm that Advanced Credit and Debit Card Payments are enabled for you, check your sandbox business account as follows:

1. Log into the [PayPal Developer Dashboard](#), go to Apps & Credentials. In the REST API apps, select the name of your app.
2. Go to Features > Accept payments.
3. Select the Advanced Credit and Debit Card Payments checkbox and select Save Changes.

Important: If the advanced credit and debit card payment status for the account is disabled, the merchant needs to ask for permission from PayPal.

NOTE: If you as a merchant are making any changes in Credit/Debit cards settings (Merchant tools > Payment methods > Manage Credit/Debit Cards), to go smoothly with checkout, make sure the type of credit card matches one of the following below:

- Visa
- MasterCard
- American Express
- Diners Club*
- Discover
- JCB
- UnionPay*
- Maestro
- Elo
- Hiper
- Hipercard

2.1.5 Limit of saved credit cards

Allows setting the limit for saving credit cards to the buyer's wallet. The value of the property is equal to the number of credit cards which a logged-in user is allowed to save. By default, it is Unlimited.

NOTE: If you decrease the value, already saved credit cards will not be deleted.

2.1.6 Transactions post processing

Orders paid with integration can be post-processed from a separate Business Manager module. Post-processing includes transaction capture, refund, and void. Captured orders can be connected to PayPal Tracking API. Also, you can create a new transaction.

Check [Business Manager](#) chapter for more details.

2.1.7 PayPal-provided Billing Address and Phone Number

To retrieve a buyer's billing address and phone number from the PayPal Checkout flow, please contact your PayPal representative or PayPal Support to enable them.

Important: Retrieving billing address and phone number should be enabled in merchant account before PayPal button will be enabled on a cart page.

2.1.8 Connect with PayPal

'Connect with PayPal' enables users to create an account for your website and log in using their PayPal credentials.

NOTE: In case user logs in via Connect with PayPal for the first time they will also receive the credentials for login via a Single Authentication (Demandware standard login).

Also, the user can access their pre-existing user's account (if it exists) via Connect with PayPal if the email of the pre-existing user account matches with the PayPal email.

When a customer links or unlinks their PayPal account, they will receive an email notification.

NOTE: we do not recommend using Login with PayPal functionality if you have custom options, namely the possibility of changing customers' emails.

2.1.9 Webhook Support

The purpose of the webhooks is to reflect changes which were done manually or automatically on PayPal console level.

For now, only the following hooks are supported: **Payment authorization voided**, **Payment capture refunded**, and **Payment capture completed**, **Vault payment-token deleted**, **Customer dispute created**, **Customer dispute resolved**, **Customer dispute updated**.

2.1.10 PayPal Vault Flow

Starting from 24.2.0 plugin version, **Billing Agreement** is deprecated, and new **Vault flow** is introduced.

To Enable Vault flow "Store PayPal account in Vault" Site Preference has to be set to "Enabled".

In order to migrate from Billing Agreement to Vault flow the 'PayPalMigrateBillingAgreementIDsToVaultTokens' job should be run. This job converts saved PayPal accounts, namely Billing Agreement IDs to Vault tokens. If the job was not run and user open My Account page or login from Checkout page with saved BA accounts, they will be automatically converted to the Vault tokens in real time.

NOTE: we strongly recommend to run 'PayPalMigrateBillingAgreementIDsToVaultTokens' job before the migration.

2.1.11 Google Pay

Google Pay is a mobile payment and digital wallet service provided by Alphabet Inc.

Buyers can use Google Pay on PayPal to make payments on the web using a web browser.

Sellers can use PayPal with Google Pay to sell physical goods, such as clothes and electronics, and intangible professional services, such as concerts or gym memberships.

For more details on compatibility and availability please read [Google Pay support article](#).

2.1.12 Fastlane by PayPal

Fastlane by PayPal is a new one-click guest checkout experience that merchants utilizing PayPal's platform can offer to their shoppers. This allows customers to make quick and hassle-free purchases. With Fastlane, customers can save their information for a checkout process that can be as quick as a single tap. There is no need to remember a username or password, no need to update personal information, and no need to share a credit card with multiple businesses over the internet.

NOTE: Fastlane is available for the US-based merchants only and only in the scope of int_paypal cartridge. On storefront Fastlane is available only to guest users and as an alternative Credit Card payment method.

2.1.13 PayPal App Switch

App Switch allows PayPal customers to start a transaction in a browser or app and complete it in the PayPal app, streamlining checkout with strong multi-factor authentication. The buyer can use biometrics and passkey to log in instead of entering their password.

The buyer experience falls back to the web flow in cases where App Switch isn't available.

NOTE: It is recommended to use native device browser as a default browser to avoid any issues.

2.1.14 Contact Module

The Contact Module helps buyers view and modify the email and phone number shared with merchants for a given order. It offers greater flexibility and control to buyers, particularly for gift orders where buyers need to specify alternative contact details.

Please see detailed information by the link:

<https://developer.paypal.com/docs/checkout/standard/customize/contact-module/>

The screenshot displays the PayPal Contact Module interface. At the top, the PayPal logo is visible. Below it, there are three columns of information: the total amount (\$293.98 today), payment options (4 interest-free payments of \$73.50), and financing plans (Plans as low as \$28.08/mo for 12 months). Each column has a 'Terms' link. Below this, the 'Pay with' section shows 'CREDIT UNION 1' with a dropdown arrow. The 'Ship to' section shows 'Joy Gray' with a dropdown arrow. The 'Contact' section is expanded, showing 'Email' with 'test_billing@g.com' and 'Phone' with '+1 02025550162'. Both contact fields have a 'To change, return to Store1' link. At the bottom, there is a 'Review Order' button and a link to 'About payment methods'.

Figure 2.1.2.1 Billing Flow

PLEASE NOTE: To ensure that the contact module displays correctly, please configure the phone number input field to support entry of a valid country code and allow a minimum of ten digits for the phone number.

Country code required

Min 10 digits

+1 1234567890

Figure 2.1.2.2 Phone number formatting

PL **PayPal**

for 12 months. [Terms](#) [Terms](#)

Pay with **CREDIT UNION 1** [▼](#)
Checking **6425

Ship to **Joy Gray** [▼](#)
473 Wiseman Street, Sevierville, CT 37862

Contact [^](#)

Email

☒ test_billing@g.com
To change, return to Store1

Phone

☒ (408) 425-9805

☐ +380 484049633

[Review Order](#)

[About payment methods](#)

Figure 2.1.2.3 Express Checkout Logged-In Customer

Figure 2.1.2.4 Express Checkout Guest

NOTE: The Contact Module is currently **available in the US only**.

2.1.15 BNPL (Buy Now Pay Later) Configurator

The Pay Later Messaging Configurator is a drop-in JavaScript module that added to Business Manager. Merchants can access the admin panel and use the Messaging Configurator to set up, customize, and preview Pay Later messaging. Based on the merchant’s configuration, you can display the Pay Later messages to buyers.

2.2 Privacy and Payment

This integration requires access to the following customer data elements: Shipping Address, Order Details, Customer Profile.

3 Implementation Guide

3.1 Overview

Two cartridges support this integration:

1. int_paypal - Commerce Cloud storefront extension
2. bm_paypal - Business Manager extension “Transactions”, “Disputes”, “Preference Configuration” and “Styles Configuration”

NOTE: Starting from 23.4.0 release cartridge paypal_credit_financing_options is removed and no longer supported as deprecated.

NOTE: Latest supported OCAPI (int_paypal_ocapi) is 24.1.0 and it is not maintained starting from 24.2.0

3.2 Installation

3.2.1 Import metadata archive

Before installing the cartridge, make sure that you have a working sandbox with a site based on SFRA installed.

1. Create metadata.zip from the metadata folder then upload and import it. To do so, go to Business Manager > Administration > Site Development > Site Import & Export. Upload the archive using ‘Local’ option in the Upload Archive section.
2. After uploading choose metadata.zip in the list and click on import button.

Site Import & Export

This page allows you to export the current configuration of your organization including all of its sites. To download an archive, just click its file name.

Import

Upload Archive:
☒ Local ☐ Remote

No file selected.

Select	Name	Location	File Size	Last Modified
☒	metadata.zip	local	6.08 KB	8/27/20 11:26:08 am
☐	SiteGenesis Demo Site			
☐	Storefront Reference Architecture Demo Sites			

Figure 3.2.1. metadata.zip archive import

3.2.2 Add cartridge to cartridge path

Add int_paypal into the cartridge path of a target site. To do so, go to Business Manager > Administration > Sites > Manage Sites > Your Target Site > Settings and insert int_paypal before your cartridges record, as shown in Figure 3.2.2.

RefArch - Settings

Click Apply to save the details. Click Reset to revert to the last saved state.

Instance Type:

Deprecated. The preferred way of configuring HTTP and HTTPS hostnames is by using new features of the site aliases configuration ("SEO > Aliases Configuration"). The HTTP/HTTPS h configuration and are intended only to support an older configuration style.

HTTP Hostname:

HTTPS Hostname:

Instance Type: All

Cartridges:

Effective Cartridge Path:

- int_paypal
- app_storefront_base
- plugin_apple_pay
- plugin_facebook
- plugin_pinterest_commerce
- plugin_web_payments
- bc_content
- core

Figure 3.2.2. Storefront cartridge path

3.2.3 Adding API Credentials

Services:

- int_paypal.http.rest - is used for the main plugin logic (Smart Button, manipulations with transactions, etc.).

To access credentials, go to Business Manager > Administration > Operations > Services > Credentials.

You need to fill the next credentials with the identical data:

- NAME: Paypal_Sandbox_Credentials
SB URL: <https://api.sandbox.paypal.com/>

Prod URL: <https://api.paypal.com/>

User: Client Id

Pass: Secret

For both credentials use Client Id as **User** and Secret as **Password**. Client Id and Secret can be obtained in the app details at the PayPal developer portal.

For more details about REST apps and credentials please visit [Get credentials](#) page at the PayPal Developer Portal.

UPDATE: Current integration supports only one credentials usage at a time. If you wish to use multiple credentials at one sandbox you should customize cartridge at PayPalRestService.js file.

(Optional) For non-production activities we recommend enabling the communication log for the int_paypal.http.rest service. The communication log will log every request and response to the log files. To do this navigate to the Business Manager > Administration > Operations > Services and click on int_paypal.http.rest. Check Communication Log Enabled checkbox (Figure 3.2.3)

PLEASE NOTE: By enabling the communication log, some details such as customer's personal information including address, phone, and email will be logged in cleartext. This should only be used for debugging purposes. We recommend, if possible, to only use this in a sandbox; however, when used in production, ensure you are disabling / unchecking the communication log box immediately after you are done with your debugging. Further, some credentials will also be saved in cleartext.

[Administration](#) > [Operations](#) > [Services](#) > int_paypal.http.rest - Details

int_paypal.http.rest[?]

Fields with a red asterisk (*) are mandatory. Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

Name:	*	int_paypal.http.rest
Type:		HTTP
Enabled:		☒
Service Mode:		Live
Log Name Prefix:		PayPalRest
Communication Log Enabled:		☒
Force PRD Behavior in Non-PRD Environments:		☐
Profile:		PayPal_Profile
Credentials:		paypal_sandbox_credentials

Figure 3.2.3. Service Example

3.2.4 PayPal Transaction and PayPal Styles Configuration Business Manager modules installation

Add **bm_paypal** into the record of the Business Manager cartridge path. Go to **Business Manager > Administration > Sites > Manage Sites > Manage the Business Manager Site > Settings** as shown in Figure 3.2.4.

[Administration](#) > [Sites](#) > [Manage Sites](#) > Business Manager - Settings

Settings Cache Hostnames

Business Manager - Settings

Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

Instance Type: Sandbox/Development ▾

Deprecated. Up to two instance specific hostname aliases for Business Manager can be configured here.
⚠ Setting a different hostname here will make some Business Manager modules unreachable. Manage additional hostnames in the 'Hostnames' tab instead.

HTTP Hostname:

HTTPS Hostname:

Instance Type: All

Cartridges: bm_paypal

Effective Cartridge Path: app_business_manager:plugin_apple_pay:plugin_facebook:plugin_payments:plugin_pinterest_comr

Figure 3.2.4. Business Manager Cartridge Path

Grant access to Business Manager modules:

- Go to Business Manager > Administration > Organization > Roles & Permissions and select the target role that needs to have access to PayPal Transaction management (e.g., the Administrator role)
- Select the Business Manager Modules tab in the drop-down list on top. You must select your target site as the active context.
- Scroll to the PayPal Plugin module. Grant access to it by selecting the checkbox (Transactions, Styles Configuration, Preference Configuration, Disputes) and clicking Update.

NOTE: before granting assets to your Business Manager modules, the **bm_paypal** cartridge must be uploaded, and the correct cartridge path must be entered.

Once you do this, every Business Manager user with that role will be able to manage PayPal Transactions by selecting Business Manager > Merchant Tools > PayPal Plugin > Transactions, change Smart Button styling under Business Manager > Merchant Tools > PayPal Plugin > **Styles Configuration** and change Custom site preferences under Business Manager > Merchant Tools > PayPal Plugin > **Preference Configuration**, manage PayPal Disputes by selecting Business Manager > Merchant Tools > PayPal Plugin > **Disputes**.

3.2.5 Custom cache configuration

Custom cache required for fast and stable work on the integration. Navigate to Business Manager > Administration > Operations > Custom Caches and ensure that the "Enable Caching" checkbox is selected.

There are 1 custom cache objects used by the integration:

1. paypalRestOAuthToken - OAUTH token for communication with PayPal REST API. Out of the box, the integration supports single site usage and the token retrieved from PayPal is cached. For multiple credentials usage we recommend removing the cache definition from caches.json file in the cartridge.

Access tokens have a finite lifetime. In general, access tokens have a lifespan of 900 seconds (15 minutes) or 32400 seconds (nine hours), depending on the associated scopes. You can find more information about the access token here: [Access token](#)

'paypalRestOAuthToken' cache definition has an additional value:

'expireAfterSecond' - a cache finite lifetime with **900** default value.

There are cases when 'expireAfterSecond' value can be higher than the access token lifetime and it can cause an error during PayPal REST API server usage.

To avoid such errors, we recommend investigating the access token's 'expires_in' value by following this link: [Access token](#) and setting an appropriate value into 'paypalRestOAuthToken' cache. To do this, follow the next steps:

1. Go to 'int_paypal/caches.json' file in the project repository.
2. Adjust 'expireAfterSecond' value with one you got during investigation



```
{
  "caches": [
    {
      "id": "paypalRestOAuthToken",
      "expireAfterSeconds": 900
    }
  ]
}
```

Adjust this value

Figure 3.2.5. Caches.json file

UPDATE: Custom Cache could be disabled for developing or testing purposes. we do not recommend disabling it in production and staging ENVIRONMENTS to avoid performance issues and http calls quota violation.

3.3 Configuration

3.3.1 Updating Cartridge Custom Site Preferences

Go to Business Manager > Merchant Tools > PayPal Plugin > Preference Configuration. You will see a list of available tabs (Figure 3.3.1)

Figure 3.3.1 Preference Configuration

The list of plugin site preferences:

PayPal General tab

1. **Payment Model**, which determines whether the funds are captured immediately (Sale) or later (Authorize).
 - a. *Authorization* - authorize a payment by placing a hold on the customer's card for the funds.' Capture funds immediately' site pref. should be set to 'no'.
 - b. *Sale (capture)* - settling a purchase immediately, it means that authorization and capture occurs at the same time as the sale. Capture funds immediately' site pref. should be set to 'yes'.

NOTE: When the Authorize Payment Model is enabled, but transactions placed via LPMs would still follow the Sale flow.
2. **3DS verification** , which defines if Strong Customer Authentication should be triggered for buyers always, only when it is required or disabled.
 - a. *No Verification(DISABLED)* - disable 3DS.
 - b. *Smart Authorization (SCA_WHEN_REQUIRED)* - returns a 3D Secure contingency when it is a mandate in the region where you operate. This is the default when neither parameter is explicitly passed.
 - c. *3DS Verification Always (SCA_ALWAYS)* - trigger 3D Secure for every transaction, regardless of SCA requirements.
3. **Authorization and Capture webhook id**, which is Webhook id of payment authorization voided, payment capture refunded, and payment capture completed events.
4. **List of enable funding methods**, which allows setting up a list of additional enable funding methods for SDK URL creation (example: "paylater").
5. **List of disable funding methods**, which allows setting up a list of disable funding methods for SDK URL creation (example: "paylater" or/and "card").
6. **Enable Transaction Logs** - enables logging of the requests and responses from PayPal while creating transaction. You can review these requests\responses in the Transaction Details.

7. **OCAPI Configuration**, configuration for OCAPI. JSON format. List of fields: clientId, clientPassword, apiVersion, bmUserLogin, bmUserPassword.
8. **WebDAV Configuration**, configuration for WebDAV. JSON format. List of fields: host, username, password.
9. **Merchant Name**, which display the name of the business or individual receiving the payment in the Apple Pay, Google Pay and PayPal payment modal.

PayPal Configuration tab

1. **PayPal Button Location**, which allows setting up an additional PayPal button location on the website.
2. **PayPal Button messages location**, which shows messaging directly with buttons to promote Pay Later offers and other PayPal value propositions. Messaging is currently only supported for US merchants and US customers. Merchants must be eligible for Pay Later to display Pay Later offers with buttons.
3. **Store PayPal account in Vault**, regulates buyers ability to manage their vault flow:
 - Enabled - buyers are allowed to save new payment methods and use previously saved ones.
 - Partially Disabled - buyers can use only previously saved payment methods.
 - Disabled - buyers cannot save new payment methods or use previously saved ones.
4. **PayPal Debit Credit Button Enable**, which allows enabling PayPal Debit Credit Button on the storefront which will be displayed next to the PayPal button. Please note that the PayPal Debit Credit Button location will be the same as the location of the PayPal button. Pay attention that PayPal Debit Credit Button will not be rendered when PayPal Credit Card Payment Method is enabled.
5. **Pay Later Button Enable** which allows enabling PayPal Pay Later Button on the storefront which will be rendered next to the PayPal button. Please note that PayPal Pay Later Button location will be the same as the location of the PayPal button. Depending on the vault mode and client-id one of the buttons will be rendered.
6. **Digital Goods enabled**, the flow is designed for selling only Digital Goods, which do not require delivery to the shipping address. In case you enable this flow, the shipping options will be hidden on the storefront.
7. **PayPal Fraudnet Enabled**, which allows collecting buyer's browser-based data and passing it to PayPal Risk Service to help reduce fraud.
8. **Connect with PayPal button enabled**, which identifies whether the Connect with PayPal (CWPP) button is enabled on the Login and the Checkout pages.
9. **Connect with PayPal button url**, the field for entering a URL which is used for Connect with PayPal feature.
10. **Connect with PayPal agent login**, part of Connect with PayPal functionality that requires agent credentials to perform "Login On Behalf" action.
11. **Connect with PayPal agent password**, part of Connect with PayPal functionality requires agent credentials to perform "Login On Behalf" action.

12. **Account Linking Security Layer enabled**, which enables additional verification step (enter credentials from the pre-existing user account) during the account linking while your buyers use Connect with PayPal for the first time and the user account email matches with the PayPal primary account email.
13. **Automatic payment method adding enabled**, which enables automatic adding of the PayPal account and address to the user account if this site pref. is set to 'yes', PayPal Vault and Connect with PayPal button is enabled.
14. **Account Linking Security Layer enabled**, which enables additional security step where users should enter credentials for the existing account during first login via Connect with PayPal.
15. **Simplified PayPal Dispute Page**, which enables activating of the simplified UI of the PayPal Dispute page.

NOTE: If Simplified PayPal Dispute Page is enabled, PayPal disputes will not be stored on the Demandware side. In addition, PayPal disputes will be retrieved and updated from PayPal in real-time by using dispute API.

If Simplified PayPal Dispute Page is disabled, the full functionality of the Dispute page will be activated, i.e., search dispute section and dispute history will be shown. In this case PayPal disputes will be stored in the custom object on the Demandware side and will be retrieved and updated via PayPal Dispute webhooks.
16. **PayPal Returning Customer Experience Enabled**, activates redesigned PayPal button with returning customer experience for logged-in customers, namely saved PayPal account and the preferred funding source will be displayed on the PayPal button and it's possible to Check out without re-entering a payment method.

PayPal Credit Card tab

1. **Store Credit Card in Vault**, which regulates vault flow which allows saving credit cards for logged-in buyers:
 - Enabled - buyers are allowed to save new payment methods and use previously saved ones.
 - Partially Disabled - buyers can use only previously saved payment methods.
 - Disabled - buyers cannot save new payment methods or use previously saved ones.
NOTE: If you disable this site preference and the logged-in user has a saved credit card in the wallet, then they will not be able to use it on the checkout.
2. **Store Credit Card in Vault via My Account**. This site preference regulates vault options of saving Credit Cards only on My Account and becomes active only when "Store Credit Card in Vault" is Enabled. When set to "Yes" value, it allows User to save new Credit Cards from My Account.
3. **Verify Credit Card on My Account**, which enables verification while saving credit cards on the My Account page.

NOTE: 3DS verification can be requested automatically if it is required.
4. **Limit of saved credit cards**, which allows you to set the maximum number of cards that logged-in user can save.

NOTE: If you decrease the value, already saved credit cards will not be deleted.

5. **Credit Card Expire Notification Configuration**, which allows notifying and highlighting (show the respective label) logged-in buyers if their credit cards are expired or are going to expire.

Apple Pay tab

1. **Apple Pay Button Location**, which allows setting up an additional Apple Pay button location on the website.

Google Pay tab

1. **Google Pay Button Location**, which allows setting up an additional Google Pay button location on the website.
2. **Google merchant id**, which identifier issued after your website is approved by Google. Required when PaymentsClient is initialized with an environment property of PRODUCTION, but may be omitted in TEST environment.

Venmo tab

1. **Venmo Button Location on Express Checkout**, which allows setting up an additional Venmo button location on the website.

Pay Later Banners/Messaging tab

1. **Pay Later Cross-Border Messaging Enabled**, which enables Cross-border messages for Pay Later. It means that the text for Pay Later messages can be displayed according to your website location.

Fastlane by PayPal tab:

1. **Enable Fastlane**, which offers an accelerated checkout experience that recognizes guest shoppers and auto fills their details so they can pay in seconds.
2. **Fastlane Privacy**, which allows to control whether Fastlane by PayPal branding is shown by email field. Terms & Conditions: PayPal powers this accelerated checkout solution from Fastlane. Since you'll share consumers' email address with PayPal, please consult your legal advisors on the appropriate privacy settings for your business.
3. **Use Fastlane Payment UI**, which allows to control whether the PayPal Payment UI is used or the native UI of the partner.
4. **Cardholder Name**, which allows to control whether or not the cardholder name is displayed in the card field's UI. Note: Is not applicable for Payment UI component.
5. **List of domains**, which allows to add your domain name for Fastlane. You can use up to 5 domains, such as example.com

3.3.2 Job configuration for removal of outdated transaction

RemoveOutdatedPayPalTransaction job is deprecated in actual release.

3.3.3 Static values configuration

All configuration options are available in `cartridge/config/sdkConfig.js` file. You could change:

1. Allowed currencies. By default, all currencies are allowed. The currency list is available under `allowedCurrencies` variable. More details [Currency](#) (currency section)
2. Disable funds. By default, only alternative payment methods are disabled. More details [Disable funding](#) (disable-funding section) and [Alternative Payment Methods](#)
3. Also, in `cartridge/config/paypalConstants.js` file you can setup `SOFT_DESCRIPTOR` variable which will be processed as `soft_descriptor` field in transaction's `purchaseUnit` object. The soft descriptor is the dynamic text used to construct the statement that appears on a payer's card statement. More info on `soft_descriptor` can be found on [PayPal Order Documentation](#).

3.3.4 PayPal Credit Card Configuration

To enable Card Fields:

1. Activate **PayPal Credit Card** payment method. Navigate to Business Manager > Merchant Tools > Ordering > Payment Methods and enable **PayPal Credit Card** payment method with payment processor **PAYPAL**

2. Activate **PayPal Debit Credit Button** custom preference. Navigate to Business Manager > Merchant Tools > PayPal Plugin > Preference Configuration > PayPal and set value Yes for **PayPal Debit Credit Button Enable**

NOTE: Card Fields are applicable for **Sale** and **Authorize** flows

To add new credit type:

1. Navigate to Business Manager > Merchant Tools > Ordering > Payment Methods and enable **PayPal Credit Card** payment method with payment processor **Credit Card <PAYPAL_CREDIT>**
2. Go to Credit/Debit Cards tab and add new card with name and Type

The screenshot shows the 'Payment Methods' interface in the PayPal Business Manager. The 'Credit/Debit Cards' tab is selected. A table lists various payment methods, with 'CREDIT_CARD' highlighted. Below the table, the 'CREDIT_CARD Details' form is visible. The 'Payment Processor' dropdown is set to 'Credit Card <PAYPAL_CREDIT>'. Red boxes and numbers 1 and 2 highlight the 'Credit/Debit Cards' tab and the 'Payment Processor' dropdown respectively.

ID	Name	Enabled	Sort Order
ApplePay	ApplePay	Yes	11
BANK_TRANSFER	Bank Transfer	No	3
BML	Bill Me Later	No	5
CREDIT_CARD	Credit Card	Yes	8
DW_ANDROID_PAY	Android Pay	No	2
DW_APPLE_PAY	Apple Pay	No	1
GIFT_CERTIFICATE	Gift Certificate	No	4
GooglePay	GooglePay	Yes	6
PAYPAL_CREDIT_CARD	Paypal Credit Card	Yes	13
PayPal	PayPal	Yes	10
SRV	Secure Remote Commerce	Yes	7
Venmo	Venmo	Yes	12

CREDIT_CARD Details

Description:

Image: [HTML Editor](#) [Select](#)

Payment Processor: **Credit Card <PAYPAL_CREDIT>**

Countries: [All](#) [Edit](#)

Figure 3.3.2 Adding new type of Credit Card.

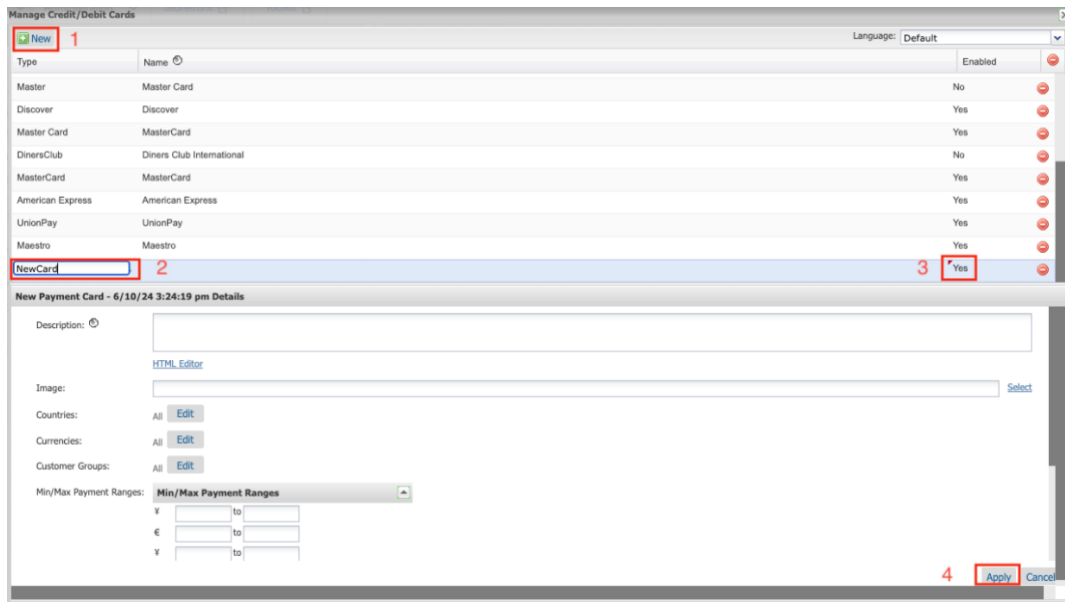


Figure 3.3.3 Adding new type of Credit Card.

3.3.5 Limit of saved Credit Cards

The Limit of Saved Credit Cards option allows to set a limit on the number of credit cards a customer can store in their wallet. The value of the property is equal to the number of credit cards that the logged in user can store.

(PP_Save_CC_On_My_Account) (Boolean)	Yes ☐ No ☐	Enables Credit Card functionality on My Account page.
Limit of saved credit cards	3 credit cards (3) unlimited	Allows setting the limit for saving credit cards to the buyer's wallet. The value of the property is equal to the number of credit cards which are possible to save.
(PP_Save_CC_Limit) (Enum of Integers)		
3DS verification for Credit Card		Allows triggering the following flows with 3D Secure on the Checkout page: 'always' (which triggers an authentication for every transaction), 'never' (which bypasses an authentication for every transaction), 'if-risky' (which triggers an authentication for risky transactions only).

Figure 3.3.4 Credit card limit preference

By default, is Unlimited. Other options are also available:

- 1 credit card
- 2 credit cards
- 3 credit cards
- 4 credit cards
- 5 credit cards
- Unlimited

NOTE: If you decrease the value, already saved credit cards will not be deleted.

3.3.6 Limit of saved PayPal accounts

The Limit of PayPal accounts option allows to set a limit on the number of PayPal accounts a customer can store in their wallet. The value of the property is equal to the number of PayPal accounts that the logged in user can store.

Limit of saved PayPal accounts

3 accounts (3)
1 account

Allows setting the limit for saving PayPal accounts to the buyer's wallet. The value of the property is equal to the number of PayPal accounts that are possible to save. Note: The recommended value is '1'.

(PP_Saved_Accounts_Limit)
(Enum of Integers)

Figure 3.3.5 PayPal Account limit preference

By default, is 1 account. Other options are also available:

- 1 account
- 2 accounts
- 3 accounts
- Unlimited

NOTE: If you decrease the value, already saved PayPal accounts will not be deleted.

3.3.7 Alternative Payment Methods Configuration

By default, all local payment methods are disabled. To enable payment method, navigate to Business Manager > Merchant Tools > Ordering > Payment Methods

Choose a payment method and set Enabled to Yes

The following are the IDS of alternative payment methods that are supported by the integration:

- bancontact
- eps
- ideal
- mybank
- p24
- blik
- ApplePay
- GooglePay

New Sort Order Credit/Debit Cards Import/Export				Language: Default	
ID	Name	Enabled	Sort Order		
DW_ANDROID_PAY	Android Pay	No	2		
DW_APPLE_PAY	Apple Pay	No	1		
GIFT_CERTIFICATE	Gift Certificate	No	4		
GooglePay	GooglePay	Yes	8		
PAYPAL_CREDIT_CARD	Paypal Credit Card	Yes	13		
PayPal	PayPal	Yes	10		
SRG	Secure Remote Commerce	Yes	9		
Veramo	Veramo	Yes	12		
bancontact	Bancontact	No	14		
eps	EPS	No	15		
ideal	IDEAL	No	16		
mybank	MyBank	No	17		
p24	P24	No	18		
sofort	Klarna Pay Now / SOFORT	No	6		

Figure 3.3.6. Payment Methods configuration

UPDATE: Alternative Payment Methods, except for Apple Pay and Google Pay, are available only on the billing page and are always using Sale Flow (Capture funds immediately) even when Payment Model in BM is set to Authorize flow.

3.3.8 Service Profile Configuration

By default, integration does not have any service profile configuration. We recommend putting limitation and timeout values according to storefront traffic to prevent fraud or potential attack. You can read more about configuration values [here](#).

3.3.9 SendFailedTransactionEmail Job Configuration

SendFailedTransactionEmail job checks transaction statuses for a specified period of time. In case it finds appropriate status, it will send an email to the specified address with the transaction statuses and amounts of these transactions. If there are no such transactions, then email will not be sent, and respective log will be recorded to the job log. The purpose of this job is to give merchants the ability to receive emails regarding the failed transaction in order to be aware of it and quickly make a certain decision.

This job can accept the following job parameters:

- **Email** - email address that is used for sending alerts.
- **DaysToRetrieve** - the period for which you need to check transaction statuses (number of days).
- **StatusesToCheck** - the statuses that will be considered to be checked, namely. For example: DENIED, DECLINED, FAILED.
- **AlertsInBM** - activates alerts for failed transactions in Business Manager (default: true)

Possible statuses list:

CREATED, PARTIALLY_CAPTURED, CAPTURED, VOIDED, PENDING, AUTHORIZED, COMPLETED, REFUNDED, PARTIALLY_REFUNDED, CANCELLED, EXPIRED, DENIED, DECLINED, FAILED

NOTE: Statuses in StatusesToCheck parameter must be specified as it is shown in Figure 3.3.7: in uppercase and with commas between the statuses.

NOTE: Parameter settings are not shipped with the job. Only DaysToRetrieve parameter has default value '7' and AlertsInBM value is 'true'. Other parameters must be filled in manually using the Business Manager interface (as shown in Figure 3.3.7).

Select and Configure Step

Custom.PaypalSendFailedTransactionEmail

Context: Site

ID*

PayPalSendFailedTransactionEmail

Description

Email

user@salesforce.com

Job Parameters

DaysToRetrieve*

7

Job Parameters

StatusesToCheck*

DENIED, DECLINED, FAILED

Job Parameters

☒ AlertsInBM

Job Parameters

Figure 3.3.7 sendFailedTransactionEmail parameters

NOTE: In the case of a large number of orders, it is better to run the job more often to prevent a possible sandbox overload and performance decrease.

NOTE: To see alert notifications in BM, you need to configure it. Go to **Administration > Operations > Notification Settings**. Find section **PayPal Plugin > Transactions** and choose where you want your alerts to appear (as shown in Figure 3.3.8).

Administration > Operations > Notifications

Configure Business Manager Notifications and Channels

Keep your systems running smoothly by notifying your teams when something needs their attention. You can configure the types of alerts that your teams receive and where they receive these alerts, such as email or Slack.

Settings

Slack

Event	Business Manager	Slack Channel
PayPal Plugin		
> Transactions 2 options selected Header X Homepage X		Select a channel
Checks the Transactions Module to see if there are any transactions with failed statuses.		

Figure 3.3.8 Notification settings

3.3.10 PayPalMigrateBillingAgreementIDsToVaultTokens Job Configuration

PayPalMigrateBillingAgreementIDsToVaultTokens job convert saved PayPal accounts (Billing Agreement IDs) into Vault tokens, enabling logged-in buyers to use their saved PayPal accounts that were saved via the Billing Agreement flow.

This job can accept the following job parameters:

- **sourceFolderPath:** Target folder path related to IMPEX/src
- **sourceFileName:** Source file name

NOTE: If the merchant does not use the job to convert BA IDs into Vault tokens, the system will perform this conversion in "real-time" when a user successfully logs into their storefront account.

NOTE: By default, value for saving folder is Impex/src/PayPalMigrateBillingAgreementIDsToVaultTokens. To change saving folder or name of the file parameters should be filled in manually using the Business Manager interface (as shown in Figure 3.3.9).

The screenshot shows a 'Select and Configure Step' dialog box. At the top, it says 'Custom.MigrateBillingAgreementIDsToVaultTokens' with a help icon. Below this, 'Context: Site' is displayed. The 'ID*' field contains 'MigrateBillingAgreementIDsToVaultTokens'. The 'Description' field is empty. The 'sourceFolderPath' field contains 'PayPalMigrateBillingAgreementIDsToVaultTokens' with a 'Job Parameters' link. The 'sourceFileName' field contains 'TestName' with a 'Job Parameters' link. At the bottom, there is a checkbox labeled 'Always execute on restart.' which is currently unchecked.

Figure 3.3.9 PayPalMigrateBillingAgreementIDsToVaultTokens parameters

NOTE: To see alert notifications in BM, you need to configure it. Go to **Administration > Operations > Notification Settings**. Find section **Operations > Jobs** and choose where you want your alerts to appear (as shown in Figure 3.3.10).

The screenshot shows the 'Operations > Jobs' notification settings interface. It includes a header with '2 options selected' and a 'Select a channel' dropdown. Below the header, there is a red-bordered box containing the message: 'Some of the Billing Agreement(s) not converted, for more details see the output file. Try to run job once again in order to convert all the BA IDs to Vault tokens.'

Figure 3.3.10 Notification settings

Profile system object must be added to the job resource lock, since the job modifies customer profile data.

3.3.11 Apple Pay Configurations

To enable Apple Pay, perform the following steps:

- 1) Complete sandbox onboarding steps:
<https://developer.paypal.com/docs/checkout/apm/apple-pay/#link-setupyoursandboxaccounttoacceptapplepay>
- 2) Complete Go live steps for production:
<https://developer.paypal.com/docs/checkout/apm/apple-pay/#link-golive>
- 3) Enable Apple Pay Payment Method in BM.
- 4) Configure Apple Pay Site Preference section to make Apple Pay button visible on pdp/pvp/cart/minicart pages.

NOTE: Apple Pay is available only in Safari on the following platforms:

- iOS 12.1 and later
- macOS 10.14.1 and later

3.3.12 Apple Pay Address fields validation

From Apple Pay API version 3 and later Apple Pay JS API supports customized errors. It gives an ability to create validation rules for Billing and Shipping addresses in Apple Pay payment sheet.

To read about the feature, please follow [the link](#).

The file for the setup validation is located at [cartridges/int_paypal/cartridge/scripts/paypal/helpers/applePayAddressValidation.js](#)

This file provides basic US locale validation rules for Apple Pay address fields. Based on those rules, Apple Pay pop-up will throw a customized error:

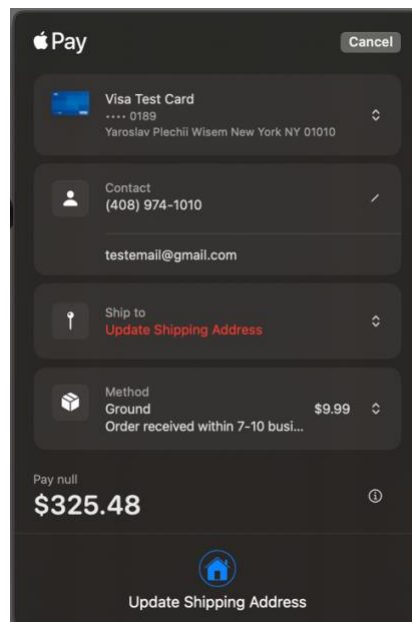


Figure 3.3.11 Apple Pay pop-up window.

These validation rules can be easily adjusted to meet the specific requirements simply by changing the appropriate regular expressions. It is also possible to create detailed errors by passing `errorMessage` for some specific field mismatch. For example:

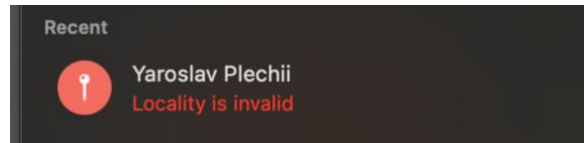


Figure 3.3.12 Apple Pay pop-up window Error message example.

The validation result can be passed to `completePayment` or `completeShippingContactSelection` methods. Some contact fields can be passed only to `completePayment` method, some can be passed to both. For more detailed information, check [Apple Pay documentation](#).

3.3.13 L2 L3 Data Processing

PayPal cartridge supports Level 2 and Level 3 data for Apple Pay, Credit Cards, PayPal, Venmo and LPM payments.

In Business Manager navigate to Merchant Tools -> PayPal Plugin -> Preference Configuration -> General tab and set Level 2 and 3 Processing preferences to Yes.

Some fields require additional configuration on code-level, and some are supported out of the box.

Level 2 and Level 3 data mapping:

L2 Fields

Invoice ID: Passed

Tax Total: Passed

L3 Fields

Shipping Amount: Passed

Discount Amount: Passed

Ships From Postal Code: **Not passed**

Shipping Address: Passed

Line Items

- Name: Passed
- Description: Passed
- Quantity: Passed
- Unit Amount: Passed (This value cannot be negative or zero.)
- SKU: Passed
- URL: Passed
- UPC: Passed
- Unit of Measure: **Not passed**

- Total Amount: Passed
- Tax: Passed
- Discount Amount: Passed
- Commodity Code: **Not passed**

3.4 Webhook support

The purpose of the webhooks is to reflect changes which were done manually or automatically on PayPal console level.

To configure webhook support, follow [this](#) PayPal guide.

Steps for configuration:

1. Go to the [PayPal Developer](#) and log in to the Dashboard with your account.
2. Navigate to "Apps & Credentials" and select or create your REST API app.
3. Scroll to the Webhooks and add new or select existing.
4. Enter a URL to your site with an endpoint Paypal-WebHooks.
The general rule for creation URL is:
{your environment}.
dx.commercecloud.salesforce.com/on/demandware.store/{your site}/{your local}/{controller}
(Ex. https://xxx-278.dx.commercecloud.salesforce.com/on/demandware.store/Sites-RefArch-Site/en_US/Paypal-WebHooks).
5. Only include events that are needed.
6. Save and copy the **Webhook ID**.
7. Enter it to the Custom Site Preference with ID: PayPal Webhooks ID (*PP_WH_ID*) (which is located inside PayPal Plugins -> Preference Configuration > tab "General").

Sandbox Webhooks

Configure webhooks to notify your app when certain events occur. To configure a webhook, define your webhook listener URL and a list of events for which to listen. You can configure up to ten webhooks. Each webhook can subscribe to either specific events or all events. To learn more about webhooks, see [webhooks notifications](#).

Webhook	Webhook ID	Events Tracked	
https://dx.commercecloud.salesforce.com/on/demandware.store/Sites-RefArch-Site/en_US/Paypal-WebHooks	*****	Customer dispute created, Customer dispute resolved, Customer dispute updated, Payment authorization created, Payment authorization voided, Payment capture completed, Payment capture refunded, Vault payment-token created, Vault payment-token deleted	 

Figure 3.4.1 Example of created Webhook in PayPal Developer account

NOTE: It does not have additional notifications regarding changes.

NOTE: you may subscribe only to the events listed below.

No	Webhook name	Description
1	Payment authorization voided	It is triggered when a payment has been voided.
2	Payment capture refunded	It is triggered when a merchant refunds a payment capture.
3	Payment capture completed	It is triggered when a payment capture is completed.
4	Vault payment-token deleted	It is triggered when a payment method is deleted from the customer's account.
5	Customer dispute created	It is triggered when a dispute is created
6	Customer dispute resolved	It is triggered when a customer dispute is resolved.
7	Customer dispute updated	It is triggered when a customer dispute is updated.

Important:

The Order Access Settings (Allow List) is a specific configuration within our system designed to enhance the security and management of order access permissions. By populating this list with specific values, administrators can limit order access to only those entities (endpoints) included in the Allow List. PayPal specific webhooks must be added to the list to work properly.

Navigate to Merchant Tools > Site Preferences > Order

Validate Limit Storefront Order Access preference. If your storefront configured to “No”, no changes is required by PayPal Web-hooks integration. If the setting is set to “Yes” or to “Allow List”, the PayPal web-hooks will be not able to update order information. To enable webhooks to modify order information, please set the setting to “Allow List” and add PayPal specific web-hooks endpoint: Paypal-WebHooks

3.5 PayPal Shipping Module

The shipping module presents shipping details to a buyer during the Pay with PayPal and Pay with Venmo flow. The merchant has several options available for controlling how shipping addresses and shipping options are handled. The server-side shipping callbacks allow you to update the shipping and order amount information as buyers make changes on the review page. Please see <https://developer.paypal.com/docs/checkout/standard/customize/shipping-module/> for details.

To configure Shipping module, please ensure that Preference Configuration -> General -> OCAPI Configuration contains correct values for OCAPI connection.

The example of configuration:

```
{
```

```

"clientId": "CLIENT_ID_VALUE",
"clientPassword": "CLIENT_ID_PASSWORD",
"apiVersion": "23.2",
"bmUserLogin": "salesforce@salesforce.com",
"bmUserPassword": "PASSWORD"
}

```

Open Administration -> Site Development -> Open Commerce API Settings and make sure that following resources implements read and write attributes for post method.

- "resource_id": "/customers/auth"
- "resource_id": "/baskets/*/shipments/*/shipping_methods"
- "resource_id": "/baskets/*/shipments/*/shipping_method"
- "resource_id": "/baskets/*/shipments/*/shipping_address"
- "resource_id": "/baskets/{basket_id}"

The valid configuration must look like:

```

{
  "_v" : "23.2",
  "clients": [{
    "client_id": "e38a01d6-a752-4eed-8388-8cfada2e196c",
    "resources":
    [
      {
        "resource_id": "/customers/auth",
        "methods": ["post"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
      },
      {
        "resource_id": "/customers/{customer_id}/auth",
        "methods": ["post"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
      },
    ]
  }
]
}

```

```

{
  "resource_id":"/baskets",
  "methods":["post"],
  "read_attributes": "(**)",
  "write_attributes": "(**)"
},
{
  "resource_id":"/baskets/{basket_id}/items",
  "methods":["post"],
  "read_attributes": "(**)",
  "write_attributes": "(**)"
},
{
  "resource_id":"/baskets/{basket_id}/shipments",
  "methods":["post"],
  "read_attributes": "(**)",
  "write_attributes": "(**)"
},
{
  "resource_id":"/baskets/{basket_id}/billing_address",
  "methods":["put"],
  "read_attributes": "(**)",
  "write_attributes": "(**)"
},
{
  "resource_id":"/baskets/{basket_id}/payment_instruments",
  "methods":["post"],
  "read_attributes": "(**)",
  "write_attributes": "(**)"
},
{
  "resource_id":"/baskets/{basket_id}/customer",
  "methods":["put"],
  "read_attributes": "(**)",

```

```

        "write_attributes": "(**)"
    },
    {
        "resource_id": "/orders",
        "methods": ["post"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/orders/{order_no}",
        "methods": ["put"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/baskets/{basket_id}/shipments/{shipment_id}",
        "methods": ["patch"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/baskets/*/shipments/*/shipping_methods",
        "methods": ["get"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/baskets/*/shipments/*/shipping_method",
        "methods": ["put"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/baskets/*/shipments/*/shipping_address",

```

```

        "methods": ["put"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/customers/{customer_id}/payment_instruments",
        "methods": ["get"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/customers/{customer_id}/payment_instruments/{payment_instrument_id}",
        "methods": ["get"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/baskets/{basket_id}/customer",
        "methods": ["put"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/baskets/{basket_id}/payment_methods",
        "methods": ["get"],
        "read_attributes": "(**)",
        "write_attributes": "(**)"
    },
    {
        "resource_id": "/baskets/{basket_id}",
        "methods": ["patch", "delete", "get"],
        "read_attributes": "(**)",
        "write_attributes": "(**)",
        "cache_time": 0
    }

```

```

    },
    {
      "resource_id":"/customers",
      "methods":["post"],
      "read_attributes": "(**)",
      "write_attributes": "(**)"
    },
    {
      "resource_id":"/customers/{customer_id}/payment_instruments",
      "methods":["post", "get"],
      "read_attributes": "(**)",
      "write_attributes": "(**)"
    },
    {
      "resource_id":"/customers/{customer_id}/payment_instruments/{payment_instrument_id}",
      "methods":["delete"],
      "read_attributes": "(**)",
      "write_attributes": "(**)"
    },
    {
      "resource_id":"/sessions",
      "methods":["post"],
      "read_attributes": "(**)",
      "write_attributes": "(**)"
    }
  ]
}]
}

```

3.6 PayPal App Switch

The plugin allows enabling App Switch flow (see feature app switch section for details)

To enable App Switch follow PayPal Plugin -> Preference Configuration -> PayPal and set PayPal App-switch enabled to Yes.

3.7 BNPL Configurator and Pay Later

Please use PayPal BNPL Visual configurator to adjust messaging appearance (PayPal Plugin -> Styles Configuration -> PayPal Pay Later Configurations). Allows setting up PayPal Pay Later Message\Banner location on the storefront.

To configure Pay Later messaging, please follow PayPal Plugin -> Preference Configuration -> Pay Later Banners/Messaging and configure messaging according to the requirements.

- Pay Later Cross-Border Messaging Enabled - Enables Cross-border messages for Pay Later. The Pay Later messages will be displayed according to your website location.

3.8 Plugin configuration exports (optional)

A feature that allows merchants to download PayPal plugin configurations, such as PayPal custom preferences, PayPal services, enabled payment methods/processors and PayPal plugin logs. Additionally, it tests PayPal service connection. (Read more in chapter 4.8 Plugin configuration exports (optional))

To configure the plugin exports feature:

- first set up **WebDAV and OCAPI access keys** if they weren't set up before. To do so go to your profile and click on Manage Access Keys and generate access keys (double check that keys are enabled)
- then go to **Merchant tools > PayPal plugin > Preference configuration General Tab** and paste WebDAV and OCAPI configurations in JSON format for the **OCAPI Configuration and WebDAV Configuration preferences**, e.g.

- **OCAPI Configuration preference (PP_OCAPI_Config):**

```
{
  "clientId": "OCAPI client id",
  "clientPassword": "OCAPI client password",
  "apiVersion": "23.2",
  "bmUserLogin": "SFRA Business manager login email",
  "bmUserPassword": "Agent User Login and OCAPI access key"
}
```

- **WebDAV Configuration preference (PP_WebDAV_Config):**

```
{
  "host": "****.dx.commercecloud.salesforce.com",
```

```

    "username": "SFRA Business manager login email",
    "password": "WebDAV File Access and UX Studio access key"
  }

```

- save and refresh the page. After that go to OCAPI configs, that are in **Administration > Site development > Open Commerce API Settings**, select Data type and add the following JSON (if JSON exists there already, just add the two new resource objects for the jobs):

```

{
  "_v": "23.2",
  "clients": [
    {
      "client_id": "OCAPI client id",
      "resources": [
        {
          "methods": ["post"],
          "read_attributes": "(**)",
          "write_attributes": "(**)",
          "resource_id": "/jobs/*/executions"
        },
        {
          "methods": ["get"],
          "read_attributes": "(**)",
          "write_attributes": "(**)",
          "resource_id": "/jobs/*/executions/*"
        }
      ]
    }
  ]
}

```

4 Operations and Maintenance

4.1 Data Storage

The int_paypal integration requires System Objects Extension to store payment-related information.

OrderPaymentInstrument custom properties:

- **paypalOrderId** - Result of a payment action (Auth, Order, Sale).
- **paypalVaultEmail** - email address of vaulted PayPal Account
- **paypalBillingAddress** - paypal billing address that is associated with payment instrument
- **paymentId** - generally is used for collecting Funding source name (paypal, mybank, venmo...).
- **paypalLpmAccountHolderName** - collects an account holder name provided by Local payment method's response.
- **paypalPaymentStatus** - collects a PayPal payment status.
- **paypalRequest** - collects a PayPal request.
- **paypalResponse** - collects a PayPal response.
- **paypalToken** - collects a PayPal token needed for proper NVP PayPal Transactions.

Profile custom properties:

- **isExternalProfile** - Identifies whether the customer's profile is external (via Connect with PayPal).
- **isDisabledFeatureAPMA** - Identifies whether the Automatic Payment Method Adding feature is disabled.
- **flashMessages** - contains messages to show.
- **isMergedAccount** - Identifies whether user accounts are merged between PayPal and Single Auth.

Order custom properties:

- **paypalPaymentMethod** - Property to differentiate PayPal-related orders.

4.2 Logs

This integration introduces three new custom logs:

1. **PayPal Storefront Custom logs** - starts with prefix custom-PayPal-blade2-2-appserver-20150722.log. This Log file contains all errors-related information in the int_paypal cartridge.
2. **PayPal Business manager custom logs** - starts with prefix custom- PayPal-BM-blade. This Log file contains all errors-related information in the bm_paypal cartridge.
3. **Service communication logs** - starts with service-PayPalRest. These logs contain every request and response to the PayPal endpoint. To enable these logs, check [Adding API Credentials](#) **Optional** section.

4.3 HTTP Service Availability

You can track availability and downtime by service status in the Commerce Cloud Business Manager. Go to Administration > Operations > Service Status > int_paypal.http.rest.

You can configure options for HTTP calls to REST API related to this PayPal integration via the Commerce Cloud Service Profile Interface. To do this, go to Administration >

Operations > Services > Service Profiles -> **PayPal_Default_Profile**. You can set a timeout for all requests, enable the Circuit Breaker mechanism, and adjust the Rate Limit.

[Administration](#) > [Operations](#) > [Services](#) > [Service Profiles](#) > PayPal_Default_Profile - Details

PayPal_Default_Profile

Fields with a red asterisk (*) are mandatory. Click **Apply** to save the details. Click **Reset** to revert to the last saved state.

This profile is used by 1.00 service.

Name: *	PayPal_Default_Profile
Timeout (ms):	60,000
Enable Circuit Breaker:	☐
Max Circuit Breaker Calls:	0
Circuit Breaker Interval (ms):	0
Enable Rate Limit:	☐
Max Rate Limit Calls:	0
Rate Limit Interval (ms):	0

Figure 4.3.1. PayPal Service Profile Settings

4.4 Testing

You must obtain your own test account on the [PayPal Developer Portal](#). If you can't create your own PayPal Sandbox account, contact PayPal support.

4.5 SFRA core templates overlay list

The following templates from app_storefront_base cartridges are changed by int_paypal

cart/cart.isml

account/dashboardProfileCards.isml

account/components/oauth.isml

account/login.isml

account/profileCard.isml

checkout/billing/paymentOptions/paymentOptionsContent.isml

checkout/billing/paymentOptions/paymentOptionsSummary.isml

checkout/billing/paymentOptions/paymentOptionsTabs.isml

checkout/billing/billingSummary.isml

checkout/confirmation/confirmationDetails.isml

checkout/cart/miniCart.isml

checkout/customer/customerOAuthCard.isml
checkout/checkout.isml
common/scripts.isml
components/pageFooter.isml
home/homePage.isml
product/components/addToCartButtonExtension.isml
product/components/addToCartGlobal.isml
product/components/addToCartProduct.isml
search/searchResult.isml
Templates are up to date with SFRA 7.0.0

4.6 Support

To get help and support from PayPal:

- PayPal Business support – Go to PayPal’s [Contact Us](#) and log in to your PayPal account.
- Technical Support – Go to the [Merchant Technical Support Help Center](#).

4.7 Plugin service connection testing

To test whether PayPal service (int_paypal.http.rest) has been configured successfully, go to **Merchant Tools > PayPal plugin > Preference configuration** and click the **Test connection with PayPal** button.

The screenshot shows the PayPal plugin configuration interface. At the top, there are three buttons: 'Generate Plugin Configuration' (green), 'Test connection with PayPal' (blue, highlighted with a red underline), and 'Self-Check' (grey). Below these buttons is a horizontal tab bar with the following tabs: 'General' (selected), 'PayPal', 'Credit Card', 'Apple Pay', 'Google Pay', 'Venmo', 'Fastlane by PayPal', and 'Pay Later Banners/Messaging'. Under the 'General' tab, the heading 'PayPal General' is displayed in green. Below this heading is the 'Instance Type' section, which contains a dropdown menu currently set to 'Sandbox'. At the bottom of the interface, there is a search bar labeled 'Search by ID' with a magnifying glass icon, and three buttons: 'Cancel', 'Apply to Other Sites', and 'Save'.

Figure 4.7.1. Test connection with PayPal button

If the service has been configured correctly, you will get a success message, otherwise an error will be shown.

4.8 Plugin configuration exports (optional)

To properly troubleshoot the PayPal plugin, you might need to collect plugin configurations and this feature allows you to do so in a few clicks and have needed data zipped and downloaded to your device.

The downloads will include:

1. List of PayPal custom preferences
2. List of PayPal services
3. Enabled payment methods and payment processors
4. PayPal plugin logs
5. General data like SFRA version, plugin version, PayPal Partner attribution id, instance type and compatibility mode

Additionally, PayPal service connection will be tested to ensure that PayPal service is configured correctly.

After the feature is set up (details in chapter 3.8 3.8Plugin configuration exports (optional)), click on **Generate Plugin Configuration** button in **Merchant tools > PayPal plugin > Preference configuration**.

In the popup check the items you would like to download. If none is selected, only general data will be generated.

The zip file will be downloaded into the location chosen in your browser downloads settings.

Once everything is downloaded, you will see a success message, otherwise an error will be shown.

NOTE: Downloaded files might contain sensitive data, thus before sending them to a 3rd party, look through the files.

The screenshot shows a web interface for PayPal configuration. At the top, there are three buttons: 'Generate Plugin Configuration' (green, highlighted with a red underline), 'Test connection with PayPal' (blue), and 'Self-Check' (grey). Below these is a horizontal menu with tabs: 'General' (selected), 'PayPal', 'Credit Card', 'Apple Pay', 'Google Pay', 'Venmo', 'Fastlane by PayPal', and 'Pay Later Banners/Messaging'. Under the 'General' tab, the heading 'PayPal General' is displayed. Below this, there is a section for 'Instance Type' with a dropdown menu currently set to 'Sandbox'. At the bottom, there is a search bar labeled 'Search by ID' with a magnifying glass icon, and three buttons: 'Cancel', 'Apply to Other Sites', and 'Save'.

Figure 4.8.1. Generate Plugin Configuration button

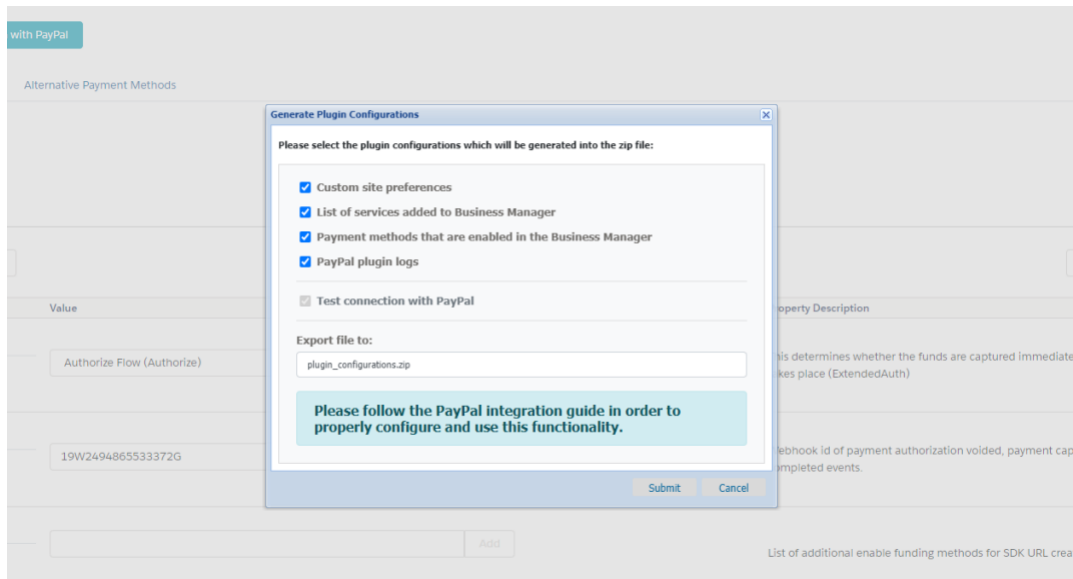


Figure 4.8.2. Generate Plugin Configuration popup

4.9 Plugin Configuration Self-Check

To perform a self-check of the configured business flows in the plugin you can use Plugin Self-Check feature by clicking Self-Check button in the Business Manager (PayPal Plugin->Preference Configuration page->Self-Check button (Figure 4.9.1.)).

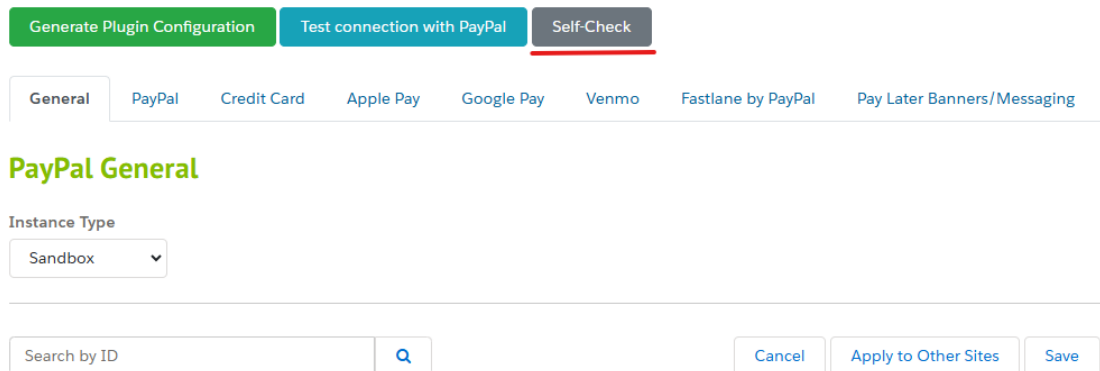


Figure 4.9.1. Self-Check button

You can check the following plugin's configurations of the business flows (Figure 4.9.2.):

- Plugin's service configurations
- Payment Method activation
- Payment Processors configurations
- Custom preferences configurations

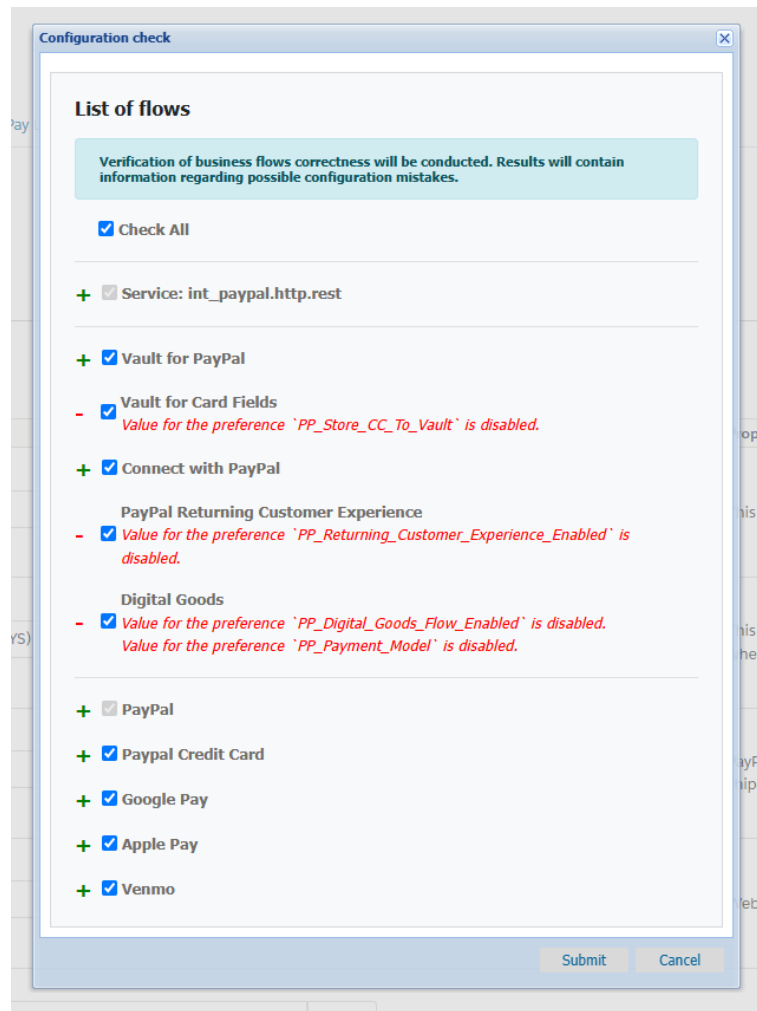


Figure 4.9.2. Self-Check of the business flows configured in the plugin

5 User Guide

5.1 Business Manager modules

5.1.1 PayPal Transactions

This integration introduces a new Business Manager function, PayPal Transactions. See [PayPal Transaction Business Manager module configuration](#) for a detailed description of how to grant access to the PayPal Transactions page.

The PayPal Transactions extension allows you to manage PayPal transactions assigned to Commerce Cloud Orders by the Commerce Cloud Business manager. PayPal Transactions module supports reauthorization (after 3 days), capture and partial capture, void, refund.

The main PayPal Transactions page displays all orders in a site that were paid or partially paid by PayPal. You can access this page in the Ordering menu by selecting Business Manager > Merchant Tools > PayPal Plugin > Transactions.

This page displays various information from the order record:

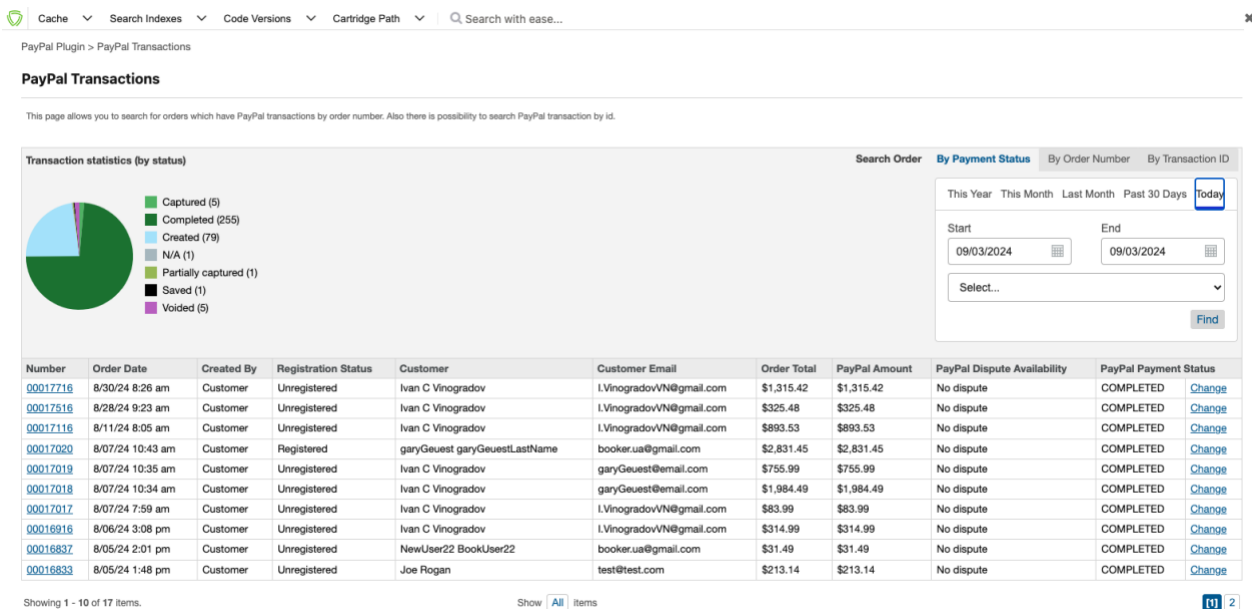


Figure 5.1.1. PayPal Transactions Business Manager Main Page Interface

The transaction statistics block shows the total number of all transactions sorted by payment status (as shown in Figure 5.1.2).

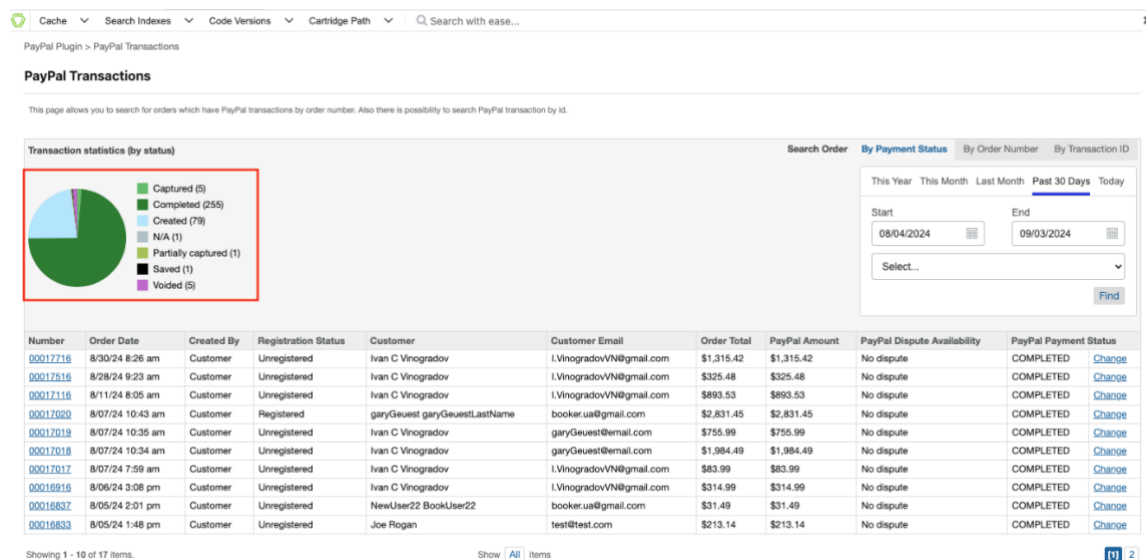


Figure 5.1.2. PayPal Transactions Business Manager Transactions statistics

You can search Orders placed within PayPal order number (as shown in Figure 5.1.3) or by PayPal Transaction ID (as shown in Figure 5.1.4) or by PayPal Payment Status (as shown in Figure 5.1.5).

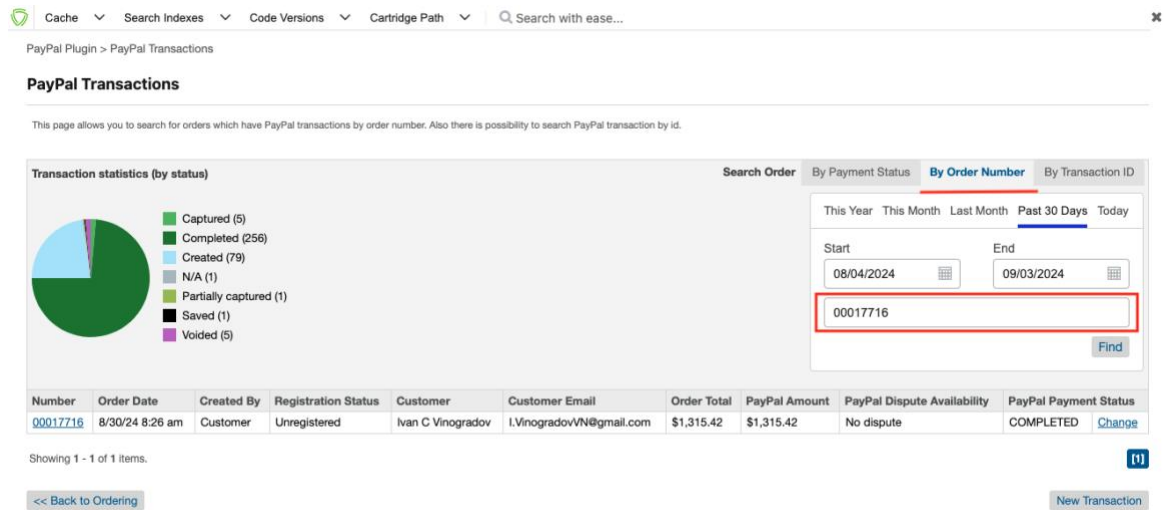


Figure 5.1.3. Search Orders with PayPal Transaction by Commerce Cloud Order Number

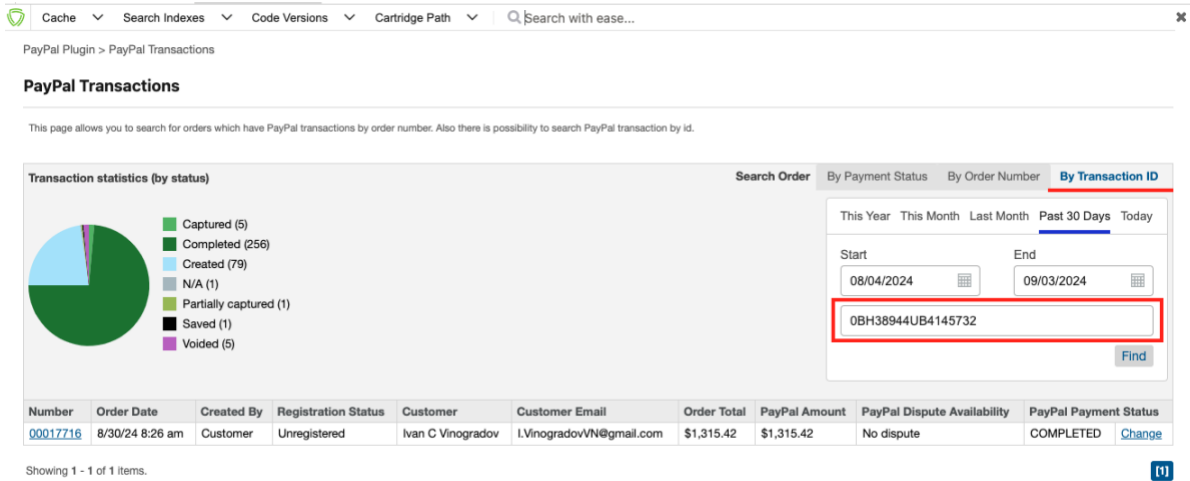


Figure 5.1.4. Search Orders with PayPal Transaction by Transaction ID

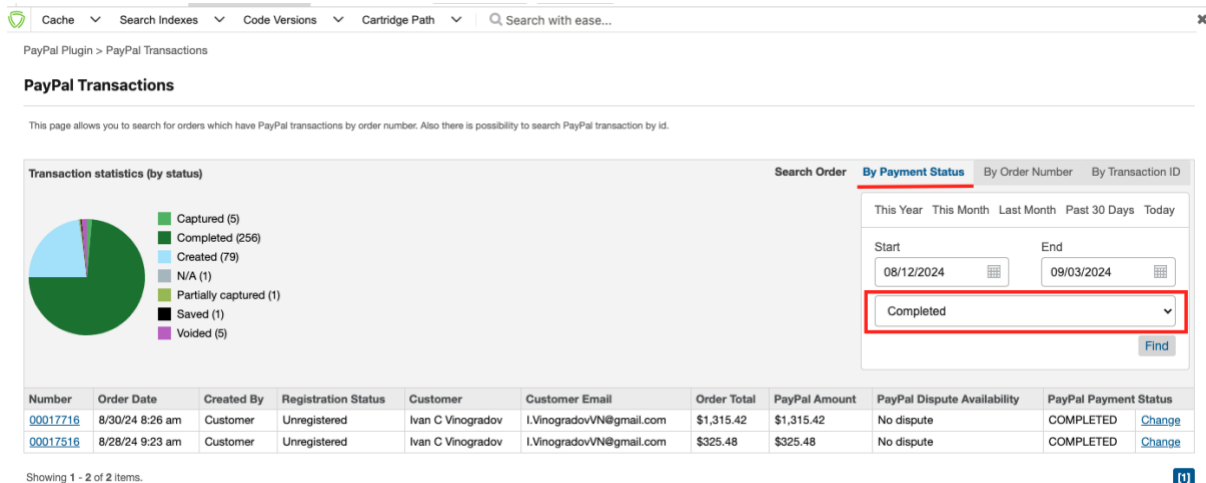


Figure 5.1.5. Search Orders with PayPal Transaction by Payment Status

You can see Transaction Details from the PayPal Payment Transaction of the actual order by clicking Order Number or the Change link on the right. If an order has more than one related transaction, you will see a selection box with all the transactions that are related to the current order (as shown in Figure 5.1.6).

The screenshot shows a 'Transactions of order: 00000714' popup window. It contains the following information:

- Transaction ID:** 00L53809R21778442. Other transactions: 00L53809R21778442 (dropdown).
- Name:** Paul Lorens
- Transaction Email:** p.lorenstest@gmail.com
- Amount:** 261.97 USD
- Shipping:** 34.99 USD
- Sales Tax:** 12.48 USD
- Order/Invoice ID:** 00000714
- Order Date:** 7/12/23 1:07 pm
- Transaction Payment Status:** VOIDED
- Shipping Address:** Paul Lorens, 2253 Hudson St, Denver, CO 80207\ -3846 US

Transaction History

Timestamp	Status	Amount
7/12/23 1:09 pm	Voided	261.97
7/12/23 1:07 pm	Created	261.97

Transaction Logs

- > [Request to PayPal](#)
- > [Response from PayPal](#)

Figure 5.1.6. Transaction Details Popup

The screenshot shows a 'Transactions of order: 00015207' popup window. It contains the following information:

- Transaction ID:** 7AE52170XT8052505. Other transactions: 7AE52170XT8052505, Captured: 7GX80784LM544511C, Captured: 3D495099BU1827429 (dropdown).
- Name:** Ivan C Vinogradov
- Transaction Email:** I.VinogradovVN@gmail.com
- Amount:** 244.63 USD. Captured: 244.63. Refunded: 243.63
- Shipping:** 0.00 USD
- Invoice ID:** 00015207
- Order Date:** 7/22/20 1:45 pm
- Payment Status:** CAPTURED
- Shipping Address:** Ezio Auditore, Rome, Rome, Rome, US 12345 DE

A red box highlights the 'Other transactions' dropdown menu, which contains the following options:

- ✓ 7AE52170XT8052505
- Captured: 7GX80784LM544511C
- Captured: 3D495099BU1827429

Figure 5.1.7. PayPal Transactions Selected for Their Relationships to Commerce Cloud Order Transactions

Depending on the transaction type and status, the order may have the following action buttons:

- Capture (Figure 5.1.8)
- Void (Figure 5.1.8 and Figure 5.1.9)
- Issue Refund (Figure 5.1.10)
- Reauthorize (Figure 5.1.9)

After click on a button pop-up with details will appear:

- Void Form (Figure 5.1.11)
- Capture Form (Figure 5.1.12)
- Issue Refund Form (Figure 5.1.13)
- Reauthorize Form (Figure 5.1.14)

The screenshot shows a PayPal transaction details pop-up window titled "Transactions of order: 00002001". The window displays the following information:

- Transaction ID:** 78S38612V2334723T
- Other transactions:** 78S38612V2334723T (dropdown menu)
- Name:** test buyer
- Transaction Email:** Denys_Lieukhyn-buyer@epam.com
- Amount:** 44.09 USD
- Shipping:** 15.99 USD
- Sales Tax:** 2.10 USD
- Order/Invoice ID:** 00002001
- Order Date:** 8/09/23 10:35 am
- Transaction Payment Status:** CREATED
- Buttons:** Capture, Void
- Shipping Address:** test buyer, 1 Main St, San Jose, CA 95131, US
- Transaction History:**

Timestamp	Status	Amount
8/09/23 10:35 am	Created	44.09

Figure 5.1.8. Authorized transaction payment actions

Transactions of order: 00000716

Transaction ID: 9EB029467U4608838

Other transactions: 9EB029467U4608838

Name: test buyer

Transaction Email: Denys_Lieukhyn-buyer@epam.com

Amount: 235.19 USD

Shipping: 67.99 USD

Sales Tax: 11.20 USD

Order/Invoice ID: 00000716

Order Date: 7/19/23 7:13 am

Transaction Payment Status: CREATED

Reauthorize Capture Void

Shipping Address: test buyer
1 Main St,
San Jose, CA 95131
US

Transaction History

Timestamp	Status	Amount
7/19/23 7:13 am	Created	235.19

Figure 5.1.9. Authorized transaction payment actions (after 3 days)

Transactions of order: 00002001

Transaction ID: 78S38612V2334723T

Other transactions: 78S38612V2334723T

Name: test buyer

Transaction Email: Denys_Lieukhyn-buyer@epam.com

Amount: 44.09 USD *Captured: 44.09*

Shipping: 15.99 USD

Sales Tax: 2.10 USD

Order/Invoice ID: 00002001

Order Date: 8/09/23 10:35 am

Transaction Payment Status: CAPTURED

Issue Refund

Shipping Address: test buyer
1 Main St,
San Jose, CA 95131
US

Transaction History

Timestamp	Status	Amount
8/09/23 11:15 am	Completed	44.09
8/09/23 10:35 am	Created	44.09

Figure 5.1.10. Captured transaction payment actions

Transactions of order: 00000721

Transaction ID: 9EK10583A0272263L

Name: test buyer

Transaction Email: Denys_Lieukhyn-buyer@epam.com

Amount: 99.74 USD

Shipping: 5.99 USD

Void Authorization

Authorization ID: 9EK10583A0272263L

Name: test buyer

Email: Denys_Lieukhyn-buyer@epam.com

Amount to Void: 99.74 USD

Shipping Address: test buyer
1 Main St,
San Jose, CA 95131
US

Transaction History

Timestamp	Status	Amount
7/19/23 8:26 am	Created	99.74

Figure 5.1.11. Void Authorization Form

Transactions of order: 00000721

Capture Funds

Transaction ID: 9EK10583A0272263L

Name: test buyer

Email: Denys_Lieukhyn-buyer@epam.com

Authorization Amount: 99.74 USD

Capture Amount: 99.74 USD

Note to merchant: Your buyer test buyer authorized a payment of 99.74 USD. If you capture an amount different than this, you should notify your buyer of this change. PayPal strongly recommends that you explain any unique circumstances (e.g. multiple captures, changes in item availability) to your buyer in detail below. Your buyer will see this note in the Transaction Details.

Note to Buyer: (optional)

255 characters left

7/19/23 8:26 am Created

Figure 5.1.12. Capture Form

Transactions of order: 00002001

Transaction ID: 78S38612V2334723T Other transactions: 78S38612V2334723T

Issue Refund

Transaction ID: 78S38612V2334723T

Name: test buyer

Email: Denys_Lieukhyn-buyer@epam.com

Original payment: 44.09 USD Captured: 44.09 Refunded: 0.0

Refund amount: USD

Invoice ID: 00002001

Note to Buyer: (optional)

255 characters left

Submit Cancel

Timestamp	Status	Amount
8/09/23 11:15 am	Completed	44.09
8/09/23 10:35 am	Created	44.09

Figure 5.1.13. Issue Refund Form

Transactions of order: 00000716

Transaction ID: 9EB029467U4608838 Other transactions: 9EB029467U4608838

Name: test buyer

Transaction Email: Denys_Lieukhyn-buyer@epam.com

Amount: 235.19 USD

Shipping: 67.99 USD

Reauthorize Payment

Authorization ID: 9EB029467U4608838

Name: test buyer

Email: Denys_Lieukhyn-buyer@epam.com

Amount to Reauthorize: 235.19 USD

Submit Cancel

Shipping Address: test buyer
1 Main St,
San Jose, CA 95131
US

Transaction History

Timestamp	Status	Amount
7/19/23 7:13 am	Created	235.19

Figure 5.1.14 Reauthorize Form

To view request and response log data for a particular transaction, use the Transaction Details window and the Transaction Logs section. This will help you analyze the information in more detail (Figure 5.1.15).

Transaction Email: ptest4@paypal.com

Amount: 119.67 USD *Captured:* 119.67 *Refunded:* 0.0

Shipping: 7.99 USD

Sales Tax: 5.70 USD

Invoice ID: 00000502

Order Date: 9/27/22 12:54 pm

Payment Status: CAPTURED

Shipping Address: Integration Grimes
1 Main St, No address
San Jose, US 95131
CA

Transaction Logs

Request to PayPal
This section contains requests to PayPal which were made for transaction creation

```
{
  "note": "",
  "orderId": "00000502",
  "orderToken": ""
}
```

Response from PayPal
This section contains responses from PayPal which were made for transaction creation

```
{
  "id": "85652565D4939545U",
  "status": "COMPLETED",
  "links": {}
}
```

Figure 5.1.15. Reauthorize Form

If you need to create a new transaction - use the New Transaction button (see Figure 5.1.16) and log in to your Paypal account.

PayPal Transactions

This page allows you to search for orders which have PayPal transactions by order number. Also there is possibility to search PayPal transaction by id.

Transaction statistics (by status)

- Captured (5)
- Completed (256)
- Created (79)
- N/A (1)
- Partially captured (1)
- Saved (1)
- Voided (5)

Search Order By Payment Status By Order Number By Transaction ID

This Year This Month Last Month Past 30 Days Today

Start: 08/04/2024 End: 09/03/2024

Select...

Find

Number	Order Date	Created By	Registration Status	Customer	Customer Email	Order Total	PayPal Amount	PayPal Dispute Availability	PayPal Payment Status
00017716	8/30/24 8:26 am	Customer	Unregistered	Ivan C Vinogradov	I.VinogradovVN@gmail.com	\$1,315.42	\$1,315.42	No dispute	COMPLETED Change
00017516	8/28/24 9:23 am	Customer	Unregistered	Ivan C Vinogradov	I.VinogradovVN@gmail.com	\$325.48	\$325.48	No dispute	COMPLETED Change
00017116	8/11/24 8:05 am	Customer	Unregistered	Ivan C Vinogradov	I.VinogradovVN@gmail.com	\$893.53	\$893.53	No dispute	COMPLETED Change
00017020	8/07/24 10:43 am	Customer	Registered	garyGeuest garyGeuestLastName	booker.ua@gmail.com	\$2,831.45	\$2,831.45	No dispute	COMPLETED Change
00017019	8/07/24 10:35 am	Customer	Unregistered	Ivan C Vinogradov	garyGeuest@email.com	\$755.99	\$755.99	No dispute	COMPLETED Change
00017018	8/07/24 10:34 am	Customer	Unregistered	Ivan C Vinogradov	garyGeuest@email.com	\$1,984.49	\$1,984.49	No dispute	COMPLETED Change
00017017	8/07/24 7:59 am	Customer	Unregistered	Ivan C Vinogradov	I.VinogradovVN@gmail.com	\$83.99	\$83.99	No dispute	COMPLETED Change
00016916	8/06/24 3:08 pm	Customer	Unregistered	Ivan C Vinogradov	I.VinogradovVN@gmail.com	\$314.99	\$314.99	No dispute	COMPLETED Change
00016837	8/05/24 2:01 pm	Customer	Unregistered	NewUser22 BookUser22	booker.ua@gmail.com	\$31.49	\$31.49	No dispute	COMPLETED Change
00016833	8/05/24 1:48 pm	Customer	Unregistered	Joe Rogan	test@test.com	\$213.14	\$213.14	No dispute	COMPLETED Change

Showing 1 - 10 of 17 items.

Show [All](#) items

[New Transaction](#)

Figure 5.1.16. "New Transaction" button

5.1.2 PayPal Styles Configuration

Styles Configuration allows to choose styling of the Smart Button from the Business Manager. After installation, a new menu item will appear under PayPal Plugin in the Business Manager (Figure 5.1.17).

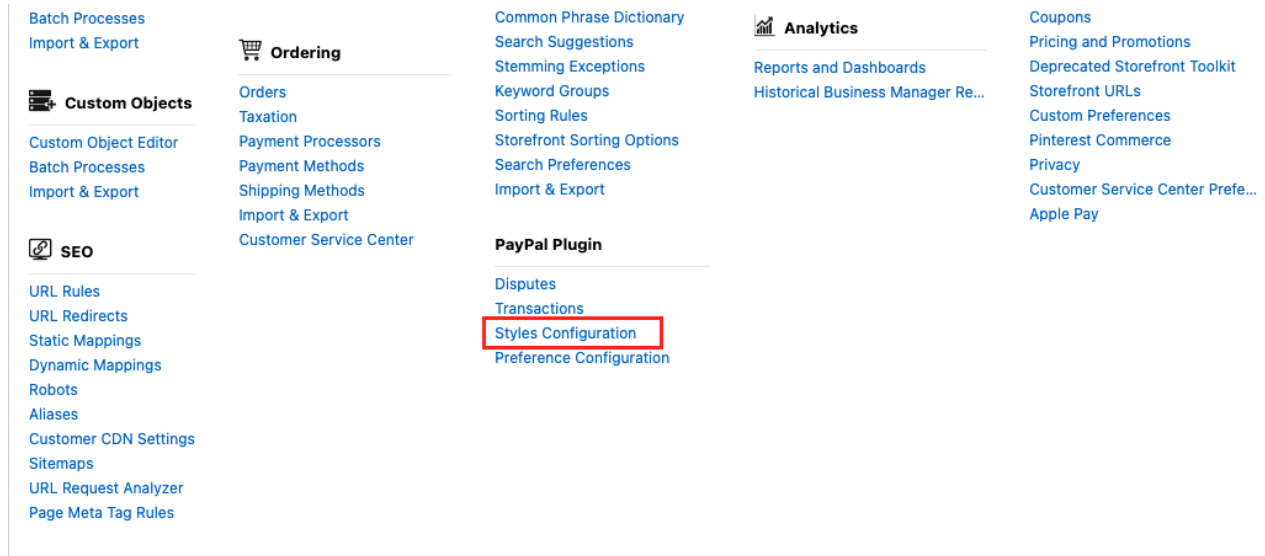


Figure 5.1.17 PayPal Styles Configuration in the Business Manager

If `bm_paypal` cartridge was installed properly you should see styling options for the PayPal and Venmo buttons, PayPal Pay Later Configurations, Connect with PayPal, Apple Pay, Goggle Pay and Card Fields. The buttons style configurations can be applied only to active button locations. To read more about styling options visit [Customize the PayPal Buttons](#). After configuration is done click **Save** to update styling.

Note: Due to the limitation the tagline and layout have constant values: layout - vertical, tagline - false.

The image shows the 'PayPal and Venmo Buttons Style Configurations' form. At the top, there are tabs for different button types: 'PayPal and Venmo', 'PayPal Pay Later Configurations', 'Apple Pay', 'Google Pay', 'Connect with PayPal', and 'Card Fields'. The 'PayPal and Venmo' tab is selected. The form contains several configuration options: 'Choose a location:' (dropdown menu with 'billing' selected), 'Choose a color:' (dropdown menu with 'black' selected), 'Choose a shape:' (dropdown menu with 'pill' selected), 'Choose a label:' (dropdown menu with 'buynow' selected), and 'Choose a height:' (range slider with 'min: 25', '46', and 'max: 55'). There is also an unchecked checkbox for 'Apply to all button locations' and a 'Save' button at the bottom left. On the right side of the form, there are three preview buttons: 'PayPal Buy Now', 'venmo Buy Now', and 'Pay Later'.

Figure 5.1.18. PayPal and Venmo styling options

- **PayPal Button Messaging:** depending on Button Messaging locations, the styling will be available for Billing, Cart, Mini-Cart, PVP and PDP pages (Figure 5.1.19). After configuration is completed, click **Save** to update styling.

PayPal Button Messaging Configuration

Choose a location:

Choose an align:

Choose a color:

Choose a position:

☐ Apply to all Messaging locations

Save

Note: Messaging is currently only supported for US merchants and US customers. Merchants must be eligible for Pay Later to display Pay Later offers with buttons.



Pay in full or 4 interest-free payments of \$25.00. [Learn more](#)

Figure 5.1.19 PayPal Button Messaging configurations

- **Apple Pay:** depending on button locations, the styling will be available for cart and billing pages (Figure 5.1.20). To read more about styling options visit [Customize the Apple Button](#). After configuration is completed, click **Save** to update styling.

PayPal **Apple Pay** Google Pay Connect with PayPal Card Fields

Apple Pay Button Configuration

Choose a location

Choose a color

Choose a type

☐ Apply to all Apple Pay button locations

Save



Figure 5.1.20 Apple Pay Styles Configuration view

- **Connect with PayPal:** depending on the button locations, the styling will be available for login and billing pages (Figure 5.1.21) To read more about styling options visit [Customize the Connect with PayPal Button](#). After configuration is completed, click **Save** to update styling.

PayPal
Apple Pay
Google Pay
Connect with PayPal
Card Fields

Connect with PayPal Button Configuration

Choose a PayPal Button Location:

login

Choose a label:

Connect with PayPal

Choose a shape:

pill

Choose a size:

large

Choose a theme:

blue

☐ Apply to all PayPal button locations

Save



Figure 5.1.21. Connect with PayPal Button Configuration

- Card Fields could be customize buy merchant([Figure 5.1.22](#)), the styling will be available on billing page. After configuration is completed, click **Save** to update styling.

PayPal
Apple Pay
Google Pay
Connect with PayPal
Card Fields

Card Fields Style Configuration

Color



Invalid Color Highlight



Valid Color Highlight



Font Size

min 8 12 max 16

Save

Note: To test error cases, you should use card (1111 1111 1111 1111) and date (11 / 11)

* Cardholder name

* Card number

4111 1111 1111 1111

* CVV

...

* Expiration date

MM/YY

Figure 5.1.22. Card Fields Style Configuration

- PayPal Pay Later Configurations: The styling will be available for Cart, Category, Product, Product preview Checkout pages ([Figure 5.1.23](#)). After configuration is completed, click **Publish Changes** to update styling.

PayPal Pay Later Messaging

Publish Changes

Your customers can already buy now and pay later with PayPal — add messaging to your site to let them know. PayPal's Pay Later helps boost merchants' conversion rates and increases cart sizes by 39%.¹ You get paid in full up front. [More about Pay Later](#)

Customize your messaging

Preview

Dark

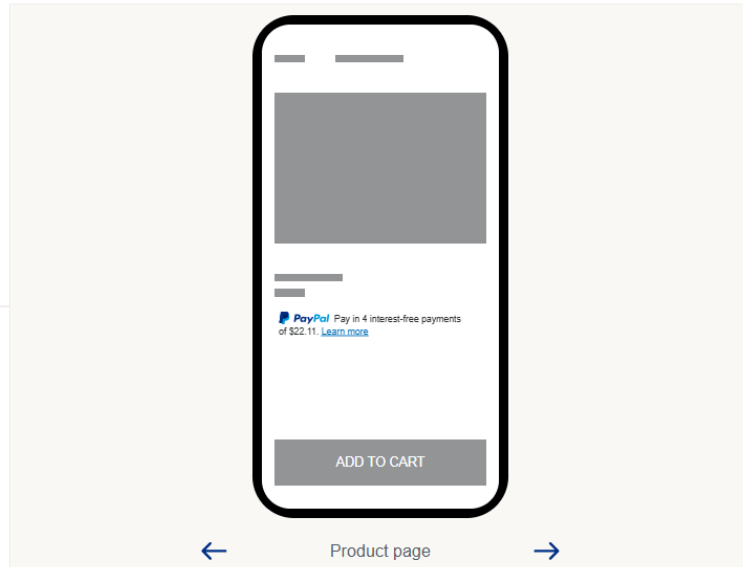
Decide where you want dynamic messaging to show up and how you want it to look on your site.

- ☒ Product page
- ☒ Cart
- ☒ Checkout
- ☒ Product preview
- ☒ Category page

[Reset to our recommendations](#)

Need more help? [Go to the developer docs](#)

¹PayPal Q2 Earnings-2021.



Customers see the Pay Later offer most relevant to their purchase.

Figure 5.1.23 PayPal Pay Later Messaging

- Google Pay: depending on the button locations, the styling will be available for Billing, Cart, Mini-Cart, PVP and PDP pages (Figure 5.1.24). To read more about styling options visit [Customize the Google Pay Button](#). After configuration is completed, click **Save** to update styling.

PayPal

Apple Pay

Google Pay

Connect with PayPal

Card Fields

Google Pay Button Configuration

Choose a location

billing

Choose a color

default

Choose a type

buy

Choose a corner radius

min: 0 4 max: 100

Custom button size

static

☐ Apply to all Google Pay button locations

Save

Figure 5.1.24 Google Pay Styles Configuration view

Limitations:

The styling configuration comes with limitations:

1. When selecting the horizontal buttons view for the PVP location in the "BM - > Button Style Configurations", the storefront will display only one button instead of multiple buttons. This behavior is due to resolution constraints imposed by the SDK.
2. On certain screens where the Tagline is active, it may appear truncated due to SDK limitations. Additionally, it does not display on the PVP screen at all.
3. On certain resolutions, the labels for PayPal and Venmo buttons may not be visible based on the configured button height. E.G PDP Screen: Labels are not displayed when the button height is configured to a maximum of 55px.

5.1.3 PayPal Disputes

This integration introduces a new Business Manager function, PayPal Disputes. See [PayPal Transaction Business Manager module configuration](#) for a detailed description of how to grant access to the PayPal Disputes page.

The PayPal Disputes extension allows you to manage PayPal disputes assigned to Commerce Cloud Orders by the Commerce Cloud Business manager.

The main PayPal Disputes page displays all opened disputes on a site. You can access this page by selecting Business Manager > Merchant Tools > PayPal Plugin > Disputes. (Figure 5.1.25) This page displays various information from the dispute record:

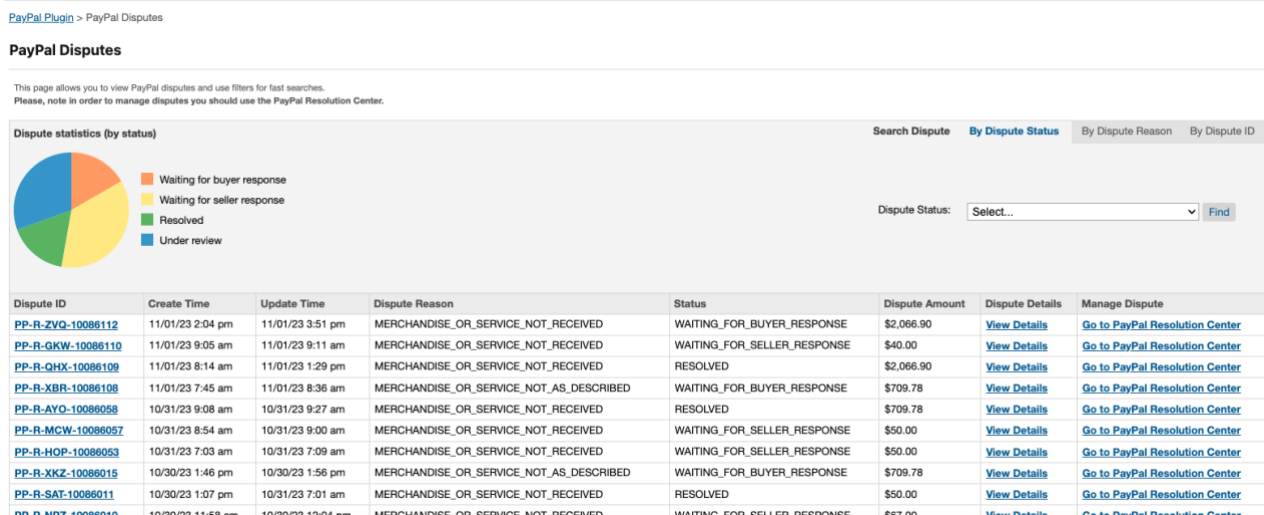


Figure 5.1.25 PayPal Disputes Business Manager Main Page Interface

The disputes statistics block shows number and statutes of all disputes (as shown in Figure 5.1.26).

PayPal Disputes

This page allows you to view PayPal disputes and use filters for fast searches.
Please, note in order to manage disputes you should use the PayPal Resolution Center.

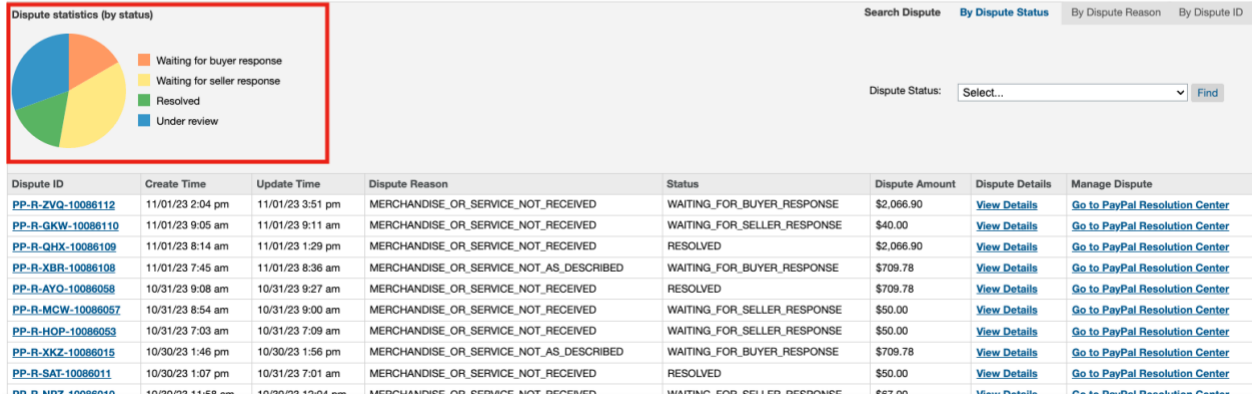


Figure 5.1.26 PayPal Disputes Business Manager Dispute statistics

You can search Disputes placed within PayPal Dispute Status (as shown in Figure 5.1.27) or by PayPal Dispute Reason (as shown in Figure 5.1.28) or by PayPal Dispute ID (as shown in Figure 5.1.29).

PayPal Disputes

This page allows you to view PayPal disputes and use filters for fast searches.
Please, note in order to manage disputes you should use the PayPal Resolution Center.

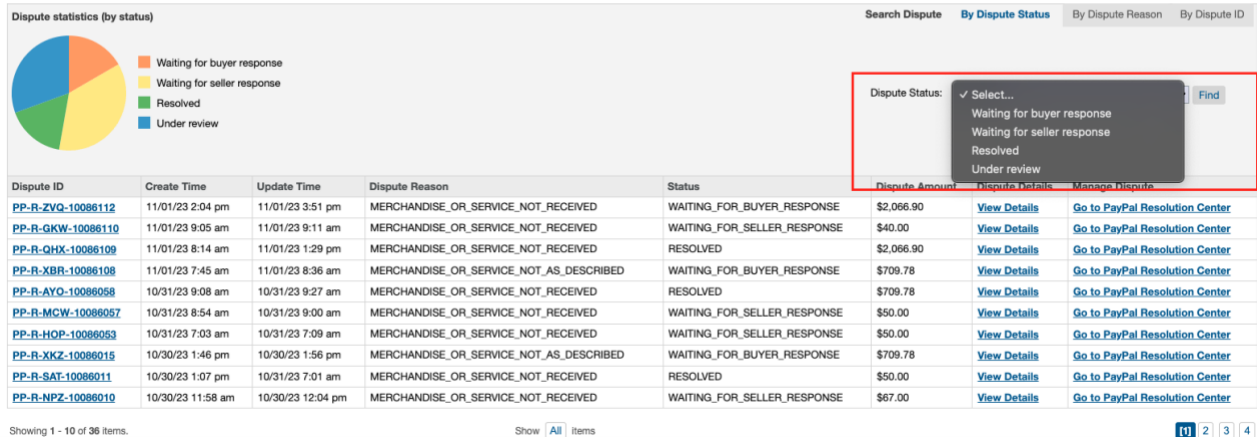


Figure 5.1.27 Search Disputes by Dispute Status

PayPal Disputes

This page allows you to view PayPal disputes and use filters for fast searches.
Please, note in order to manage disputes you should use the PayPal Resolution Center.

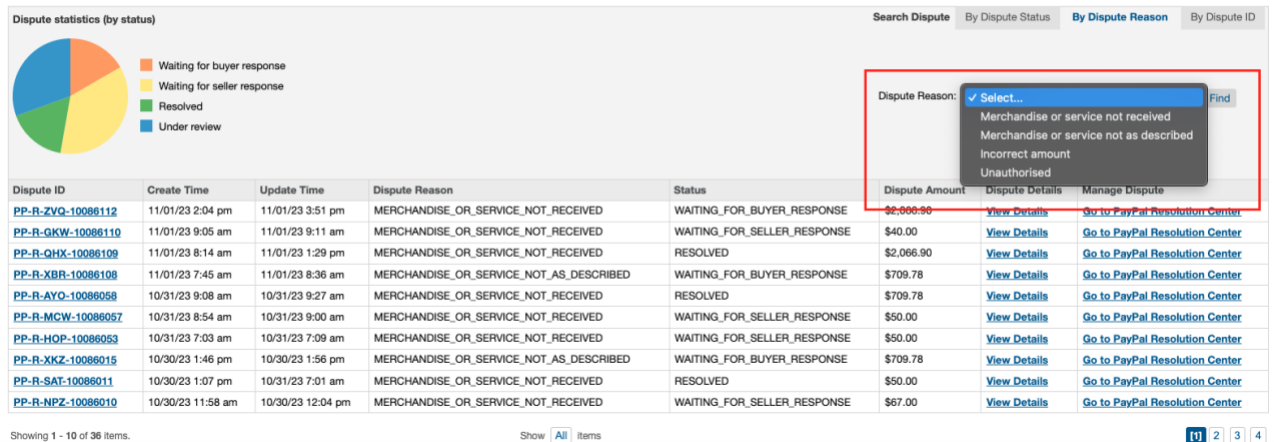


Figure 5.1.28 Search Disputes by Dispute Reason

PayPal Disputes

This page allows you to view PayPal disputes and use filters for fast searches.
Please, note in order to manage disputes you should use the PayPal Resolution Center.

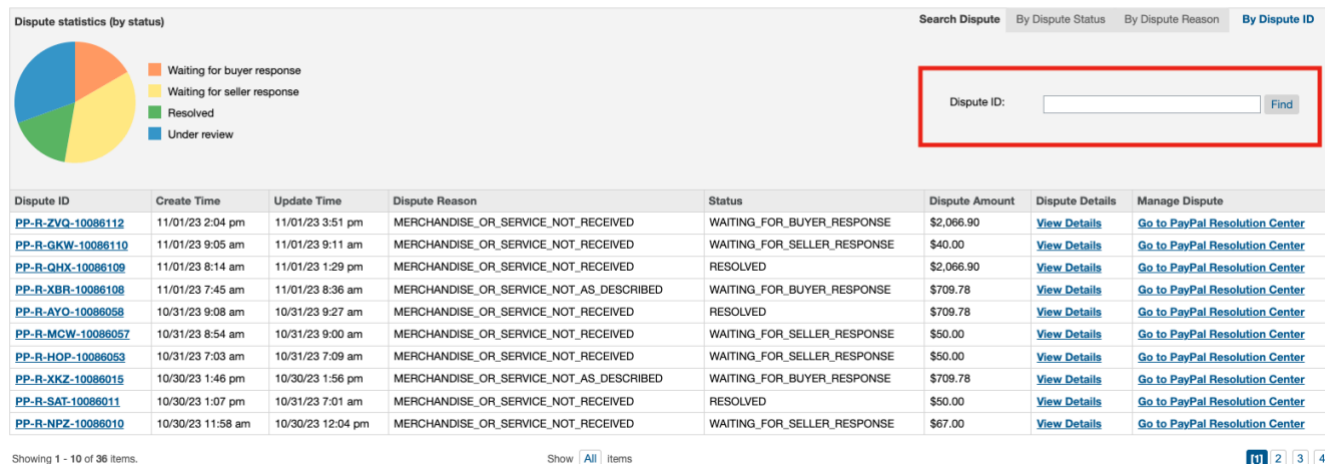


Figure 5.1.29 Search Dispute by Dispute ID

You can see Dispute Details from the PayPal Disputes by clicking **View Details** of the actual dispute (as shown in Figure 5.1.30)

Dispute: PP-R-ZVQ-10086112

Dispute ID: PP-R-ZVQ-10086112

Create Time: 11/01/23 2:04 pm

Update Time: 11/01/23 3:51 pm

Disputed Amount:	2066.90 USD
Dispute Reason:	MERCHANDISE_OR_SERVICE_NOT_RECEIVED
Dispute Status:	WAITING_FOR_BUYER_RESPONSE
Buyer Requested Amount:	2405.53 USD
Buyer Response Due Date:	11/21/23

Dispute Lifecycle Stage:	INQUIRY
Dispute Channel:	INTERNAL

Seller Transaction ID:	4TH704686V554092T
Buyer Transaction ID:	1GG94214C2233361J
Transaction Payment Status:	HELD
Transaction Create Time:	11/01/23 2:02 pm
Transaction Amount:	2405.53 USD
Invoice Number:	00004916
Seller Name:	Test Store
Seller Email:	sb-wc47gs19877838@business.example.com

Dispute Evidences:

Type: CREATE

Source: SUBMITTED_BY_BUYER

Notes: JPG GIF PNG PDF TXT | Maximum of 10 files up to 10 MB each, 50 MB in total

Type: PROOF_OF_FULFILLMENT

Source: REQUESTED_FROM_SELLER

[Messages Related to the Dispute](#)

Figure 5.1.30 Dispute Details Popup

You can see Transaction details from the Dispute popup by clicking Seller Transaction ID. (as shown in Figure 5.1.31)

Transactions of order: 00004821

Transaction ID:

Other transactions: [Captured: 61418468D3778552B](#)

61418468D3778552B

Capture ID: 61418468D3778552B

Name:	Ivan C Vinogradov
Transaction Email:	I.VinogradovVN@gmail.com

Amount:	709.78 USD	Captured: 709.78	Refunded: 684.52
Shipping:	15.99 USD		
Sales Tax:	33.80 USD		

Order/Invoice ID:	00004821
Order Date:	10/30/23 1:52 pm
Transaction Payment Status:	PENDING
Payment Method:	PayPal

Shipping Address:

Ivan C Vinogradov
2253 Hudson Street,
Denver, CO 80207
US

Transaction History

Figure 5.1.31 Transaction details of Dispute

To see Message History of Dispute press to **Messages Related to the Dispute** menu in Dispute details popup (as shown in Figure 5.1.32)

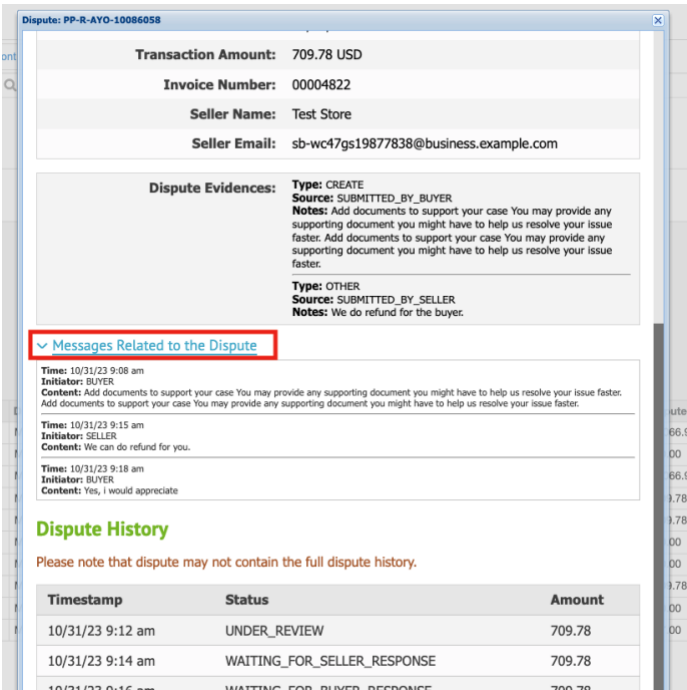


Figure 5.1.32 Messages Related to the Dispute

You can go to PayPal Resolution Center to manage Dispute by clicking Dispute ID or the Resolution Center on the right (redirect to Resolution Center as shown in Figure 5.1.33).

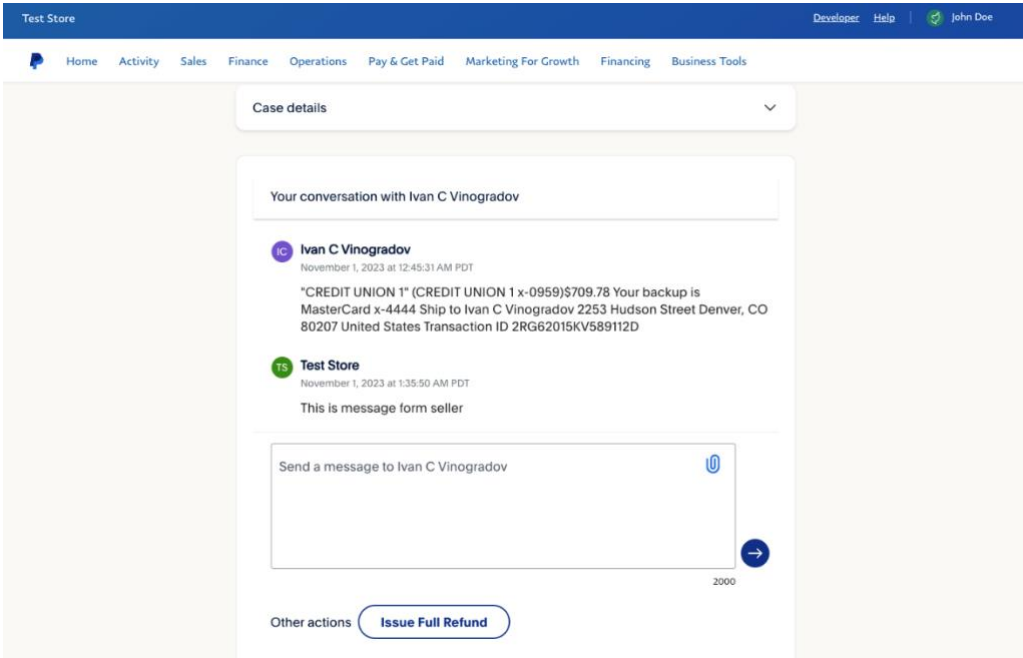


Figure 5.1.33 PayPal Resolution Center

5.1.4 PayPal Tracking API

If you sell tangible goods, you should send tracking information to PayPal as soon as you ship items to your customers.

Providing tracking information and item-level details can also help you improve the customer experience and reduce your operational costs. You can achieve it using the following techniques:

- Reduce dispute costs by allowing payers to dispute individual items instead of the entire order.
- Quicker access to money in payment and dispute holds. Adding tracking to an order qualifies holds for the early release.
- Real-time package tracking updates inside the PayPal app, which reduces the need for third-party apps to provide order tracking on PayPal orders.
- Improved seller risk profiles built from shipping and order status, which helps lower reserve requirements.

[More information](#)

We have created a job for automatically searching for suitable orders and creating connections with Tracking API. Also, you can use the function itself.

How to prepare order:

1. Made full capture for the order transaction. (**PayPal Payment Status** should be **COMPLETED**)
2. Navigate to the captured order (Merchant Tools > Ordering > Orders > Order: [ID]).

Merchant Tools > Ordering > Orders > Order: 00002702(RefArch) 

General | Attributes | Payment | Notes | History

Details for Order '00002702'

Information:	Contains 1 line item to 1 shipping location The total price is \$99.74.						
Date Received:	12/21/23 11:51:00 am Etc/UTC						
Site:	RefArch						
Created By:	Customer						
Customer:	Paul Lorens (unregistered customer)						
Customer No.:	n/a						
IP Address:	195.56.119.209						
Email:	p.lorenstest@gmail.com 						
Phone:	4084259805						
Order Status:	New	Confirmation Status:	Confirmed				
Shipping Status:	Not Shipped	Export Status:	Ready for Export				

Shipment 00012002

Qty	Product ID	Name	Manufacturer	Tax Rate	Unit Sales Price	Tax Basis	Item Total
1	701644391737M	Woven Trimmed Cardigan.		5.00 %	\$89.00	\$89.00	\$89.00
Shipment Shipping Cost:							\$5.99
Total Shipping Cost (001):							\$5.99
Shipping Total:							\$5.99
Tax Total:							\$4.75
Total:							\$99.74

[Send Email](#) [Print Order](#)

Figure 5.1.34 BM Order Page

3. Open Shipment and enter the Tracking Number

General Attributes

Shipment '00012002'

Shipping Method: 001

Shipping Address: Paul Lorens
2253 Hudson Street
Denver CO 80207
US

Tracking Number:

Shipping Status: Not Shipped

Qty.	Product ID	Name	Manufacturer	Tax Rate	Unit Sales Price	Item Total
1	701644391737M	Woven Trimmed Cardigan.		5.00 %	\$89.00	\$89.00
Shipment Shipping Cost:						\$5.99
Shipping Cost (001):						\$5.99
Tax:						\$4.75
Subtotal:						\$99.74

Print Packing Slip

<< Back

Figure 5.1.35 BM Shipment Page

How to use Tracking API with job:

NOTE: Job is searching for fully captured orders from the previous number of days.

1. Go to Administration > Operations > Jobs and choose **PaypalUseOrderTrackingApi** job.
2. Go to Job Steps to enter needed number of days (default 7)

Cache Search Indexes Code Versions Cartridge Path Search with ease...

Administration / Operations / Jobs /

PaypalUseOrderTrackingApi

General Schedule and History Resources Job Steps Failure Handling Notification

Job Parameters

Scope: RefArch

PaypalUseOrderTrackingApi

Select and Configure Step

Custom.PaypalUseOrderTrackingApi

Context: Site

ID* PaypalUseOrderTrackingApi

Description

Days* 7 Job Parameters

☐ Always execute on restart.

Exit Status Rules

Add Rule

Back Assign

© 2024 Salesforce, Inc. All Rights Reserved. RefArch Time Zone: Coordinated Universal Time | Instance Time Zone: Eastern Standard Time | Version: 24.1 Last Updated: Dec 18, 2023 (Compatibility Mode: 22.7)

Figure 5.1.36 BM Job Steps configuration

3. Run the Job

How to use Tracking API with function:

Use function **addTrackinAPI**

(bm_paypal/cartridge/scripts/paypal/paypalApi/bmPaypalApi.js).

You need to pass an order to create tracking, but make sure that the order is captured.

Important: by default, we pass OTHER as the shipment carrier.

Add a specific shipment carrier:

1. Navigate to the Merchant Tools > Ordering > Shipping Methods and press NEW or edit existed.
2. Enter carrier name Into **PayPal Carrier Name** field (carrier must be supported by PayPal).

The screenshot shows a 'General' configuration window for a shipping method. It includes fields for ID, Name, Description, Enabled, Default, Based On, Tax Class, and Currency. Below these are sections for Store Pick Up, Shipping Method Attributes, and Online Pick Up. The 'Paypal' section at the bottom is highlighted with a red box, showing the 'Paypal Carrier Name' field.

Figure 5.1.37 BM New Shipping Method configuration

Important: PayPal Carrier Name must be taken from [carriers list](#)

5.2 Storefront Functionality

Integration has 3 types of checkouts with PayPal. All of them are available from the payment page or cart page.

UPDATE: Checkout from cart requires customer data from PayPal. Review [PayPal-provided Billing Address and Phone Number](#) BEFORE ENABLING PAYPAL BUTTON ON A CART PAGE.

1. One-time checkout. Available for both guest and registered user types.
2. Checkout with vault creation. Available for registered user types if PayPal Vault Flow is enabled in Custom Preferences.
3. Checkout using saved PayPal Account. Available for registered users that already passed checkout with PayPal at least once and saved it.

5.2.1 Email, Phone, Shipping and Billing Address behavior

When a customer proceeds to checkout using PayPal button on the Express Checkout page, their phone, email, shipping and billing address will be taken from customer's PayPal

account. During Express Checkout the shipping address validation will be performed to avoid having an order created with non-applicable shipping address (Figure 5.2.1).

To checkout with a saved account, the default account values will be taken as shipping address and phone.

If a customer leaves the email field empty during checkout from payment page, last used email from the session will automatically be set in a form or PayPal account email, in case the session value is empty.

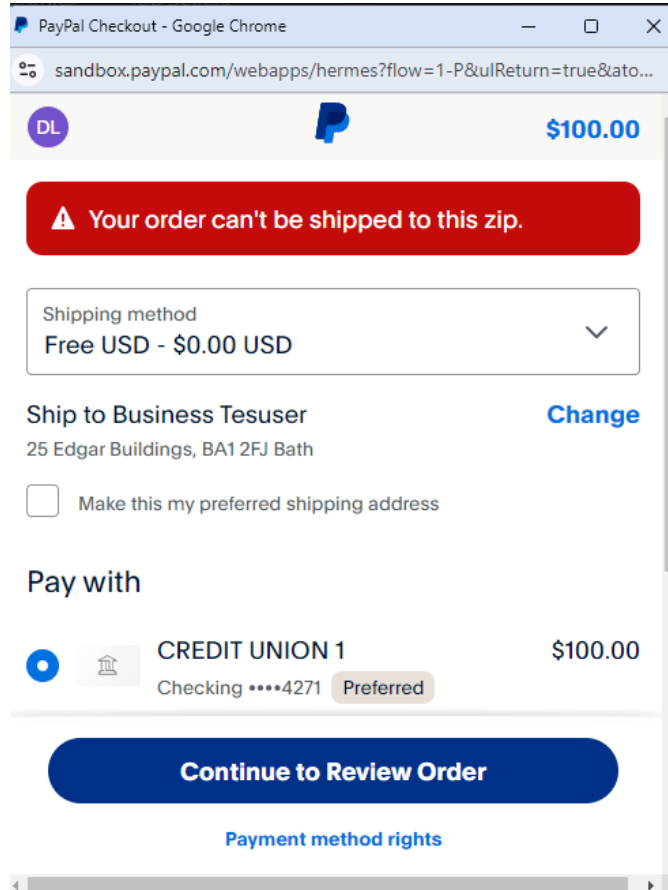


Figure 5.2.1 Error message of failed shipping validation.

5.2.2 Connect with PayPal

Ability for a buyer to create a storefront account by clicking on just one button.

This functionality is available on the following pages: Account Login, Page before Checkout (for guest buyers).

Connect with PayPal provides customers with the option to access your website or app using their PayPal credentials or establish a connection between their PayPal account and your platform, facilitating the exchange of basic information. This solution utilizes only the primary email address obtained from PayPal and grants login access only for the validated email address.

You can read more about this feature on [this](#) page.

Enhanced functionality:

1. Additional security layer - user who is logged in via PayPal for the first time and

has a pre-existing SF user account with the same email as the primary email in My PayPal account, will be asked to put in the credentials from pre-existing SF user account before the account linking with PayPal.

2. Email notifications:

- In case a user logs in via Connect with PayPal for the first time, they will also receive an email notification for login via a Single Authentication (Demandware standard login).
- A logged-in user who is linked via PayPal to their pre-existing SF user account, will be notified via email about account linking.
- A logged-in user who is linked via PayPal to their pre-existing SF user account, will be notified via email about account unlinking.

3. Automatic payment adding - adds account and address from PayPal to the user account.

To use this feature, you have to go through [these](#) steps. During configuration on PayPal side, select all fields to successfully create an account on storefront:

Information requested from customers

Select scope attributes for OpenID connect (OAuth2) protocol. You are requesting your customers to share this data with you.

Personal Information

☒ Full Name

☒ Email

Address

☒ Street Address

☒ City

☒ State

☒ Country

☒ Postal Code

Account information

☒ Account verification status

☒ PayPal account ID (payer ID)

Figure 5.2.2 Connect with PayPal configurations

After user clicks on the “Connect with PayPal” button and proceeds through PayPal authentication form successfully an account on storefront will be created with the data from PayPal buyer account.

Home

Login

Create Account

* Email

* Password

☐ Remember me

[forgot password?](#)

Login

Login with Google

Login with Facebook

Connect with PayPal

Figure 5.2.3 Connect with PayPal button

When deploying to production, do not forget to change Site Pref with ID: PP_CWPP_Button_Url from <https://www.sandbox.paypal.com/connect?> To <https://www.paypal.com/connect?> (Check site pref description).

Also, you should create and configure an agent user with specified permissions to use the full “Connect with PayPal” functionality:

- go to “Administration > Organization > Roles & Permissions” and choose “New”
- please fill in all the fields according to the example in the screenshot below (Figure 5.2.4) and click “Apply”

Administration > Organization > Roles > New Access Role

General Users Business Manager Modules Functional Permissions WebDAV Permissions Locale Permissions Price Adjustment Limits Customer Service Center Permissions

New Role

This page allows you to create a new access role. Please type in an ID that uniquely identifies the access role. An error is displayed if no ID is provided or if an access role with such an ID already exists. Click **Apply** to create the access role. Click **Cancel** to abort.

ID:

Description:

[<< Back to List](#)

Figure 5.2.4. New role creation

- choose tab “Functional Permissions” (Administration > Organization > Roles > Agent Role - Functional Permissions). Select necessary site/sites to use your changes (Figure 5.2.5):

Select Context

Select a context to view and edit permissions. **Organization** and **Sites** can't be selected at the same time.

☐ Organization

☒ Sites

☒ RefArch

☒ RefArchGlobal

Cancel

Apply

Figure 5.2.5. Select Context

- after “Apply” choose permissions as on screenshot below (Figure 5.2.6):

Administration > Organization > Roles > AgentRole - Functional Permissions

General Users Business Manager Modules **Functional Permissions** WebDAV Permissions Locale Permissions Price Adjustment Limits Customer Service Center Permissions

AgentRole - Functional Permissions

This list shows all functional permissions available in the system. Click **Select Context** to specify for which context you'd would like to modify permissions. Available context options are the organization, one site, or multiple sites. Select the checkboxes and use the **Update** button at the bottom of the page to grant certain functional permissions. Deselect the checkboxes and use the **Update** button to revoke specific functional permissions.

Selected Context: RefArch, RefArchGlobal
[Select Context](#)

Permission	Description	
Login_On_Behalf	Allows administrators to log into the Storefront on behalf of a customer.	☒
Replication_Run_For_Site	Allows to manage and start data replication processes for site-specific replication groups (i.e. search indexes).	☐
Login_Agent	Allows to log in as an agent user to the Storefront. Restrict the access to only those Business Manager users that have the permission 'Login_Agent'.	☒
Manage_Site_Catalog	Allows to manage the catalog and its assets in the selected site.	☐

Figure 5.2.6. Select Permissions

- After that, you need to select an existing or create a new user and give him the “AgentRole” created in the previous steps (example Figure 5.2.7):

Administration > Organization > Users > John_Doe@gmail.com - Roles

General **Roles** Permissions Access Keys

John_Doe@gmail.com - Roles

The list shows all roles to which this user belongs. Click **Assign** to add a role. Use the checkboxes and the **Unassign** button to unassign the user from the selected roles. Roles marked with have permission to view or manage users, access roles or access keys. They are therefore security-sensitive roles. Please be careful when changing these roles so as not to unintentionally give access privileges to certain users.

Assignment to the 'Administrator' role for this user must be managed in Account Manager. Any changes there will be effective upon next login into Business Manager.

Select All	ID	Description
☐	Administrator	The administrator has the rights to perform tasks related to the overall administration of the merchant organization and its users and roles. This access role is not site-specific and will grant the user access to the entire organization.

<< Back to List

Assign Unassign

Figure 5.2.7. User Role assigning

- click “Assign” on your user, choose “AgentRole” and click “Assign” one more time (Figure 5.2.8):

Administration > Organization > Users > Anatoli.Silusarenko@epam.com - Roles > Assign Roles

Select Roles

Select the roles you want to assign to the user.
Roles marked with  have permission to view or manage users, access roles or access keys. They are therefore security-sensitive roles. Please be careful when changing these roles so as not to unintentionally give access privileges to certain users.

 Assignment to the 'Administrator' role for this user must be managed in Account Manager. Any changes there will be effective upon next login into Business Manager.

Select All	ID	Description
☒	AgentRole	This role provides a possibility for the BM users to log in on behalf of customers.

[Assign](#) [Cancel](#)

Figure 5.2.8. User Role selecting

- after that fill all necessary Preference Configuration settings in Merchant Tools > PayPal Plugin > Preference Configuration > tab PayPal (Figure 5.2.9) enable feature, enter URL for feature and enter your User Login and Password:

PLEASE NOTE: In case an "Agent User Login and OCAPI" access key was created, you should fill the "PP_CWPP_Agent_Password" preference using this key instead of a password, as it takes precedence over the base password approach.

Connect with PayPal button enabled
(PP_CWPP_Button_Enabled) (Boolean) No It identifies whether the Connect with PayPal (CWPP) button is enabled on the Login and the Checkout pages

Connect with PayPal button url
(PP_CWPP_Button_Url) (String) https://www.sandbox.paypal.com/... This URL is used for Connect with PayPal feature. Sandbox URL: https://www.sandbox.paypal.com/connect? Production URL: https://www.paypal.com/connect?

Account Linking Security Layer enabled
(PP_CWPP_Account_Linking_Security_La...) (Boolean) No Enables additional verification step (enter credentials from the pre-existing user account) during the account linking while your buyer use Connect with PayPal for the first time and the user account email matches with the PayPal primary account email.

Automatic payment method adding enabled
(PP_Automatic_Payment_Method_Addl...) (Boolean) Yes Indicates if the 'Automatic payment method adding' feature is enabled via Connect with Paypal

Connect with PayPal agent login
(PP_CWPP_Agent_Login) (String) Connect with PayPal functionality requires agent credentials to perform "Login On Behalf" action.

Connect with PayPal agent password
(PP_CWPP_Agent_Password) (Password) Connect with PayPal functionality requires agent credentials to perform "Login On Behalf" action.

Figure 5.2.9. Custom Preferences filling

Recommendations which should be followed before moving to production:

1. Go to PayPal Sandbox configuration:
 - Enter your credentials for accessing PayPal Dev Console.
 - Go to the My Apps & Credentials.
 - Select your application.
 - Go to SANDBOX APP SETTINGS.

- Click Advanced on (bellow Log in with PayPal checkbox).
 - Unselect the ‘Enable customers who have not yet confirmed their email with PayPal to log in to your app’ checkbox.
2. Ensure that you have configured email confirmation for user account registration and MFA.

5.2.3 PayPal Digital Goods(Previous Pay Now).

PayPal ‘Digital Goods’ button helps to simplify buyer’s checkout by shortening the user journey in the PayPal checkout window.

‘Digital Goods’ is applicable to PayPal, Apple Pay, Google Pay Alternative payment methods and PayPal Credit Card (Card Fields).

This functionality is available on the following pages: PDP (Product), Mini Cart, Cart, Billing pages (for both guests and logged-in users who have their PayPal accounts.).

NOTE: Digital Goods for Alternative payment methods and PayPal Credit Card (Card Fields) is available only on the Billing page.

You can find more details by following the link: [PayPal Checkout](#)

To apply this feature, go to **Merchant Tools > PayPal Plugin > Preference Configuration > tab General, tab PayPal, tab Credit Card** and make sure the following site preferences are set to:

- Payment Model (PP_Payment_Model) - Sale Flow (Sale)

Name	Value	Default Value	Property Description
Payment Model			
(PP_Payment_Model) (Enum of Strings)	Sale Flow (Sale)	Sale Flow (Sale)	This determines whether the funds are captured immediately (Sale), later (Authorize) or Save Order Flow takes place (ExtendedAuth)

Figure 5.2.10. Payment Model site preference

- Digital Goods enabled (PP_Digital_Goods_Flow_Enabled) - Yes (By default is set to No)

Digital Goods Flow enabled			
(PP_Digital_Goods_Flow_Enabled) (Boolean)	Yes	No	PayPal Digital Goods Flow is designed for selling only digital goods which don't require delivery to the shipping address. In case you enable the flow then shipping options will be hidden on the storefront.

Figure 5.2.11. Digital Goods enabled site preference

- Store Credit Card in Vault (PP_Store_CC_To_Vault) - No (By default is set to No)

Store CC to Vault			
(PP_Store_CC_To_Vault) (Boolean)	No	No	Enables vault flow which allows saving credit cards for logged-in buyers. Note: If you disable this site preference and the logged-in user has a saved credit card in the wallet, then they will not be able to use it on the checkout.

Figure 5.2.12 Store Credit Card in Vault site preference

When everything is set correctly and the 'Digital Goods' is enabled, the PayPal button will activate the following flow:

1. PayPal 'Digital Goods' button is rendered, and the following alert message is rendered on the Cart page: 'Shipping address is not required for Digital Goods and provided address by PayPal will be ignored.'

Enter Promo Code

Promo Code

Get 10% off Orders Above \$150 **-\$225.00**

Shipping

Free USD

Shipping cost \$0.00

Sales Tax \$101.25

Order Discount **-\$225.00**

Estimated Total \$2,126.24

Shipping address is not required for Digital Goods and provided address by PayPal will be ignored.

Pay as low as \$98.88/mo. [Learn more](#)

Figure 5.2.13. PayPal button on the Cart page during 'Digital Goods' flow

The following alert message is rendered on the Billing page: 'Shipping address is not required for Digital Goods flow':

Shipping address is not required for Digital Goods flow

Payment

Billing Address

Billing details will be retrieved from the chosen payment method.

Figure 5.2.14. PayPal button on the Billing page during Digital Goods flow

2. When buyer clicks on the 'Buy Now' button, the following window is shown:

IC \$314.99 USD

Pay with ☒ **Split**

VISA The Bank Card Platinum Rewards ****1111 **PREFERRED** \$314.99 USD

☐ **CREDIT UNION 1** Checking ****0959

☐ **PayPal Credit** You have PayPal Credit! No Interest if paid in full in 6 months for your purchase of \$314.99. [See terms](#)

[See More](#)

[+ Add debit or credit card](#)

Pay Later

☐ **Pay in 4** 4 payments of \$78.75 due every 2 weeks, starting today. [Learn more](#)

☐ **Pay Monthly** **NEW** Plans as low as \$28.87 for up to 12 months. [Learn more](#)

[Complete Purchase](#)

[Payment method rights](#)

[Cancel and return to John Doe's Test Store](#)

Figure 5.2.15. PayPal window during 'Digital Goods' flow

NOTE: Shipping section does not appear since 'Digital Goods' is used only for Digital Goods that do not require shipping.

3. After clicking on the 'Complete Purchase' button, the buyer will be redirected to the Order review page. The following alert message is rendered on the Order confirmation page: 'Shipping address is not required and Shipping Method is not required'.

Receipt

Order Number: 00033304

Order Date: 1/3/23

Shipping Address is not required and Shipping Method is not required

Billing Address:

Ivan C Vinogradov

1 Main St

San Jose, CA 95131

dimatuz@gmail.com

2025550162

Payment:

I.VinogradovVN@gmail.com

paypal / \$1,314.99

1 Items

\$299.99

Black Flat Front Wool Suit



Color: Black

Size: 40

Width: Regular

Quantity

1

Total

\$299.99

Subtotal

\$299.99

Shipping

\$0.00

Sales Tax

\$15.00

Total

\$1,314.99

Figure 5.2.16. The Order confirmation page view after ‘Digital Goods’ flow is successfully processed

In case when ‘Digital Goods’ flow is disabled due site preferences, the general PayPal button flow is applied.

5.2.4 One-time checkout (non-vault flow)

One-time checkout is available from payment page (Figure 5.2.19) and from cart page (Figure 5.2.17), if custom preferences value of **Show PayPal button on the cart page** is set to ‘Yes’.

After clicking on the available payment method from Smart Payment Button the customer proceeds through checkout on PayPal side (Figure 5.2.18) and returns to the storefront to the order review page (Figure 5.2.21).

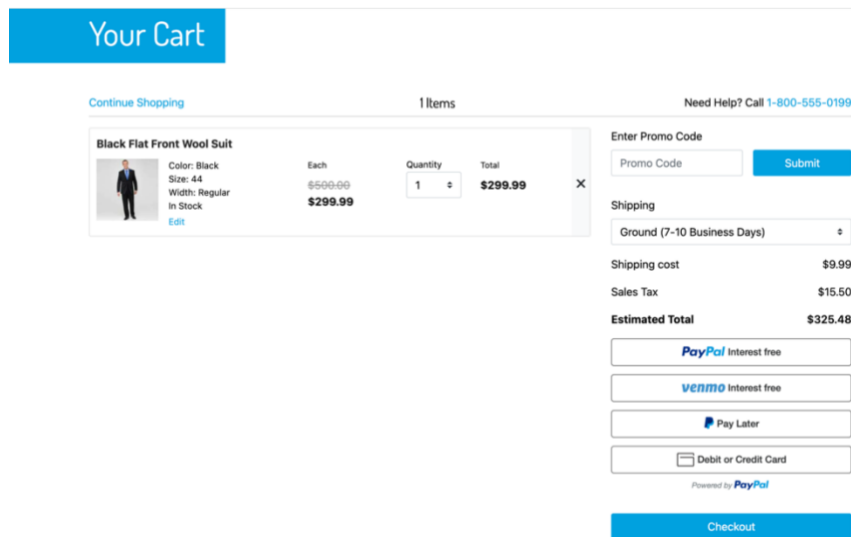


Figure 5.2.17 Checkout from the cart page using Smart Button

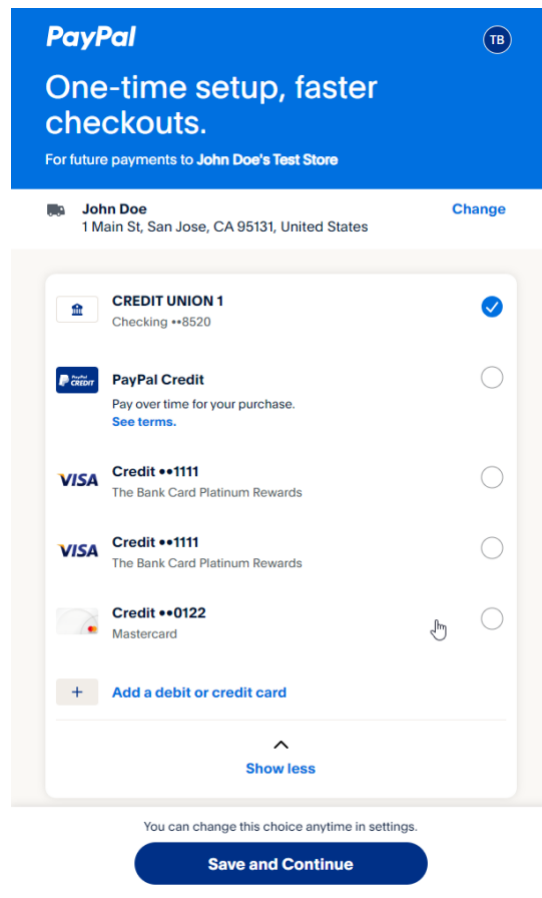


Figure 5.2.18. Basic checkout pop-up example

Payment

Billing Address

NewUser22 BookUser22 Neapol 250 Street 212 Djuno AK 12345

CREDIT

PayPal

venmo

PayPal Checkout

Pay Later

Debit or Credit Card

Powered by PayPal

Figure 5.2.19. Checkout from the payment page using Smart Button

Payment

Billing Address

Billing details will be retrieved from the chosen payment method.

CREDIT

PayPal

Select an account

✓ New account

I.VinogradovVN@gmail.com

p.lorenstest@gmail.com

Figure 5.2.20. Checkout from the payment page with already chosen account

Customer

Customer Information:

valter@re.ui

Shipping

Shipping Address:

Good Man
473 Wiseman Street
Sevierville KS 90401
8004444444

Shipping Method:

Ground (7-10 Business Days) \$5.99

Payment

Billing Address:

Monalisa Patel
1 Main St
San Jose CA 95131
valter@re.ui
8004444444

Payment:

PayPal
ptest4@paypal.com
\$49.33

Need Help? Call 1-800-555-0199

Order Summary

Subtotal \$40.99

Shipping \$5.99

Sales Tax \$2.35

Total \$49.33

1 Items \$40.99

Striped Shirt



Color: Royal Multi
Size: M
In Stock

Each	Quantity	Total
\$59.00	1	\$40.99

Place Order

Figure 5.2.21. Order review page

5.2.5 Checkout with PayPal Vault Flow

For registered user, the integration provides an ability to save PayPal Account to vault as payment method in a storefront. This allows them to proceed through checkout with PayPal faster next time.

To enable vault flow in Business Manager, navigate to **Merchant Tools -> PayPal Plugin > Preference Configuration > tab PayPal** and set Store PayPal account in Vault preference value to **Enabled**

Visually checkout with vault is similar to [Basic checkout \(non-vault flow\)](#) with 2 major differences:

- 1) On the payment page the registered customer has an additional checkbox **Save PayPal account** (Figure 5.2.22).
- 2) On PayPal side after login the customer will see **Save and Continue** instead of Save and no shipping address option (Figure 5.2.23). Check [Email, Phone, Shipping and Billing Address behavior](#) for more details.

If the registered customer had **Save PayPal account** checkbox checked after order was placed successfully, they can review saved PayPal account in **My Account** section. (Figure 5.2.24).

The screenshot shows the 'Payment' section of a checkout page. At the top, the 'Billing Address' is displayed as 'garyGeuest garyGeuestLastName 123 Main St New York City NY 10001'. Below the address, there are two tabs: 'CREDIT' and 'PayPal'. The 'PayPal' tab is selected, showing a 'PayPal Checkout' button in yellow and a 'Debit or Credit Card' button in dark grey. Below these buttons, there is a checkbox labeled 'Save PayPal account' which is checked. At the bottom, it says 'Powered by PayPal'.

Figure 5.2.22. Smart Button for registered user with vault enabled on the payment page

The screenshot shows the PayPal vault setup page. At the top, the PayPal logo is on the left, and a user icon 'PL' is on the right. The main heading is 'One-time setup, faster checkouts.' followed by 'For future payments to John Doe's Test Store'. Below this, the user's name 'Paul Lorens' and address '2253 Hudson Street, Denver, CO 80207, United States' are shown, with a 'Change' link. A card displays the selected payment method: 'CREDIT UNION 1' with a checking card ending in '6425'. A 'More ways to pay' link with a dropdown arrow is also present. At the bottom, there is a link to 'View PayPal Policies for payment method rights.' and a 'Save and Continue' button.

Figure 5.2.23. Saving to vault on the PayPal page

The screenshot displays the 'My Account' section with the following components:

- Order History:** Includes a 'View' link and a section for the 'Most Recent Order'.
 - Order Details:**
 - Image of a monitor
 - Order Number: pp_00185506
 - Date Ordered: 6/17/20
 - Order Status: NEW
 - Shipped to: Rick Flores
 - Summary:**

Total items	Total
1	\$15,766.78
- Payment:** Includes an 'Add New' link.
- PayPal:** Shows the email address 'I.VinogradovVN@gmail.com (Default)' as the saved payment method.

Figure 5.2.24. My Account section with saved PayPal account

5.2.6 Checkout using saved PayPal Account

After the registered customer saves PayPal account to vault (Figure 5.2.24), they can pay for an order using vaulted payment method and proceed through the checkout without Smart Button loading.

Review [Email, Phone, Shipping and Billing Address behavior](#) for more information about addresses behavior for saved account.

The screenshot shows the 'Payment' page with the following elements:

- Billing Address:** A dropdown menu showing 'garyGeuest garyGeuestLastName 123 Main St New York City NY 10001'.
- Payment Method:** Two buttons are visible: 'CREDIT' and 'PayPal'.
- Select an account:** A dropdown menu showing 'I.VinogradovVN@gmail.com'.
- Next Step:** A blue button labeled 'Next: Place Order'.

Figure 5.2.25. Saved PayPal account on the payment page

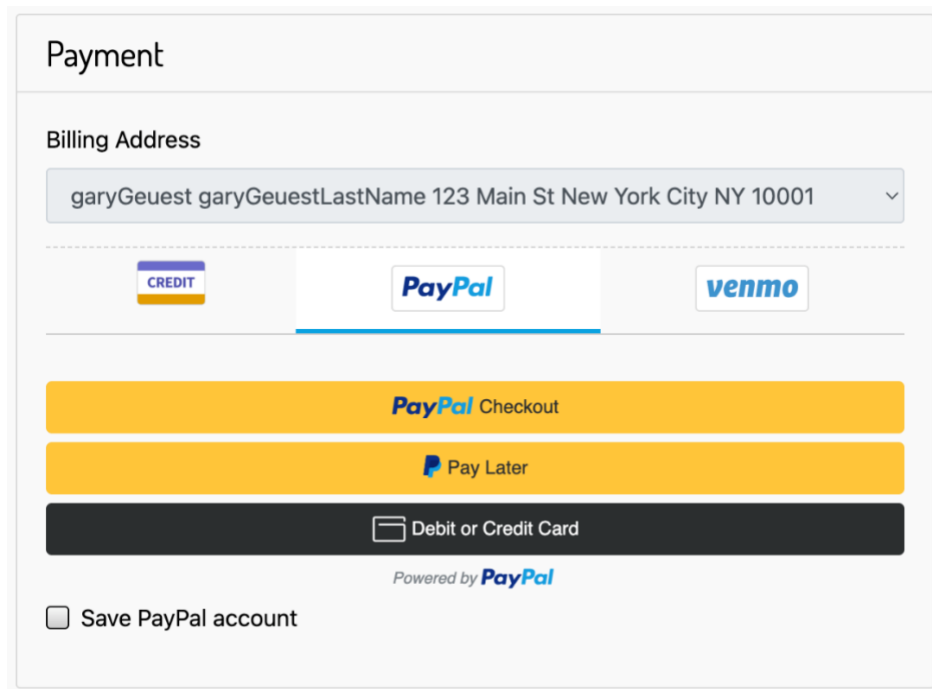
5.2.7 Checkout with multiple saved PayPal accounts

User can save new PayPal accounts till they reach the limit. Site preference “Limit of saved PayPal accounts” regulates the number of accounts that can be saved. Default value is “1”.

Customer can add new account from billing page. In a dropdown with saved account customer must choose **New PayPal** account option and load Smart Button (Figure 5.2.26).

After an order is placed, the customer can observe the new account in **My Account** section (Figure 5.2.27) and in the dropdown on the payment page (Figure 5.2.28).

Logged-in users can save a PayPal account via the Checkout page (Payment step) to the wallet only once. It means in case the user doesn't have a saved PayPal account in the buyer's wallet/PayPal Vault, then the Save my PayPal account to the wallet checkbox is shown. If the user has a saved PayPal account in his buyer's wallet/PayPal vault then the Checkbox is hidden on the UI level and as a result, the buyer cannot save a new PayPal account via the Checkout.



The screenshot displays the 'Payment' section of a checkout page. At the top, there's a 'Billing Address' dropdown menu showing 'garyGeuest garyGeuestLastName 123 Main St New York City NY 10001'. Below this, three payment method buttons are visible: 'CREDIT', 'PayPal' (which is selected and highlighted with a blue underline), and 'venmo'. Underneath these buttons, there are three large, horizontal buttons for checkout options: 'PayPal Checkout' (yellow), 'Pay Later' (yellow), and 'Debit or Credit Card' (dark grey). Below the 'Debit or Credit Card' button, it says 'Powered by PayPal'. At the bottom of the section, there is a checkbox labeled 'Save PayPal account'.

Figure 5.2.26. New PayPal account option on payment page

Home

Profile Edit First Name Good Last Name Man Email goodman@test.com Phone 8004444444	Order History View Most Recent Order  Order Number: 00013602 Date Ordered: 10/6/22 Order Status: NEW Shipped to: Harry Garcia Total items 7 Total \$199.42
Password Edit Password *****	Payment View Credit Visa *****1111 Ending 1/2024 Add New
Address Book View Default Address 4347 Kincheloe Road - Tigard - 97223 Harry Garcia 4347 Kincheloe Road Tigard, OR 97223 4087809636 Add New	PayPal ptest4@paypal.com (Default) Remove epamtester@pptest.com Remove Add New

Figure 5.2.27. Multiple saved accounts in My Account section

Payment

Billing Address

Billing details will be retrieved from the chosen payment method.

CREDIT

PayPal

Select an account

✓ New account

I.VinogradovVN@gmail.com

p.lorenstest@gmail.com

Figure 5.2.28. Multiple account view on payment page

5.2.8 Alternative Payment Methods

Alternative payment methods allow customers across the globe to pay with their bank accounts, wallets, and other local payment methods. The Relevant alternative payment method is automatically presented in a separate tab by using unbranded flow. See an example here: [Unbranded APM](#)

For example, a customer in the Netherlands might want to pay using iDEAL, which is used by more than half of consumers in the Netherlands for online purchases, whereas a customer in Belgium on the same website might want to pay using Bancontact, a popular payment method there.

Alternative Payment Methods are available on the billing page only. Shipping address and phone number are always provided by the customer. Email will be pulled from the billing page.

The transaction will be completed only when the customer returns to the storefront. If a transaction was not confirmed or the customer did not finish checkout on the PayPal side, the customer's basket still will be available.

Important: after the customer confirms the transaction and returns to the storefront, the order is placed immediately, skipping the Order Review page.

UPDATE: Alternative Payment Methods appearances based on storefront locale, not on customer's locale.

Pagamento

Indirizzo di fatturazione

Billing details will be retrieved from the chosen payment me ▾

CREDIT PayPal MyBank

MyBank MyBank

Your payment method details

Full Name

Figure 5.2.29. Alternative Payment Methods appearances on the billing page



Figure 5.2.30. Alternative Payment Method transaction confirmation

5.2.9 Venmo Payment Method

It appears on the 'Product', 'Cart', 'Mini-Cart', 'Billing' pages in case if 'Venmo' payment method is available.

Important note: In release 25.3.0 PP SFRA Venmo button is available on the Billing page only.

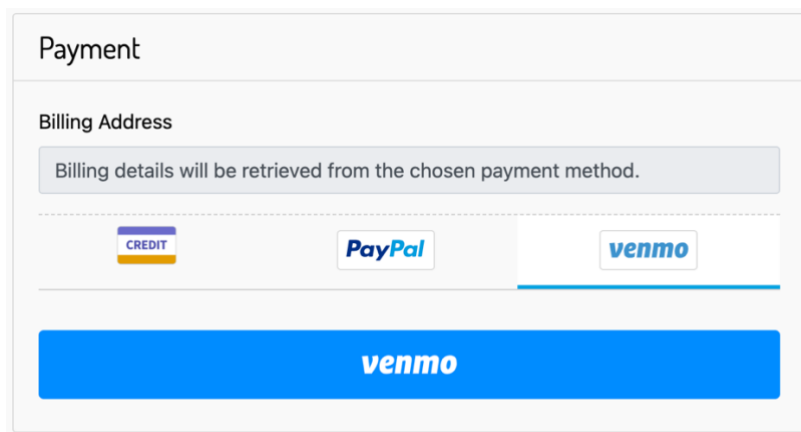


Figure 5.2.31. Venmo payment method on the Billing page

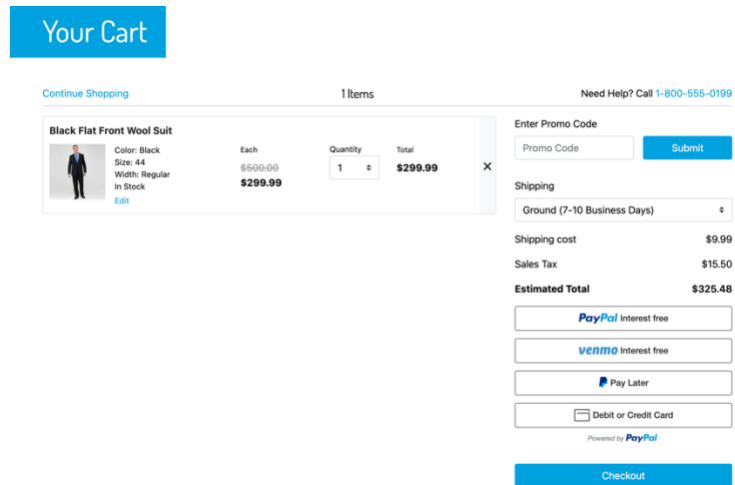


Figure 5.2.32. Venmo payment method on the express checkout page (Product, Cart, Mini-cart)

5.2.10 Manage Vault flow from My Account

If Vault flow was enabled in the Custom Preferences, the customer can save PayPal Account as payment method for faster checkout or remove saved account from My Account section at a storefront (Figure 5.2.33).

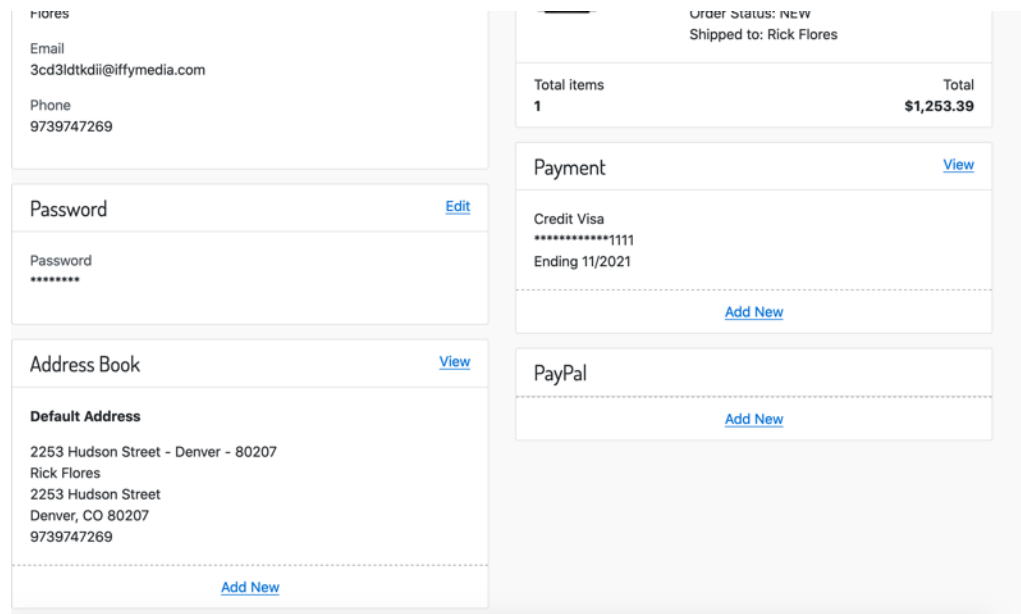


Figure 5.2.33. PayPal section in My Account

After customer clicks **Add New** button, PayPal Smart Button appears and customer must click on it to run through Vault flow. Once customer approves saving PayPal Account, the page will be reloaded and a new PayPal account email will appear under PayPal section (Figure 5.2.34).

Customer can delete PayPal account on remove button or from PayPal side. If customer deletes PayPal vault from PayPal it will be automatically removed from My Account.

The screenshot displays the 'My Account' interface. On the left, there are sections for 'Email' (3cd3ldtkdii@iffymedia.com), 'Phone' (9739747269), 'Password' (with an 'Edit' link), and 'Address Book' (with a 'View' link). The 'Address Book' section shows a 'Default Address' for Rick Flores at 2253 Hudson Street, Denver, CO 80207, with phone number 9739747269, and an 'Add New' link. On the right, there is a summary of 'Total items' (1) and 'Total' (\$1,253.39). Below this is the 'Payment' section (with a 'View' link) showing a 'Credit Visa' card ending in 1111, expiring 11/2021, with an 'Add New' link. At the bottom right is the 'PayPal' section showing the account 'I.VinogradovVN@gmail.com (Default)' with a 'Remove' link and an 'Add New' link.

Figure 5.2.34. Saved PayPal Account view in My Account

5.2.11 Credit Card functionality on the Billing page

Card fields are rendered instead of the default credit card fields on the billing page.

Available options:

- **Pay with Credit Card**

The screenshot shows the 'Card Fields form' with a PayPal logo at the top. The form contains the following fields: '* Cardholder name' (text input), '* Card number' (text input), '* CVV' (text input with three asterisks), and '* Expiration date' (text input with format MM/YY(YYYY)). At the bottom, there is a checkbox labeled 'Save this card'.

Figure 5.2.35. Card Fields form

- **Save Credit Card**

To configure Save Credit Card go to Business Manager > Merchant Tools > PayPal Plugin > Preference Configuration > Credit Card and enable Store Credit Card in Vault (PP_Store_CC_To_Vault)

NOTE: Save Credit Card on the Billing page via Card Fields works using separate from the My Account page [API](#)

- **Pay with saved Credit Card**

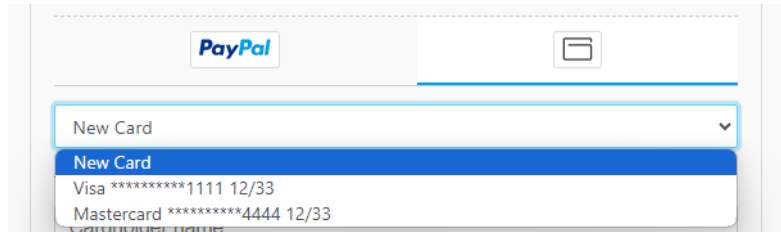


Figure 5.2.36. List of saved Credit Cards

NOTE: Save Credit Card and Pay with saved Credit Card applicable only for registered users.

5.2.12 Credit Card functionality on the My Account page

Credit Card on the My Account page gives customers the ability to manage credit card payment methods.

Available options:

- **Save Credit Card**

NOTE: Save Credit Card on the My Account works using separate from card fields [API](#)

NOTE: Enable Store Credit Card in Vault and Store Credit Card in Vault via My Account Custom Preferences to be able to save Credit Card via Card Fields on the Account Page

NOTE: The number of stored credit cards may be limited by preference

Business Manager > Merchant Tools > PayPal Plugin > Preference Configuration > Credit Card

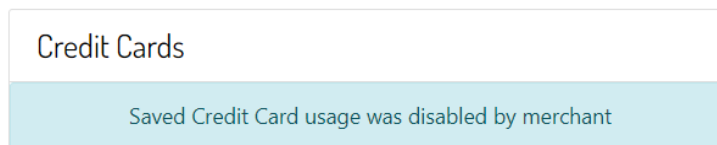


Figure 5.2.37. Credit Card section If vaulting Is not active

- Delete Credit Card
- Make default Credit Card

Credit Cards	
Joy Gray Mastercard*****0011 expiration date: 08/24	Make default Remove
John Doe Visa*****1111 expiration date: 12/25 (Default)	Remove
Add New	

Figure 5.2.38. Credit Card section

Credit Cards	
Vadim Visa*****1111 expiration date: 01/26	Make default Remove
Vadim Visa*****1111 expiration date: 01/24 (Default)	Remove
You reached the maximum number of saved credit cards. In order to save a new credit card please remove the already saved one.	

Figure 5.2.39 Limited Credit Cards

Credit Cards

* First Name

* Last Name

* Address 1

Address 2

* Country

United States

* State

* City

* ZIP Code

* Phone Number

* Cardholder name

* Card number

* CVV

* Expiration date

Cancel

Save

Figure 5.2.40. Credit Card form

5.2.13 Credit Card and Google Pay 3dSecure Fallback

The system will decide if the activation of the 3DSecure flow is mandatory for the customer's credit card and Google Pay after verification. Verification is based on requirements which include Strong Customer Authentication (SCA) - an authentication process that validates the identity of the user or a payment service or a payment transaction. The additional context, or *payment indicator*, ensures that buyer authentication and card on file, among other factors, are appropriately handled. In our plugin we cover some scenarios based on '*payment_initiator*', '*payment_type*' and '*usage*' indicators. Failure to pass these payment indicators might result in rejected transactions. You can find more info on SCA and payment indicator on [PayPal Developer portal](#).

Applicable to the following payment methods - Google Pay, saved and non-saved credit cards.

To enable go to: Merchant Tools > PayPal Plugin > Preference Configuration > General tab and set **3DS verification**.

NOTE: Even though SCA payment indicators are passed for saved credit cards, the 3DSecure verification popup will not be triggered on the Billing page. It shows up on My

Account page when credit card is being saved, if Store Credit Card in Vault via My Account site preference is enabled.

5.2.14 Credit Card Expire Notification

This functionality will notify the customer about the expiration of a credit card which is already expiring or has already expired. By default, this functionality is disabled. Other options are also available:

- **Notify when expired only** - will notify and highlight only if CC is expired
- **Notify 1 month before expiration** - will notify and highlight CC's which are going to expire in 1 month
- **Notify 2 months before expiration** - will notify and highlight CC's which are going to expire in 2 months
- **Notify 3 months before expiration** - will notify and highlight CC's which are going to expire in 3 months

To enable go to: Merchant Tools > PayPal Plugin > Preference Configuration > Credit Card tab and set **PP_Credit_Card_Expire_Notification**

An example of notifications can be seen below (Account, Checkout page):

The screenshot displays the PayPal Account page with two notification banners at the top. The first banner (red) states: "Please note that you have a credit card in your wallet that is expired. You will not be able to place the order with this credit card." The second banner (yellow) states: "Please note that you have a credit card in your wallet that is going to expire soon. Please check it." Below the banners, the page is divided into two main sections: "Profile" and "Credit Cards".

Profile Section:

- First Name:** John
- Last Name:** Doe
- Email:** test@gmail.com
- Phone:** 1234567890

Password Section:

- Password:** *****

Credit Cards Section:

Card Type	Card Number	Expiration Date	Status	Action
Visa	*****1111	08/23	Expired	Remove
Visa	*****1119	09/23	Expires in 1 month	Remove
Visa	*****0061	10/23	Expires in 2 months	Remove
Visa	*****1111	11/23	Expires in 3 months	Remove

[Add New](#)

Figure 5.2.41 Credit Card Expire Notification (Account page)

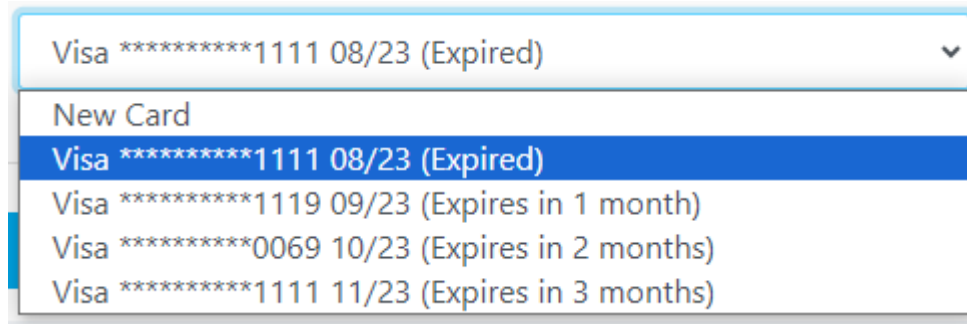


Figure 5.2.42 Credit Card Expire Notification (Checkout page)

5.2.15 Credit Messaging

Credit Messaging is present on:

- Category page - Figure 5.2.43
- Product detail page - Figure 5.2.44 and Figure 5.2.45
- Cart page - Figure 5.2.46

The visibility of each message can be changed in Custom Preferences at the Business Manager. Check [Updating Cartridge Custom Site Preferences](#) for details.

Credit option for cart and category page is calculated based on current cart total. If the cart total is lower than 150, credit messages will notify customer about it. (Figure 5.2.45)

For PDP page the product price will be used as a value for credit options. For product sets the message appears for each option and for set total.

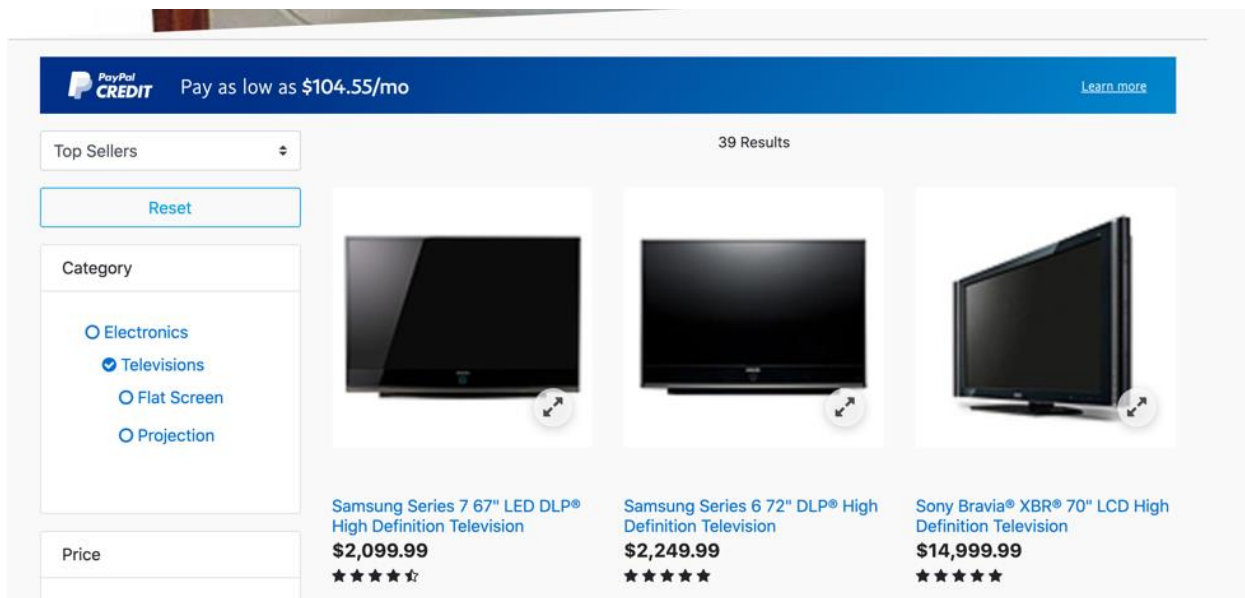


Figure 5.2.43 Credit message on category page

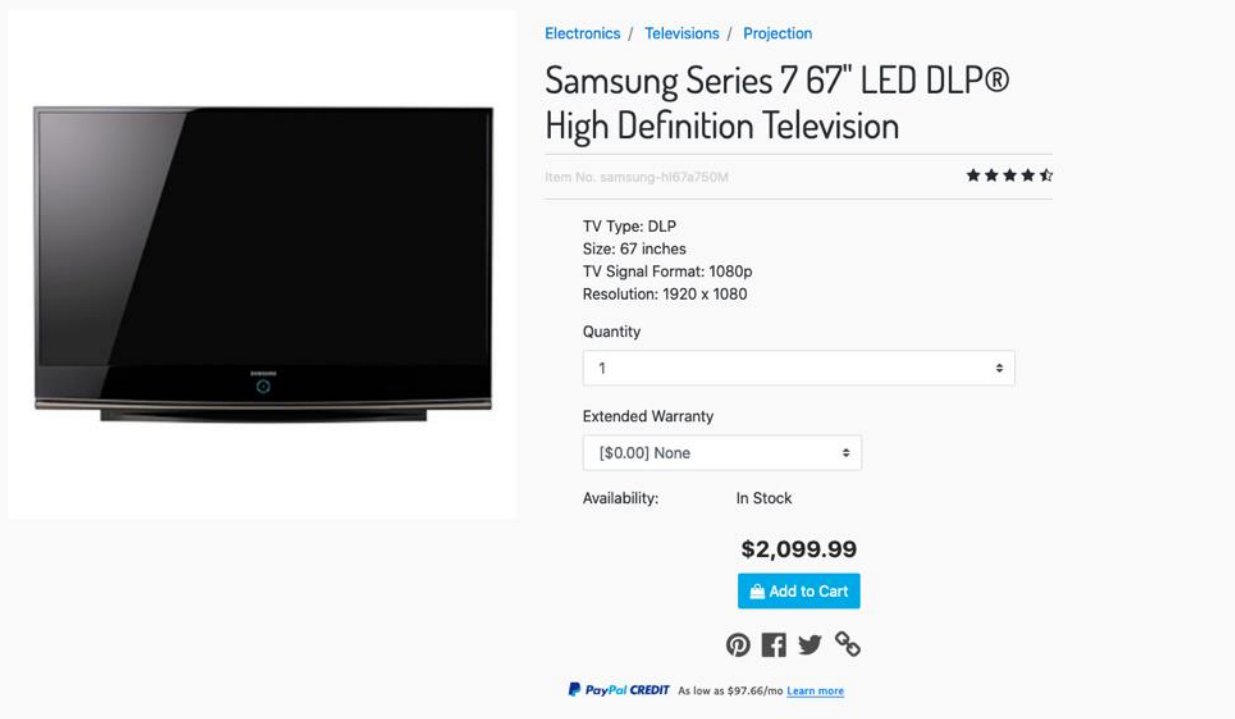


Figure 5.2.44. Credit message on PDP page

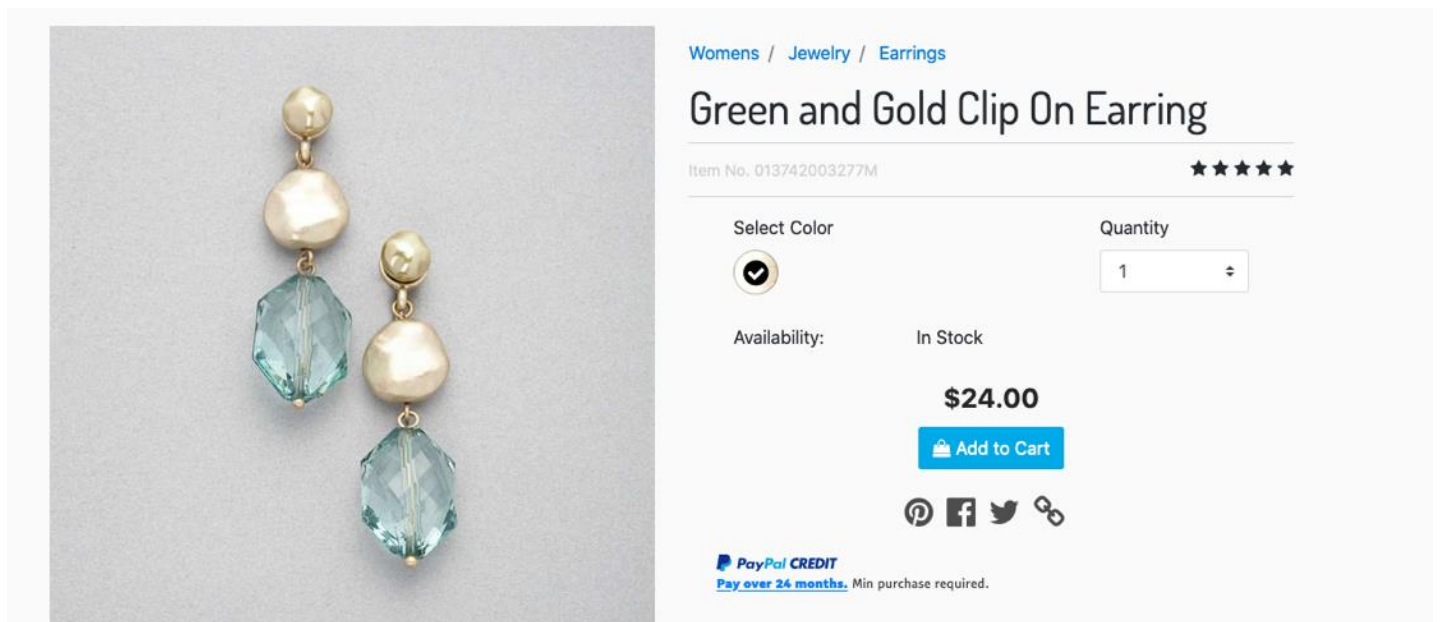


Figure 5.2.45. Credit message for product with small price

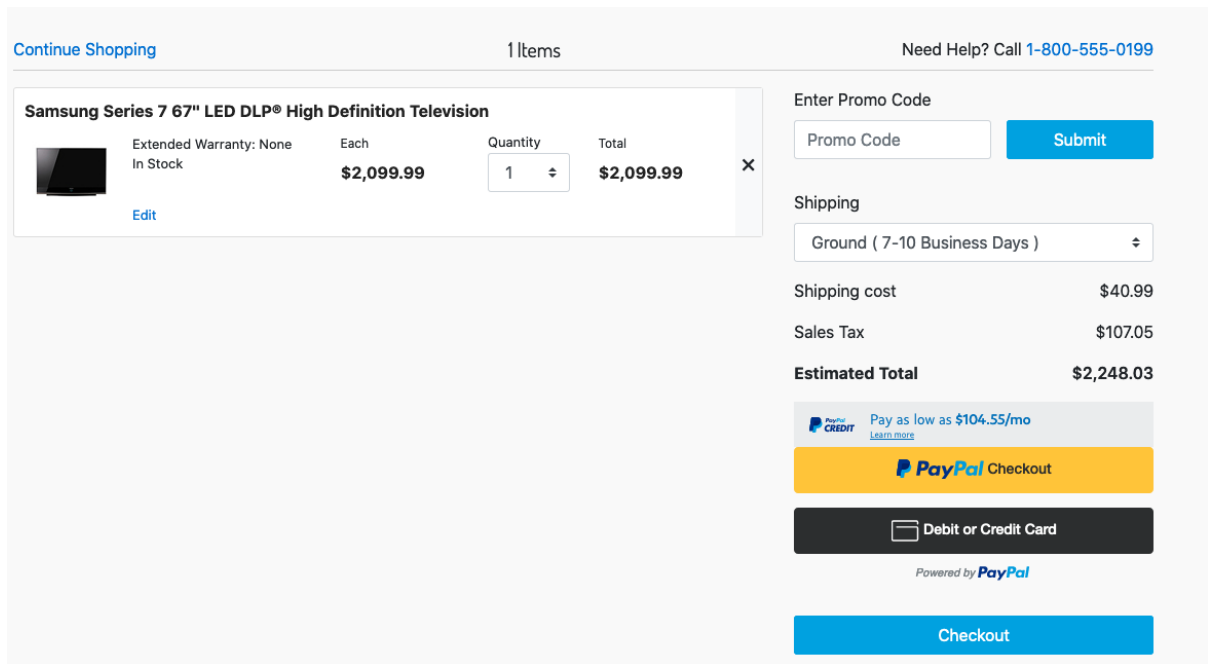


Figure 5.2.46. Credit message on cart page

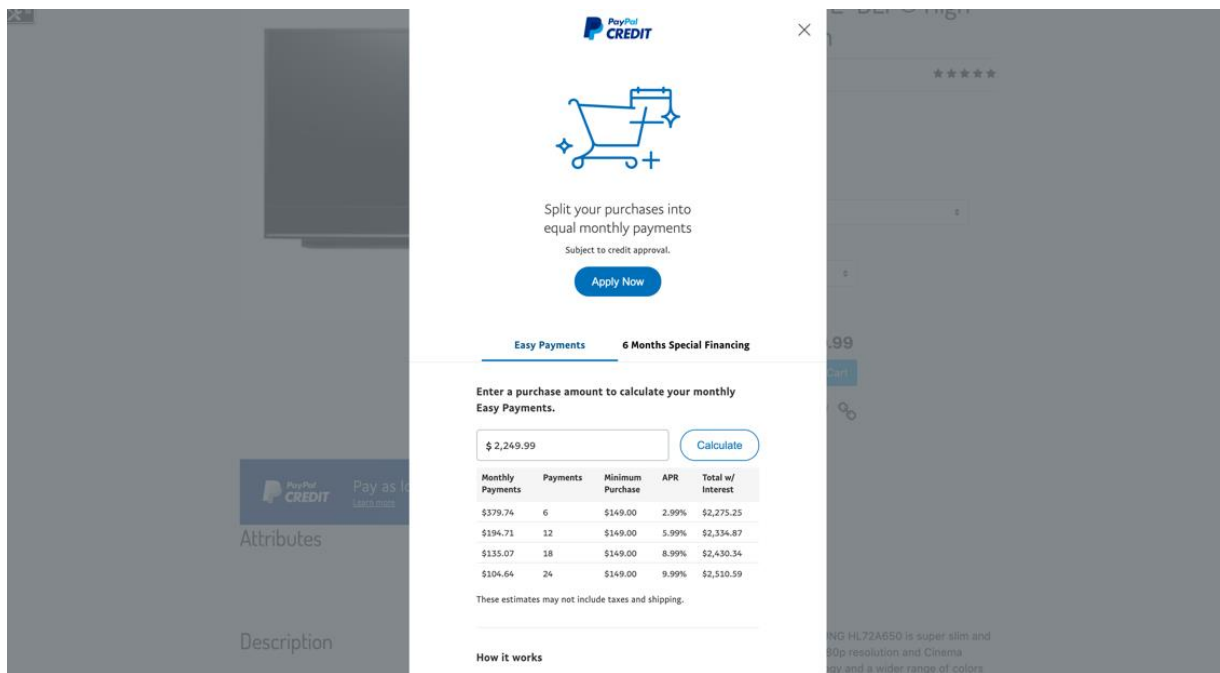


Figure 5.2.47. Credit message pop-up

You can change the location of the credit message by moving templates according to storefront design.

PDP message is represented by two templates: `bundleDetailCreditMessage.isml` and `productDetailCreditMessage.isml` which are located at `cartridges/paypal_credit_messaging/cartridge/templates/default/product/components/socialIcons.isml` template.

Cart message is represented by `cartCreditMessage.isml` which is located at `cartridges/paypal_credit_messaging/cartridge/templates/default/cart/cart.isml`

Category message is represented by `categoryCreditMessage.isml` which is located at `cartridges/paypal_credit_messaging/cartridge/templates/default/search/searchResultsNoDecorator.isml`

5.2.16 Button Messages

Button Messages is present on:

- Product view page - Figure 5.2.48
- Product detail page - Figure 5.2.50
- Mini Cart page - Figure 5.2.51
- Cart page - Figure 5.2.52Figure 5.2.52Figure 5.2.46
- Billing page - Figure 5.2.53

The visibility of each message can be changed in Custom Preferences at the Business Manager. Check Updating Cartridge Custom Site Preferences for details.

Credit option for cart and category page is calculated based on current cart total. If the cart total is lower than 150, credit messages will notify customer about it. (Figure 5.2.49)

For PDP and PVP page the product price will be used as a value for credit options. For product sets the message appears for each option and for set total.

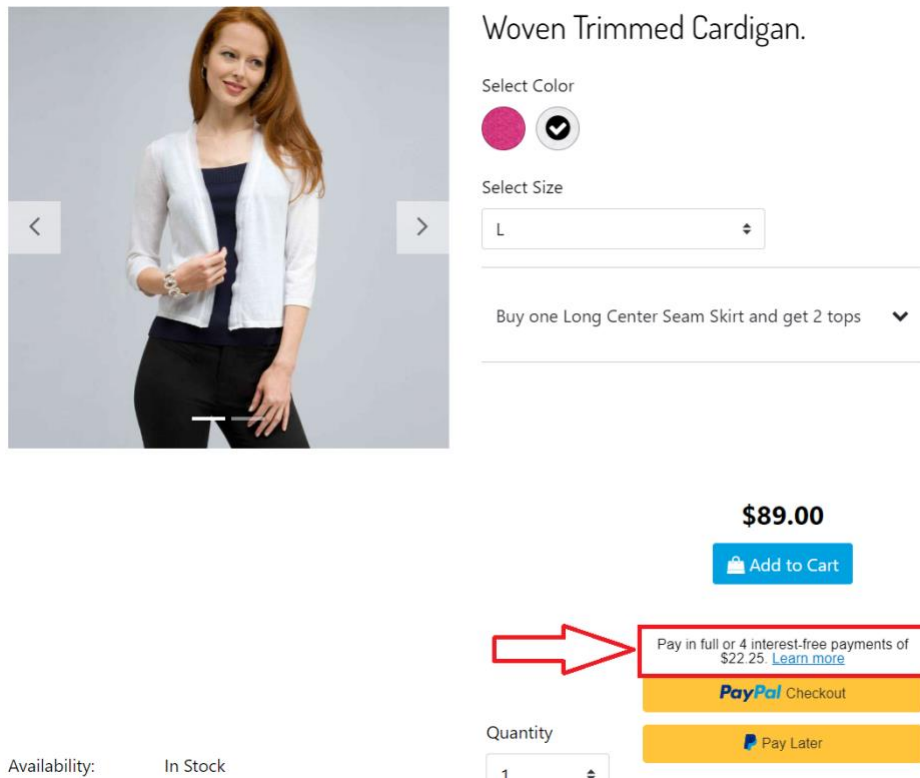


Figure 5.2.48 Button Message on Product view page

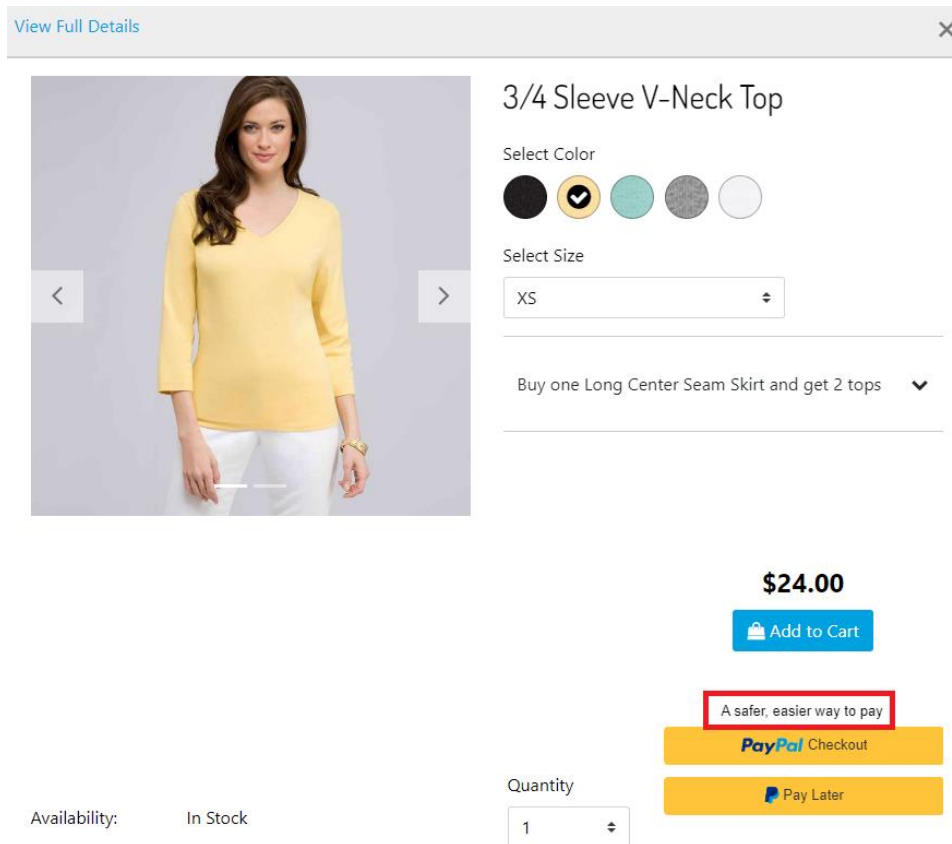


Figure 5.2.49 Button message for product with small price

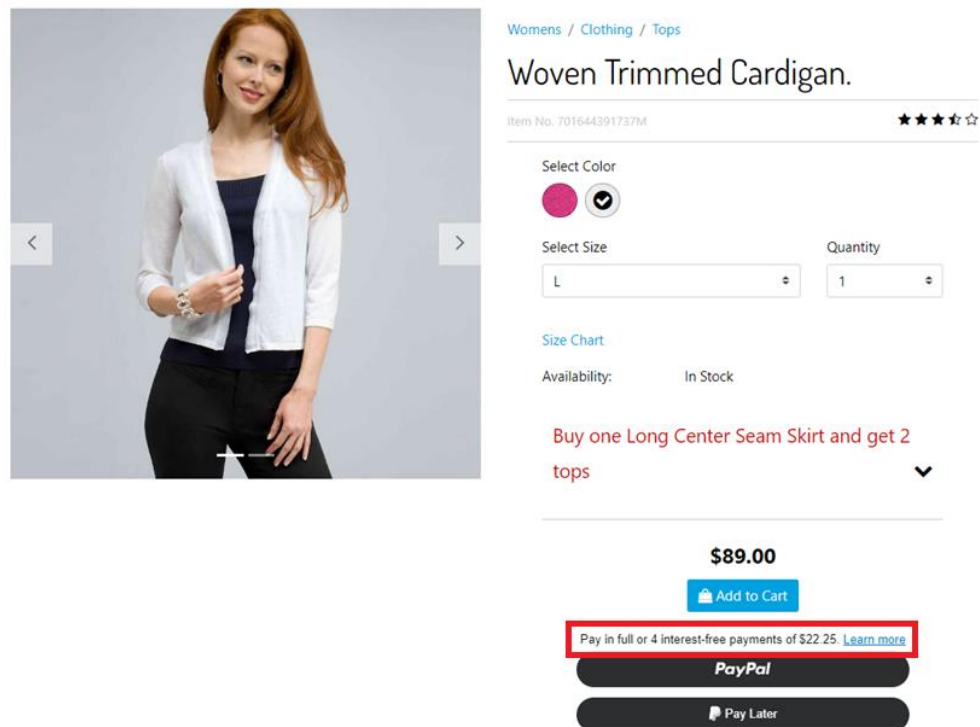


Figure 5.2.50 Button Message on Product description page

Your shopping cart [View Cart](#)

Woven Trimmed Cardigan.


Color: White
 Size: L
 In Stock

Each	Quantity	Total
\$89.00	1	\$89.00

Estimated Total **\$89.00**

[PayPal](#) Checkout

[Pay Later](#)

Pay in full or 4 interest-free payments of \$22.25. [Learn more](#)

Checkout

Figure 5.2.51 Button Message on Mini cart page

[Continue Shopping](#) 1 Items [Need Help? Call 1-800-555-0199](#)

Woven Trimmed Cardigan.


Color: White
 Size: L
 In Stock
[Edit](#)

Each	Quantity	Total
\$89.00	1	\$89.00

Enter Promo Code

Promo Code [Submit](#)

MyProm -\$50.00

Shipping

Ground (7-10 Business Days)

Shipping cost \$5.00

Sales Tax \$2.20

Order Discount - \$50.00

Estimated Total \$46.20

[PayPal](#) Checkout

[Pay Later](#)

Pay in full or 4 interest-free payments of \$11.55. [Learn more](#)

[Pay over 24 months](#)
Min purchase required. [Learn more](#)

Checkout

Figure 5.2.52 Button Message on Cart page

Payment

Billing Address

Billing details will be retrieved from the chosen payment method

PayPal venmo

Pay in full or 4 interest-free payments of \$11.00. [Learn more](#)

PayPal Checkout

Pay Later

Figure 5.2.53 Button Message on Billing page

5.2.17 Google Pay functionality

Choose a payment method and set Enabled to Yes

To enable the Google Pay, setup payment method as described in [alternative payment methods section](#), after that navigate to Business Manager > Merchant Tools > PayPal Plugin > Preference Configuration > select the **Google Pay** tab and in **Google Pay Button Location** choose location for the Google pay Button (Figure 5.2.54).

General PayPal Credit Card Apple Pay Google Pay Pay Later Banners/Messaging

PayPal Google Pay

Instance Type: Sandbox

Search by ID

Cancel Apply to Other Sites Save

Name	Value	Default Value	Property Description
Google Pay Button Location on Express Checkout	☒ Cart ☒ MiniCart ☒ PDP ☒ PVP		Activates Google Pay buttons on Cart, Mini-Cart, PDP, PVP pages.
(PP_Google_Pay_Button_Location) (Enum of Strings)			

Figure 5.2.54 Google Pay Button visibility preference.

Payment

Billing Address

Billing details will be retrieved from the chosen payment method

PayPal venmo Google Pay

Google Pay | 0652

Figure 5.2.55 Google Pay Button on Billing page

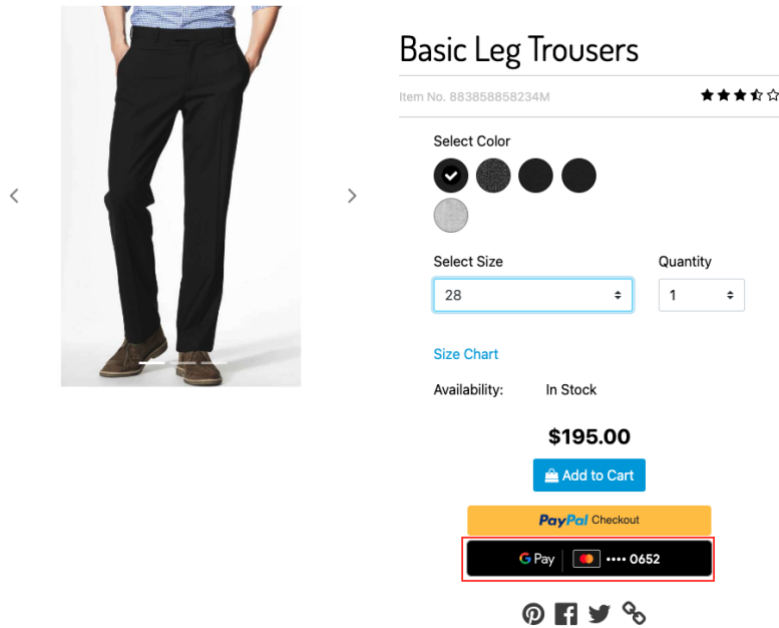


Figure 5.2.56 Google Pay Button on PDP page

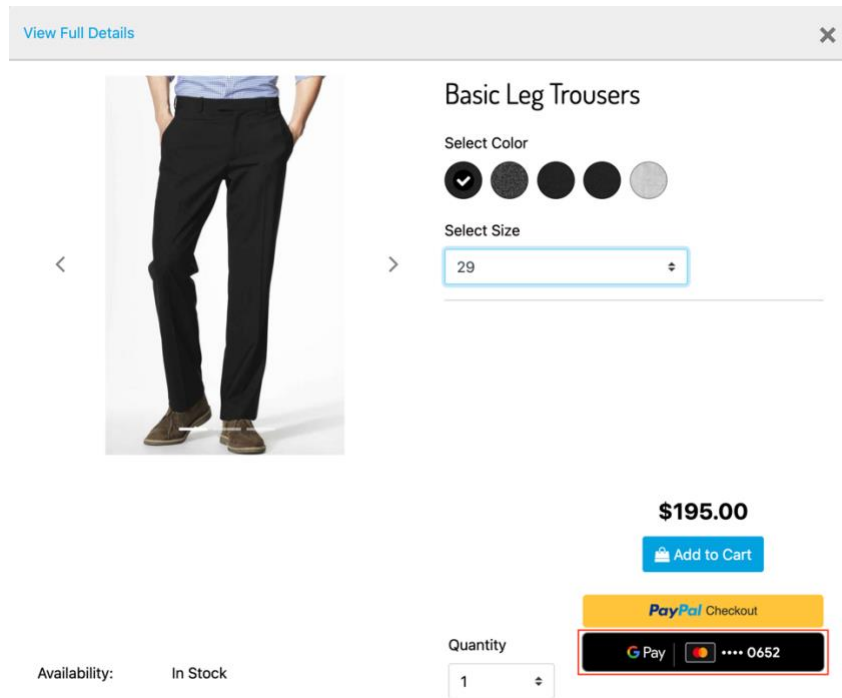


Figure 5.2.57 Google Pay Button on PVP page

Your Cart

[Continue Shopping](#) 1 Items [Need Help? Call 1-800-555-0199](#)

Basic Leg Trousers			
	Color: Black Size: 29 In Stock Edit	Each \$195.00	Quantity 1 Total \$195.00

Enter Promo Code

Shipping
Ground (7-10 Business Days)

Shipping cost \$7.99
Sales Tax \$10.15
Estimated Total \$213.14

[PayPal Checkout](#)

[G Pay](#)  **** 0652

[Checkout](#)

Figure 5.2.58 Google Pay button on Cart page

Your shopping cart [View Cart](#)

Basic Leg Trousers		
	Color: Black Size: 28 In Stock	
Each \$195.00	Quantity 1	Total \$195.00

Estimated Total \$195.00

[PayPal Checkout](#)

[G Pay](#)  **** 0652

[Checkout](#)

Google Pay button on Mini-Cart page

 pay.google.com

 **Test User**
testAccount@gmail.com

 **Visa** **** 1111

 **US User**
1600 Amphitheatre Parkway1
Mountain View, CA 94043
United States
+1 650-555-5555

 **\$7.99: Ground**
Order received within 7-10 business days

Test environment. Card and address fields are pre-filled with fake test data.
Your billing details will also be shared with the merchant

☐ Get Google Pay emails with exclusive offers, tips, and invitations to give feedback

Pay SFRA Test Shop
\$171.14

Figure 5.2.59 Google Pay pop-up window on express checkout page

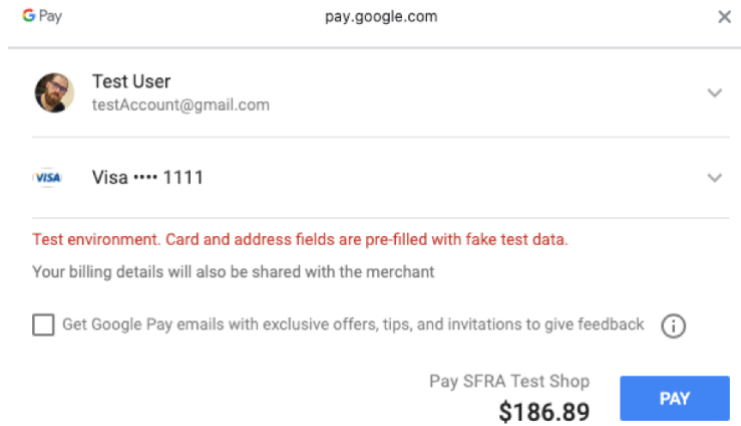


Figure 5.2.60 Google Pay pop-up window on billing page

6 Locales and addresses

6.1 General info

Out of the box PayPal is using built-in SFRA locations for checkout from billing or cart page.

Checkout is available only with one locale, multi shipping and multi locales are not supported.

7 Multi-site and multi-credential support

7.1 General info

PayPal cartridges do not come with multi-site or multi-credential support out of the box. On the service level built-in logic will use only `int_paypal.http.rest` service and credentials attached to the service to communicate PayPal REST API.

8 Gift Certificate Support

8.1 General info

PayPal cartridges do not provide the ability to add a gift certificate to a basket or redeem gift certificates during checkout.

The Customer can pay for a gift certificate using PayPal. If a gift certificate is the only item in the basket, the shipping address will not be requested from a customer and the billing address will be set as the shipping address.

If the customer adds a gift certificate as a payment option and it partially covers the order total, the customer can pay the rest using PayPal. In this case the gift certificate amount will be treated as a discount and subtracted from the order total.

8.2 Technical info

Integration supports only gift certificates from B2C Commerce API. Gift certificate as product should be present as `GiftCertificateLineItem` in a basket. Gift Certificate as payment method should be applied as `OrderPaymentInstrument` using `createGiftCertificateLineItem` function.

For more information, contact your PayPal support manager.

9 Version and SFRA Support Info

9.1 JavaScript Files and Cartridge Version Location

The JavaScript files to be added to your page are located at:

```
cartridges/int_paypal/cartridge/templates/default/home/homePage.is  
ml
```

```
cartridges/int_paypal/cartridge/templates/default/account/dashboar  
dProfileCards.isml
```

```
cartridges/int_paypal/cartridge/templates/default/account/login.is  
ml
```

```
cartridges/int_paypal/cartridge/templates/default/checkout/billing  
/paymentOptions/paymentOptionsContent.isml
```

```
cartridges/int_paypal/cartridge/templates/default/checkout/checkou  
t.isml
```

```
cartridges/int_paypal/cartridge/templates/default/search/searchRes  
ults.isml
```

```
cartridges/int_paypal/cartridge/templates/default/paypal/cart/payp  
alCartContent.isml
```

```
cartridges/int_paypal/cartridge/templates/default/paypal/cart/appl  
ePayCartContent.isml
```

```
cartridges/int_paypal/cartridge/templates/default/paypal/pdp/apple  
PayPdpContent.isml
```

The file with the current version of the PayPal cartridge is located at:

```
cartridges/int_paypal/cartridge/templates/resources/paypalVersion.  
properties
```

The file with the current version of the SFRA cartridge is located at:

```
cartridges/app_storefront_base/cartridge/templates/resources/versi  
on.properties
```

9.2 SFRA Version Support

PayPal cartridge version 25.3.0 supports SFRA 7.0.0.

9.3 Compatibility Mode

Cartridge tested with Compatibility Mode 21.2.

10 Service Errors Handling

10.1 Service Timeout Handle

If a PayPal service becomes unavailable, the error details can be found in the following logs:

- Error details can be found in the log with the custom-PayPal prefix.
- Service request/response details can be found in the log with the service-PayPalRest prefix.

On the storefront, the customer will see a PayPal button, but the pop-up will close instantly.

For more details about logs, see [Logs](#) section.

For more information about how to track service downtime and about possible options for configuration in situations when PayPal API is unavailable or throws an error, review [HTTP Service Availability](#) section.

10.2 Fallback/Recovery

If you continue to see issues where the PayPal button and/or experience is not rendering, or if you are seeing 500 or 400 errors in the logs, please contact your PayPal representative or create a request to the Merchant Technical Support so that we may investigate this further. These orders will not be processable with the PayPal payment option, and you may need to temporarily disable PayPal until the issue is resolved.

Please be aware that when contacting PayPal, we will ask for the details that may need to be extracted from the logs as well as for assistance with reproducing the issue step by step.

10.3 Handle Error Response

Error handling in the integration is based on the SFRA built-in error logic. If it is possible detailed error will be shown to a customer (Figure 10.3.1). In cases where customization is not possible, the customer will see SFRA system error (Figure 10.3.2). In this case, a detailed error from PayPal can be found at custom-PayPal log. Review [Logs](#) section for more details.

All errors text located at

`int_paypal/cartridge/templates/resources/paypalerrors.properties.`

You can add your own text for errors from REST API, using format: paypal.error + error code from paypal or error name in lower case. For example:

```
paypal.error.transaction_refused=Transaction was refused by merchant
```

```
paypal.error.unprocessable_entity=Please check billing address
```

UPDATE: Custom errors text will not appear on the order review page due to built-in SFRA logic. A System message will always appear (Figure 10.3.2).

The screenshot displays a PayPal checkout interface with a custom error message. The error message, "Order total 0 is not allowed", is shown in a red box at the top center. The page is divided into two main sections: "Shipping" and "Payment".

Shipping Section:

- Shipping Address:** NewUser22 BookUser22, Neapol 250, Street 212, Djuno AK 12345, 9234567890.
- Shipping Method:** Express (2-3 Business Days) **\$0.00**

Payment Section:

- Billing Address:** A dropdown menu shows "Billing details will be retrieved from the chosen payment me".
- Payment Method:** The "PayPal" option is selected, indicated by a blue underline.
- Checkout Button:** A button labeled "PayPal Checkout" is visible.
- Save Account:** A checkbox labeled "Save PayPal account" is checked.

Order Summary:

- 1 Items \$0.00**
- Vizio VO22LF 22" LCD High Definition T...**
- In Stock**
- Each: \$349.99, Quantity: 1, Total: \$349.99**
- \$0.00**

Figure 10.3.1. Custom error message example

We're sorry that your order could not be placed. This probably happened due to a high order volume or temporary connection errors. Please wait a few minutes and resubmit your order. We won't process your payment until you successfully place your order. If you have further questions, please contact us.

Shipping
Edit

Shipping Address:
Rick Flores
2253 Hudson Street
Denver CO 80207
9739747269

Shipping Method:
Ground (7-10 Business Days) \$0.00

Payment
Edit

Billing Address:
Dmitry Vigranovskiy
1 Main St
San Jose CA 95131
3cd3ldtkdii@ffymedia.com
9739747269

Payment:
dmitry.vigranovskiy-facilitator@epam.om
PayPal / \$944.99

Place Order

Figure 10.3.2. SFRA standard error message example

11 Automated Testing

The cartridge has unit tests included in the repository.

Please use the instructions described in the `test/unit/README.md`, `test/integration/README.md`, and `test/acceptance/README.md` files in the repository to configure and run tests.

12 Recommendations

1. Since it is possible to create a PayPal account via a verified mobile phone but with an unverified email address, it is best to unselect the **'Enable customers who have not yet confirmed their email with PayPal to log in to your app'** checkbox to reduce the possibility of security risks.

To do so, you as a merchant should access your PayPal Sandbox account at <https://developer.paypal.com/home/>, then go to **App & Credentials** and choose your app. Under **Sandbox App Settings** scroll down to **Log in with PayPal &** click **Advanced Settings**. In the opened form uncheck 'Enable customers who have not yet confirmed their email with PayPal to log in to your app' checkbox.

2. Please activate PayPal payment method in Business Manager (Merchant Tools > Ordering > Payment Methods > PayPal) so that your other activated payment methods (e.g. Google Pay, Apple Pay, etc.) can be available for customers on the Checkout.

PayPal Developer Dashboard

Home Apps & Credentials Testing Tools Event Logs

You're in sandbox mode.

My apps & credentials

REST API apps

Get started by clicking [Create App](#). [PayPal Commerce Platform for Business](#) users can get started quickly by using the **Default Application** credentials to test PayPal REST APIs in our sandb:

App name	Actions
COSTCO	
2 Default Application	System generated, no actions available
FranceApp	

Figure 12.1. Unselecting 'Enable customers who have not yet confirmed their email with PayPal to log in to your app' checkbox Step 1 Step 2

PayPal Developer Dashboard

Home Apps & Credentials Testing Tools Event Logs

You're in sandbox mode.

3 Features

Accept payments

- ☒ Advanced Credit and Debit Card Payments
PayPal payment buttons plus customized card fields.
- ☒ Apple Pay
Enable customers to pay with Apple Pay in iOS apps and Safari.
[Manage](#)
- ☐ IC++ with Gross Settlement
Set up IC++ pricing for automatic payment of processing fees.
- ☒ Vault
Save your customer's payment methods to speed up checkout.
- ☒ PayPal and Venmo
Save customer PayPal and Venmo payment methods for future transactions.
- ☒ Native Checkout SDK
Allow customers to check out without leaving your Mobile app. Available through the [Native Checkout SDK](#)
- ☒ Subscriptions
Set up recurring payments for customers.
- ☒ Invoicing
Send and manage customer invoices.
- ☒ PayPal Here
Process card transactions in person.

[Save Changes](#) [Reset](#)

Other features

- ☒ Payouts
Send payments to multiple PayPal accounts at once.
- ☒ Customer disputes
View, provide evidence, and appeal disputes.
- ☒ Log in with PayPal
Let customers log in with their PayPal logins.
- 4 ☒ Advanced Settings
- ☒ Transaction search
View your PayPal transaction history.

Figure 12.2. Unselecting 'Enable customers who have not yet confirmed their email with PayPal to log in to your app' checkbox Step 3 & Step 4

Personal Information

- ☒ Full Name
- ☒ Email

Address

- ☒ Street Address
- ☒ City
- ☒ State
- ☒ Country
- ☒ Postal Code

Account information

- ☒ Account verification status
- ☒ PayPal account ID (payer ID)

Links shown on customer consent page

Privacy policy URL
<https://example.com>

User agreement URL
<https://example.com>

Additional PayPal permissions

- ☒ Enable customers who have not yet confirmed their email with PayPal to log in to your app.

Save

[Cancel](#)

Figure 12.3. Unselecting 'Enable customers who have not yet confirmed their email with PayPal to log in to your app' checkbox Step 5

3. We recommend using a phone number format in the Shipping Address section according to the [E.164 numbering plan format](#).

13 Known Issues\Limitations

1. Venmo button is available on the Payment step of the Checkout only in plugin version 25.1.0 PP SFRA.
2. App Switch works best when native browser is a default browser (e.g Safari for iOS devices).

14 Upgrade Guide

Upgrading to 23.3.0 or newer from 23.2.0 or earlier

14.1 Manual update

List of custom attributes (should be removed)

Old attribute ID	New attribute ID
PP_API_PaymentAction	PP_Payment_Model
PP_API_APM_methods	PP_Alternative_Payment_Methods
PP_API_Button_Location	PP_Button_Location
PP_API_BA_Enabled	PP_BA_Enabled
PP_API_BA_Description	PP_BA_Description
PP_Automatic_PM_adding_enabled	PP_Automatic_Payment_Method_Adding_Enabled
PP_API_PAY_NOW_ENABLED	PP_Digital_Goods_Flow_Enabled
PP_FRAUDNET_ENABLED	PP_FraudNet_Enabled

List of attribute groups (should be removed)

Old group ID	New group ID
Paypal_Credit_Messaging	PAYPAL_PAY_LATER_BANNERS_MESSAGING
Paypal_Checkout	PAYPAL_CHECKOUT
Missing	PAYPAL_GENERAL
Missing	PAYPAL_CREDIT_CARD
Missing	removed

There are two ways to remove old attributes.

First approach:

- Remove old attributes:
 - Go to Business Manager > Administration > Site Development > System Object Types > Site Preferences - Attribute Definitions
 - Select a checkbox for the required IDs.

- Click on Delete button and you will see a message (Are you sure that you want to execute the following action for these items: delete?)
- Click on *Delete* button.
- **Remove old groups:**
 - Go to Business Manager > Administration > Site Development > System Object Types > Site Preferences - Attribute Groups
 - Select a checkbox for the required IDs.
 - Click on Delete button and you will see a message (Are you sure that you want to execute the following action for these items: delete?)
 - Click on *Delete* button.

Second approach:

PLEASE NOTE: When using this approach, you may affect the metadata from the other cartridges. We recommend using the first way to remove old attributes.

- Remove old attributes and groups:
 - Go to Business Manager > Administration > Site Development > Import & Export

[Administration](#) > [Site Development](#) > Import & Export

Import & Export

Meta Data

[Import](#) and [export](#) your system meta data (i.e., system type extensions, custom object types, custom preference definitions).

[Import](#) [Export](#)

Geolocations

[Import](#) geolocations for a country.

[Import](#)

Import & Export Files

[Upload](#) and [download](#) your import and export files.

[Upload](#) [Download](#)

Status

There are currently no import or export processes to show here.

[Refresh](#)

Figure 14.1.1. Import & Export

- Meta data > Click on *Export* button.
- Select the checkbox *System Object Type Extensions*
- Specify the file name for export (for example: bt-system-object-type-extensions.xml)

[Administration](#) > [Site Development](#) > [Import & Export](#) > Step 1 - Select Export File

System Type Extension Export - Step 1: Select Export File

Select the meta data objects that you want to export and provide a name for the export file. Fields with a red asterisk (*) are mandatory.

Export File:*

☒ System Object Type Extensions

☐ Custom Object Type Definitions

[Export](#) [Cancel](#)

Figure 14.1.2. Export System Object Type Extensions

- Click on *Export* button.

- Wait until the export status is *Success*.

[Administration](#) > [Site Development](#) > Import & Export

Import & Export

Meta Data

[Import](#) and [export](#) your system meta data (i.e., system type extensions, custom object types, custom preference definitions).

[Import](#)
[Export](#)

Geolocations

[Import](#) geolocations for a country.

[Import](#)

Import & Export Files

[Upload](#) and [download](#) your import and export files.

[Upload](#)
[Download](#)

Status

Select All	Process	Start	Duration	Status
☐	Meta Data Export <bt-system-object-type-extensions.xml>	5/16/23 8:19:04 am	00:00:00	Running
☐	Meta Data Export <bt-system-object-type-extensions.xml>	5/16/23 8:18:52 am	00:00:03	Success

[Refresh](#)
[Delete](#)

Figure 14.1.3. Successful export of meta data

- Download the exported file and delete the old attributes and groups. Be attentive.
- Upload the modified file for import. Click on *Upload* button.

[Administration](#) > [Site Development](#) > Import & Export

Import & Export

Meta Data

[Import](#) and [export](#) your system meta data (i.e., system type extensions, custom object types, custom preference definitions).

[Import](#)
[Export](#)

Geolocations

[Import](#) geolocations for a country.

[Import](#)

Import & Export Files

[Upload](#) and [download](#) your import and export files.

[Upload](#)
[Download](#)

Status

Select All	Process	Start	Duration	Status
☐	Meta Data Export <bt-system-object-type-extensions.xml>	5/16/23 8:19:04 am	00:00:00	Running
☐	Meta Data Export <bt-system-object-type-extensions.xml>	5/16/23 8:18:52 am	00:00:03	Success

[Refresh](#)
[Delete](#)

- Click on *Choose file* button and select modified file. Click on *Upload* button.

Upload Import Files

Upload File:

Choose file

bt-system-o...xtensions.xml

Upload

Left-click or drag and drop multiple files to upload them.

Manage Import Files [Go To WebDAV Folder](#)

The table below shows the import files currently available on the server.
To upload new files, click **Browse** and then **Upload** in the file upload section above.
To delete uploaded files, select them using the checkboxes and then click **Delete**.

There are currently no uploaded import files available on the server.

<< Back

Figure 14.1.4. Upload import file

Upload Import Files

Upload File:

Choose file

No file chosen

Upload

Left-click or drag and drop multiple files to upload them.

Manage Import Files [Go To WebDAV Folder](#)

The table below shows the import files currently available on the server.
To upload new files, click **Browse** and then **Upload** in the file upload section above.
To delete uploaded files, select them using the checkboxes and then click **Delete**.

Select All	Name	File Size	Last Modified
☐	bt-system-object-type-extensions.xml	432.19 KB	5/16/23 8:40:51 am
UncompressCompressDelete			

- Go to Business Manager > Administration > Site Development > Import & Export
- Meta data > Click on Export button.
- Mark file that you want to import and click *Next* button.

System Type Extension Import - Select File

Step 1 of 3. Next Step: Validate Import File
The list below shows all uploaded import files. Please select the file that you want to import.

Select	Name	File Size	Last Modified
☒	bt-system-object-type-extensions.xml	432.19 KB	5/16/23 8:40:51 am

Next >>Cancel

- Mark checkbox in Import Options section and click *Import* button.

System Type Extension Import - Validate File

Step 3 of 3.

Enable the import option if you want to remove existing attribute definitions and attribute groups that aren't contained in the import file. Start the import by clicking **Import**.

Validation Status

Current Action	Finished XML schema validation.
Validation Errors	0
Validation Warnings	0

Import File Content

System Type Extensions	63
Custom Type Definitions	0
Attribute Definitions	997
Attribute Value Definitions	508
Attribute Group Definitions	71
Custom Preference Groups	0
Custom Preference Definitions	0

Import Options

☒ Delete existing attribute definitions and attribute groups not contained in the import file, including attribute definitions marked as "externally-defined".

<< Previous

ImportCancel

- Confirm the action, click *OK* button.
- Wait until the Import status is *Success*.

Status

Select All	Process	Start	Duration	Status
☐	Meta Data Import <bt-system-object-type-extensions.xml>	5/16/23 8:52:13 am	00:00:01	Success

14.2 Migration from Billing Agreement to Vault flow.

Starting from version 24.2.0 Billing Agreement is deprecated and new Vault flow is introduced. In order to migrate, see [PayPal Vault Flow](#)..

15 Fastlane by PayPal

15.1 Overview

Fastlane by PayPal is a new one-click guest checkout experience that merchants utilizing PayPal's platform can offer to their shoppers. This allows customers to make quick and hassle-free purchases. With Fastlane, customers can save their information for a checkout process that can be as quick as a single tap. There is no need to remember a username or password, no need to update personal information, and no need to share a credit card with multiple businesses over the internet.

NOTE: The following PayPal plugin site preferences are NOT applied to Fastlane Checkout:

- Store Credit Card in Vault and Store Credit Card in Vault via My Account
- Verify Credit Card on My Account
- Credit Card Expire Notification Configuration
- Limit of saved credit cards.
- Digital Goods flow

NOTE: The Digital Goods flow will not work for guest users if Fastlane is enabled in the Business Manager.

NOTE: The Fastlane flow will not work for guest users if PayPal or PayPal Credit Card payment methods are disabled in the Business Manager.

15.2 Configuration

In Business Manager navigate to Merchant Tools -> PayPal Plugin -> Preference Configuration -> Fastlane by PayPal tab and set Enable Fastlane preferences to Yes. (Figure 15.2.1).

GeneralPayPalCredit CardApple PayGoogle PayVenmoFastlane by PayPalPay Later Banners/Messaging

Fastlane By PayPal

Instance Type

Sandbox

Search by ID

Q

CancelApply to Other SitesSave

Name	Value	Default Value	Property Description
Fastlane Checkout Enabled	Yes	No	Accelerate checkout for millions of shoppers who don't have an account with your store. Shoppers recognized by Fastlane can pay in seconds. Unrecognized shoppers enjoy sign-up for faster future checkouts.
(PP_Fastlane_Checkout_Enabled) (Boolean)			

Figure 15.2.1 Enable Fastlane.

To set watermark:

Navigate to Merchant Tools -> PayPal Plugin -> Preference Configuration -> Fastlane by PayPal tab and set Enable Fastlane Privacy preferences to Yes. (Figure 15.2.2).

Recommendation when the Fastlane Privacy is disabled:

PayPal powers this accelerated checkout solution from Fastlane. Since you will share consumers' email address with PayPal, please consult your legal advisors on the appropriate privacy setting for your business.

Fastlane Privacy	Yes	Yes	This setting will control whether Fastlane by PayPal branding is displayed in the email field. Terms & Conditions: PayPal powers this accelerated checkout solution from Fastlane. Since you'll share consumers' email address with PayPal, please consult your legal advisors on the appropriate privacy setting for your business.
(PP_Fastlane_Privacy_Enabled) (Boolean)			

Figure 15.2.2 Enable Fastlane Privacy.

Customer

Email

Fastlane by PayPal ⓘ

Continue as guest

Figure 15.2.3 Fastlane Watermark.

To set cardholder name navigate to Merchant Tools -> PayPal Plugin -> Preference Configuration -> Fastlane by PayPal tab and set Cardholder Name to Yes. (Figure 15.2.4 and Figure 15.2.5)

Please note that Fastlane by PayPal also supports a single name in the cardholder's name field. This value is used to populate the billing address form for Payment UI component integration after user enters card details. If the user provides only a single name, it will be assigned to the "first name" field, leaving the "last name" field empty. This may cause an error in the billing form if the "last name" field is required.

Cardholder Name	Yes	Yes	This setting will control whether or not the cardholder name is displayed in the card field's UI. Note: Is not applicable for Payment UI component.
(PP_Fastlane_Cardholder_Name) (Boolean)			

Figure 15.2.4 Enable Cardholder Name.

The screenshot shows a 'Payment' form with the following fields: 'Billing Address' (containing 'Gary Geuest 123 Main St New York City, NY 10001'), 'Phone Number' (containing '12125550123'), and a payment method section with 'CREDIT', 'PayPal', and 'G Pay' options. Below these are fields for 'Card number', 'Expiration (MM/YY)', 'CVV', and 'Name on card' (containing 'Gary Geuest'). The 'Name on card' field is highlighted with a red rectangular border.

Figure 15.2.5 Cardholder name field.

NOTE: Cardholder name is not applicable for Payment UI component. The cardholder name can be disabled only in the native UI (the Fastlane Payment UI site preference should be set to 'No').

15.3 Fastlane by PayPal checkout for PayPal payment UI configuration.

In Business Manager navigate to Merchant Tools -> PayPal Plugin -> Preference Configuration -> Fastlane by PayPal tab and set Payment UI preferences to No. (Figure 15.3.1)

If "Yes" is selected: Fastlane integration uses PayPal Payment UI components during the Fastlane Checkout.

The screenshot shows a configuration setting titled 'Use Fastlane Payment UI' with a sub-label '(PP_Fastlane_Payment_UI) (Boolean)'. To the right is a dropdown menu currently set to 'Yes', which is highlighted with a red rectangular border. Further right is a 'No' option. To the right of the dropdown is a descriptive text: 'This setting will control whether the PayPal Payment UI is used UI of the partner.'

Figure 15.3.1 Enable Fastlane Payment UI.

15.4 Checkout for Fastlane guest

1. Proceed to checkout flow for a storefront guest user.
2. Enter a shipping address and choose a shipping method.
NOTE: In case you use the Salesforce out-of-the-box multi-shipping option, only one shipping address related to the first item in the cart will be used for Fastlane account creation.
3. Select the Credit Card payment method.
4. The Fastlane hosted fields will be shown. (Figure 15.4.1)
5. Enter Credit Card data and proceed to place the order. (Figure 15.4.2)

Payment

CREDIT CARD PayPal G Pay

Card number

Expiration (MM/YY) CVV

Name on card
garyGeuest

☒ Billing address same as shipping

Save your info with Fastlane for faster checkouts

Mobile number
+1 (212) 555-0123

By saving your info, you agree to get codes by text to use Fastlane everywhere it's available.
You also agree to the [terms](#) and [privacy statement](#).

Fastlane by PayPal

Figure 15.4.1 Guest Fastlane fields for Payment UI.

Payment

CREDIT CARD PayPal G Pay

Card number
4111 1111 1111 1111

Expiration (MM/YY) CVV
12 / 29 256

Name on card
garyGeuest

☒ Billing address same as shipping

Save your info with Fastlane for faster checkouts

Mobile number
+1 (212) 555-0123

By saving your info, you agree to get codes by text to use Fastlane everywhere it's available.
You also agree to the [terms](#) and [privacy statement](#).

Fastlane by PayPal

Next: Place Order

Figure 15.4.2 Place order for the guest user through the Payment UI.

6. To create a Fastlane account, activate the toggle that indicates your consent to the creation of a Fastlane member account.

Payment

CREDIT

Card number
4111 1111 1111 1111

Expiration (MM/YY)
12 / 29

CVV
256

Name on card
garyGeuest

☒ Billing address same as shipping

Save your info with Fastlane for faster checkouts ⓘ

Mobile number
+1 (212) 555-0123

By saving your info, you agree to get codes by text to use Fastlane everywhere it's available.
You also agree to the [terms](#) and [privacy statement](#).

Fastlane by PayPal

Next: Place Order

Figure 15.4.3 Consent of creation Fastlane member account.

- To complete the payment process without creating a Fastlane member account, it is necessary to deactivate the toggle that indicates your consent to the creation of a Fastlane member account (Figure 17.3.5).

Payment

CREDIT

Card number
4111 1111 1111 1111

Expiration (MM/YY)
12 / 29

CVV
256

Name on card
garyGeuest

☒ Billing address same as shipping

Save your info with Fastlane for faster checkouts ⓘ

Figure 15.4.5 Save checkbox.

Session payment is applicable for a Fastlane guest user (for Fastlane Component Integration only, it means that "Yes" value should be applied to the "Use Fastlane Payment U" site preference).

- Follow checkout flow for Checkout for Fastlane guest.
- On the Order Review page move back to the previous step.
- Session payment will be shown.

4. A Fastlane guest user can select Session payment (Figure 15.4.6) or add a new Credit Card (Figure 15.4.7).

NOTE:

- Session payment also can be shown for Fastlane member in case there is no saved funding instrument in the Fastlane account, and Fastlane member returns from the Order Review to the Payment step with the previously entered card details.

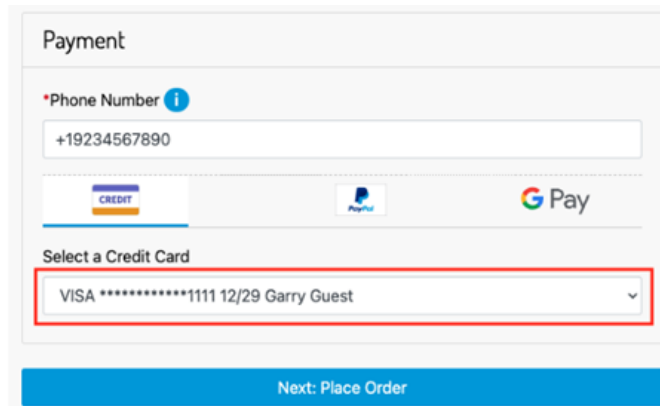


Figure 15.4.6 Session payment for Fastlane guest users.

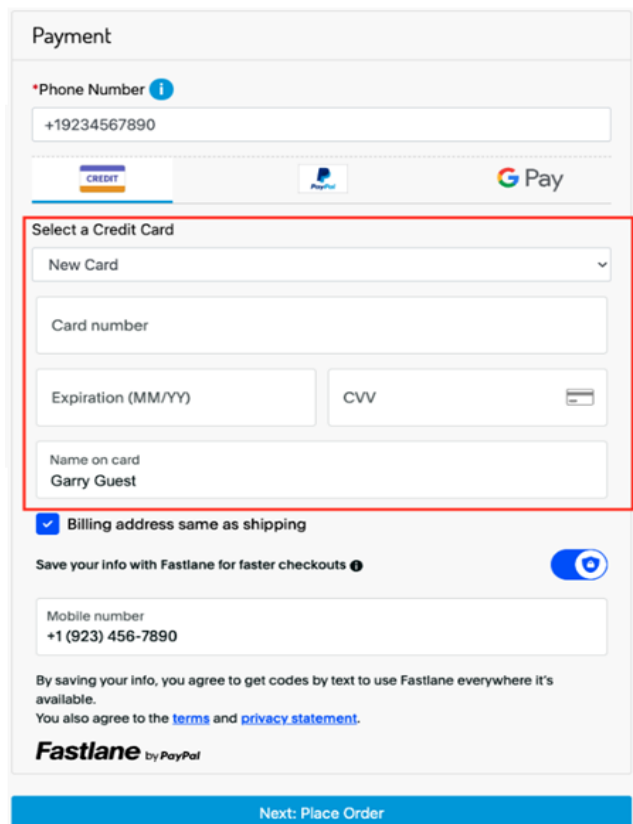


Figure 15.4.7 Adding a new Credit Card for Session payment.

15.5 Checkout for Fastlane member

1. Proceed to checkout flow for a storefront guest user.

2. Enter the email which is connected to the Fastlane network.
3. The OTP(One Time Password) popup window will be shown. (Figure 15.5.1)

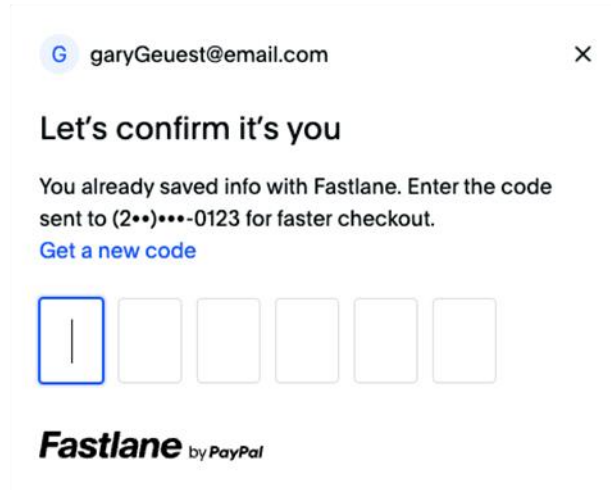


Figure 15.5.1 OTP popup window.

4. A Fastlane member can proceed to place an order with shipping and payment data, which is retrieved from their Fastlane account. (Figure 15.5.2)

A screenshot of a checkout form for Fastlane members. The form is divided into three main sections: 'Customer', 'Shipping', and 'Payment'. The 'Customer' section shows 'Customer Information: test@tst.com' with an 'Edit' link. The 'Shipping' section shows 'Shipping Address: Tania Lieberman, 2348 Rardin Drive, San Bruno CA 94066, 14026607986, Fastlane by PayPal' and 'Shipping Method: Ground (7-10 Business Days) \$5.99' with an 'Edit' link. The 'Payment' section shows 'Billing Address: Fred Flintstone, 2211 North 1st St, San Jose CA 95131, test@tst.com, 14026607986', 'Payment: Credit Card *****1111', 'Visa', 'Fastlane by PayPal', and '\$111.28' with an 'Edit' link. At the bottom, there is a blue 'Place Order' button.

Figure 15.5.2 Placing order for Fastlane members.

5. To change shipping address press the Edit button Order Review page, and then select Change. (Figure 15.5.3 and Figure 15.5.4)
The popup window for changing or adding a new shipping address will be shown. (Figure 15.5.4 and Figure 15.5.5)

Customer

Customer Information:

test@tst.com

Shipping

Shipping Address:

Tania Lieberman
2348 Rardin Drive
2348 Rardin Drive
San Bruno CA 94066
14026607986

Fastlane by PayPal

Shipping Method:

Ground (7-10 Business Days) \$5.99

Payment

Billing Address:

Fred Flintstone
2211 North 1st St
San Jose CA 95131
test@tst.com
14026607986

Payment:

Credit Card
*****1111

Visa

Fastlane by PayPal

\$69.28

Place Order

Figure 15.5.3 Changing shipping address, step 1.

Customer

Customer Information:

test@tst.com

Shipping

Shipping Address:

Tania Lieberman
2348 Rardin Drive
2348 Rardin Drive
San Bruno CA 94066
14026607986

Fastlane by PayPal

Shipping Method

☒ Ground (7-10 Business Days) \$5.99
 ☐ 2-Day Express (2 Business Days) \$9.99
 ☐ Free USD \$0.00
 ☐ Overnight (Next Day) \$15.99
 ☐ Super Saver (3-7 Business Days) \$1.99

☐ This is a Gift

Next: Payment

Figure 15.5.4 Changing shipping address, step 2.

Choose an address

San Bruno, CA 94066, US
+1 402-660-7986

Oleksandra Davydochuk
7700 Eastport Pkwy
La Vista, NE 68128, US
+1 402-660-7986

Ivan Vinogradov
2253 Hudson Street
Denver, CO 80207, US
+1 402-660-7986

Tania Lieberman ✓
2348 Rardin Drive
San Bruno, CA 94066, US
+1 402-660-7986

Fred Flintstone
10143 PayPal Road
Omaha, NE 68136, US
+1 402-660-7986

+ Add an address

Fastlane by PayPal

Figure 15.5.5 Popup window for changing or adding a new shipping address.

←

Add a shipping address

×

Country
United States

First name
Fred

Last name
Flintstone

Company name (optional)

Street address

Apt., ste., bldg. (optional)

City

State

ZIP code

Code
+1

Phone number
(402) 660-7986

Add Address

Fastlane by PayPal

Figure 15.5.6 Popup window for adding new shipping address.

6. A Fastlane member has the option to change or add a new Credit Card and Billing Address (Figure 15.5.6, Figure 15.5.7, Figure 15.5.8, Figure 15.5.9).

Payment

*Phone Number ⓘ

4084607119

PayPal Google Pay CREDIT

Mastercard •••• 1005

Fastlane by PayPal

Change

Next: Place Order

Figure 15.5.6 Change or add a new Credit Card option step 1.

Choose a card X

VISA •••• 1111 ✓

Mastercard •••• 4444

VISA •••• 0004

+ Add a card

Fastlane by PayPal

Figure 15.5.7 Credit Card list popup window.

← Add a card X

Card number

Expiration (MM/YY) CVV

Name on card
Daniel Member

Billing address ▼
123 Main St
New York City, NY 10001, US

Add Card

Fastlane by PayPal

Figure 15.5.8 Adding new Credit Card popup window.

← Add a billing address ×

Country
United States

Street address

Apt., ste., bldg. (optional)

City

State ZIP code

Add Address

Fastlane by PayPal

Figure 15.5.9 Adding new Billing Address popup window.

15.6 Checkout for Fastlane member via PayPal account

1. Proceed to checkout flow for a storefront guest user.
2. Confirm to use Fastlane Checkout in the popup window (Figure 15.6.1)

l.vin***@hotma***.com ×

Autofill your checkouts

Use your card and address from PayPal everywhere Fastlane is available.

By requesting a code by text, you agree to create a Fastlane profile and to the Fastlane [terms](#) and [privacy statement](#).

Get Code to use Fastlane

Fastlane by PayPal

Figure 15.6.1 Confirmation popup window for PayPal users.

3. After confirmation, user continues checkout as a Fastlane member (as described in [Checkout for Fastlane member](#))

15.7 Fastlane by PayPal checkout for native UI configuration.

In Business Manager navigate to Merchant Tools -> PayPal Plugin -> Preference Configuration -> Fastlane by PayPal tab and set Payment UI preferences to Yes. (Figure 15.7.1)

If "No" (default) value is selected: Fastlane integration uses native UI components during the Fastlane Checkout.

Use Fastlane Payment UI	
(BRAintree_Fastlane_Paym... (Boolean)	No ▼ No

This setting will control whether the PayPal Payment UI is used or the native UI of the partner.

Figure 15.7.1 Enable Fastlane Payment UI.

Checkout for Fastlane guest and Fastlane member with native UI is the same as for payment UI integration. The main difference is that the Payment UI utilizes Fastlane styles, which cannot be altered.

15.8 Checkout for Fastlane guest

1. Proceed to checkout flow for a storefront guest user.
2. Enter a shipping address and choose a shipping method.
NOTE: In case you use the Salesforce out-of-the-box multi-shipping option, only one shipping address related to the first item in the cart will be used for Fastlane account creation.
3. Select the Credit Card payment method.
4. Enter Credit Card data and proceed to place the order. (Figure 15.8.1)

The screenshot shows a payment form titled 'Payment'. It contains the following sections and fields:

- Billing Address:** A dropdown menu showing 'Daniel Guest 123 Main St New York City NY 10001' with 'Update Address' and 'Add New' links.
- *Phone Number:** A text input field containing '12125550123'.
- Payment Methods:** Three icons for 'Credit Card', 'PayPal', and 'Google Pay'.
- Card Information:** Fields for 'Card number', 'Expiration (MM/YY)', and 'CVV'.
- Name on card:** A text input field containing 'Daniel Guest'.
- Save your info with Fastlane for faster checkouts:** A toggle switch that is currently turned on.
- Mobile number:** A text input field containing '+1 (212) 555-0123'.
- Footer:** A note stating 'By saving your info, you agree to get codes by text to use Fastlane everywhere it's available. You also agree to the [terms](#) and [privacy statement](#).' followed by the 'Fastlane by PayPal' logo.
- Next: Place Order** button at the bottom.

Figure 15.8.1 Place order for the guest user through the native UI.

15.9 Checkout for Fastlane member

PayPal checkout for Fastlane member with Native UI configuration is the same as for Fastlane member with PayPal payment UI. (Described in section 15.4.).

15.10 Styles for Fastlane by PayPal Checkout

The Fastlane checkout styles file is located at the following path:
cartridges/int_paypal/cartridge/config/fastlaneStyleOptions.js

NOTE: The style configuration may vary depending on the Fastlane SDK version.

15.11 Fastlane by PayPal Checkout Recommendations

1. Please note that the buyer can enter any address when using Fastlane funding instruments. If the provided address is not available in the billing form on the storefront, it may result in a billing address error. Thus, please ensure that your billing address form supports all the necessary addresses.
2. Please see the [Fastlane documentation](#) in order to follow the latest recommendation regarding supported card types, and other eligibility options.